Tfh_subclustering

Load libraries

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.1     v stringr   1.5.1
v ggplot2   4.0.0     v tibble    3.2.1
v lubridate 1.9.4     v tidyr     1.3.1
v purrr     1.0.4
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(viridis)
```

```
Loading required package: viridisLite
```

```
library(viridisLite)
library(RColorBrewer)
library(Seurat)
```

```
Loading required package: SeuratObject
Loading required package: sp

Attaching package: 'SeuratObject'

The following objects are masked from 'package:base':

    intersect, t
```

```
library(SeuratObject)
library(patchwork)
library(scRepertoire)
library(reshape2)
```

```
Attaching package: 'reshape2'
```

```
The following object is masked from 'package:tidyr':

    smiths
```

```
library(pheatmap)
library(SingleCellExperiment)
```

```
Loading required package: SummarizedExperiment
Loading required package: MatrixGenerics
Loading required package: matrixStats

Attaching package: 'matrixStats'

The following object is masked from 'package:dplyr':

    count


Attaching package: 'MatrixGenerics'

The following objects are masked from 'package:matrixStats':

    colAlls, colAnyNAs, colAnys, colAvgsPerRowSet, colCollapse,
    colCounts, colCummaxs, colCummins, colCumprods, colCumsums,
    colDiffs, colIQRDiffs, colIQRs, colLogSumExps, colMadDiffs,
    colMads, colMaxs, colMeans2, colMedians, colMins, colOrderStats,
    colProds, colQuantiles, colRanges, colRanks, colSdDiffs, colSds,
    colSums2, colTabulates, colVarDiffs, colVars, colWeightedMads,
    colWeightedMeans, colWeightedMedians, colWeightedSds,
    colWeightedVars, rowAlls, rowAnyNAs, rowAnys, rowAvgsPerColSet,
    rowCollapse, rowCounts, rowCummaxs, rowCummins, rowCumprods,
    rowCumsums, rowDiffs, rowIQRDiffs, rowIQRs, rowLogSumExps,
    rowMadDiffs, rowMads, rowMaxs, rowMeans2, rowMedians, rowMins,
    rowOrderStats, rowProds, rowQuantiles, rowRanges, rowRanks,
    rowSdDiffs, rowSds, rowSums2, rowTabulates, rowVarDiffs, rowVars,
    rowWeightedMads, rowWeightedMeans, rowWeightedMedians,
    rowWeightedSds, rowWeightedVars

Loading required package: GenomicRanges
Loading required package: stats4
Loading required package: BiocGenerics

Attaching package: 'BiocGenerics'

The following object is masked from 'package:SeuratObject':

    intersect

The following objects are masked from 'package:lubridate':

    intersect, setdiff, union

The following objects are masked from 'package:dplyr':

    combine, intersect, setdiff, union
```

```
The following objects are masked from 'package:stats':

    IQR, mad, sd, var, xtabs

The following objects are masked from 'package:base':

    anyDuplicated, aperm, append, as.data.frame, basename, cbind,
    colnames, dirname, do.call, duplicated, eval, evalq, Filter, Find,
    get, grep, grepl, intersect, is.unsorted, lapply, Map, mapply,
    match, mget, order, paste, pmax, pmax.int, pmin, pmin.int,
    Position, rank, rbind, Reduce, rownames, sapply, saveRDS, setdiff,
    table, tapply, union, unique, unsplit, which.max, which.min

Loading required package: S4Vectors

Attaching package: 'S4Vectors'

The following objects are masked from 'package:lubridate':

    second, second<-

The following objects are masked from 'package:dplyr':

    first, rename

The following object is masked from 'package:tidyr':

    expand

The following object is masked from 'package:utils':

    findMatches

The following objects are masked from 'package:base':

    expand.grid, I, unname

Loading required package: IRanges

Attaching package: 'IRanges'

The following object is masked from 'package:sp':

    %over%

The following object is masked from 'package:lubridate':

    %within%

The following objects are masked from 'package:dplyr':

    collapse, desc, slice

The following object is masked from 'package:purrr':

    reduce
```

```
The following object is masked from 'package:grDevices':

    windows

Loading required package: GenomeInfoDb
Loading required package: Biobase
Welcome to Bioconductor

    Vignettes contain introductory material; view with
    'browseVignettes()'. To cite Bioconductor, see
    'citation("Biobase")', and for packages 'citation("pkgname")'.


Attaching package: 'Biobase'

The following object is masked from 'package:MatrixGenerics':

    rowMedians

The following objects are masked from 'package:matrixStats':

    anyMissing, rowMedians


Attaching package: 'SummarizedExperiment'

The following object is masked from 'package:Seurat':

    Assays

The following object is masked from 'package:SeuratObject':

    Assays
```

```
library(tidyr)
library(readxl)
library(DESeq2)
library(SummarizedExperiment)
library(GSEABase)
```

```
Loading required package: annotate
Loading required package: AnnotationDbi

Attaching package: 'AnnotationDbi'

The following object is masked from 'package:dplyr':

    select

Loading required package: XML

Attaching package: 'XML'

The following object is masked from 'package:patchwork':
```

```
    free
```

```
Loading required package: graph
```

```
Attaching package: 'graph'
```

```
The following object is masked from 'package:XML':

    addNode
```

```
The following object is masked from 'package:stringr':

    boundary
```

```
library(GSVA)
library(ggbiplot)
```

```
Attaching package: 'ggbiplot'
```

```
The following object is masked from 'package:IRanges':

    reflect
```

```
library(glmGamPoi)
```

```
Attaching package: 'glmGamPoi'
```

```
The following object is masked from 'package:dplyr':

    vars
```

```
The following object is masked from 'package:ggplot2':

    vars
```

```
library(scuttle)
library(limma)
```

```
Attaching package: 'limma'
```

```
The following object is masked from 'package:DESeq2':

    plotMA
```

```
The following object is masked from 'package:BiocGenerics':

    plotMA
```

```
library(rstatix)
```

```
Attaching package: 'rstatix'

The following object is masked from 'package:AnnotationDbi':

    select

The following object is masked from 'package:IRanges':

    desc

The following object is masked from 'package:stats':

    filter
```

```
library(data.table)
```

```
Attaching package: 'data.table'

The following object is masked from 'package:SummarizedExperiment':

    shift

The following object is masked from 'package:GenomicRanges':

    shift

The following object is masked from 'package:IRanges':

    shift

The following objects are masked from 'package:S4Vectors':

    first, second

The following objects are masked from 'package:reshape2':

    dcast, melt

The following objects are masked from 'package:lubridate':

    hour, isoweek, mday, minute, month, quarter, second, wday, week,
    yday, year

The following objects are masked from 'package:dplyr':

    between, first, last

The following object is masked from 'package:purrr':

    transpose
```

```
library(ggrepel)
# remotes::install_github("huayc09/SeuratExtend")
library(SeuratExtend)
```

```
Loading required package: SeuratExtendData
```

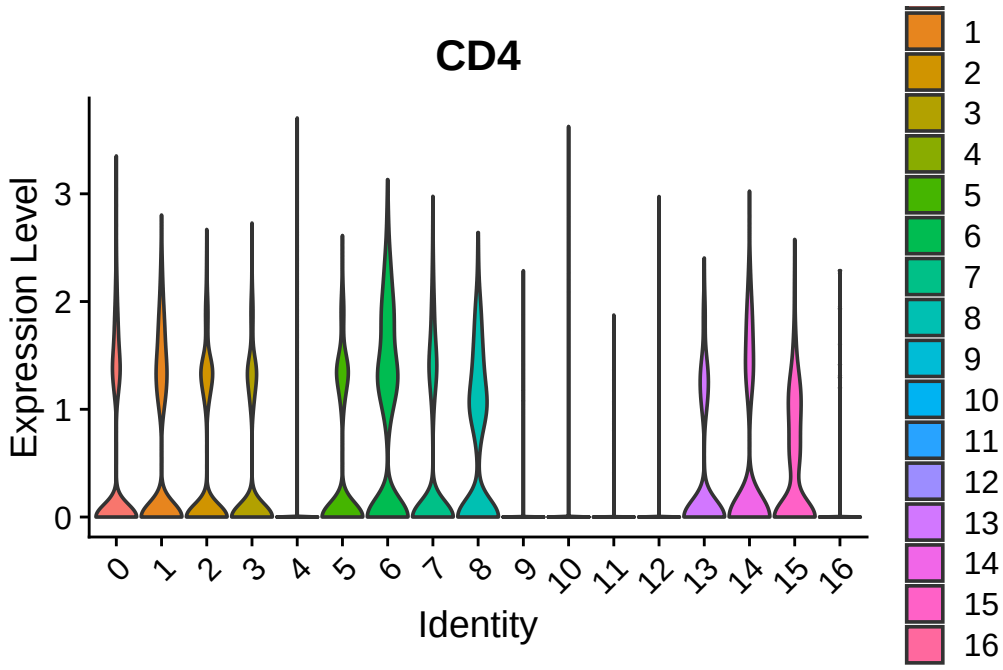
```
library(circlize)
```

```
========================================
circlize version 0.4.16
CRAN page: https://cran.r-project.org/package=circlize
Github page: https://github.com/jokergoo/circlize
Documentation: https://jokergoo.github.io/circlize_book/book/

If you use it in published research, please cite:
Gu, Z. circlize implements and enhances circular visualization
  in R. Bioinformatics 2014.

This message can be suppressed by:
  suppressPackageStartupMessages(library(circlize))
========================================


Attaching package: 'circlize'

The following object is masked from 'package:graph':

    degree
```

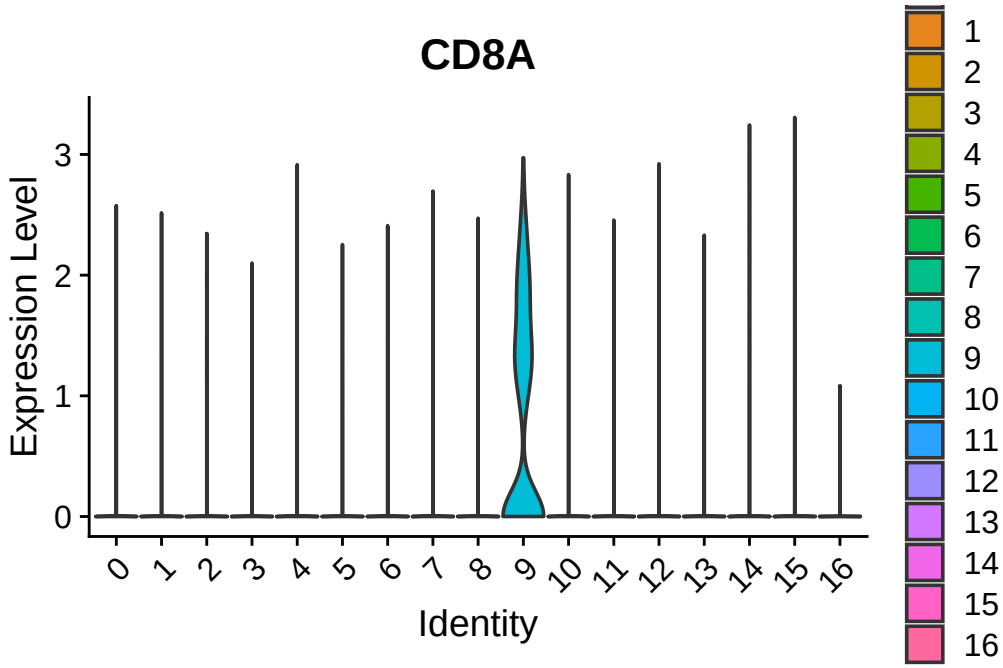
Find CD4 subsets

```
obj1 <- readRDS("lymph_node_cd4s_reclustered.rds")
VlnPlot(obj1, features = c('CD4'), pt.size = 0)  # clusters 0,1,2,3,4,5,6,7,8,10,14,15  though not sure about 4
```

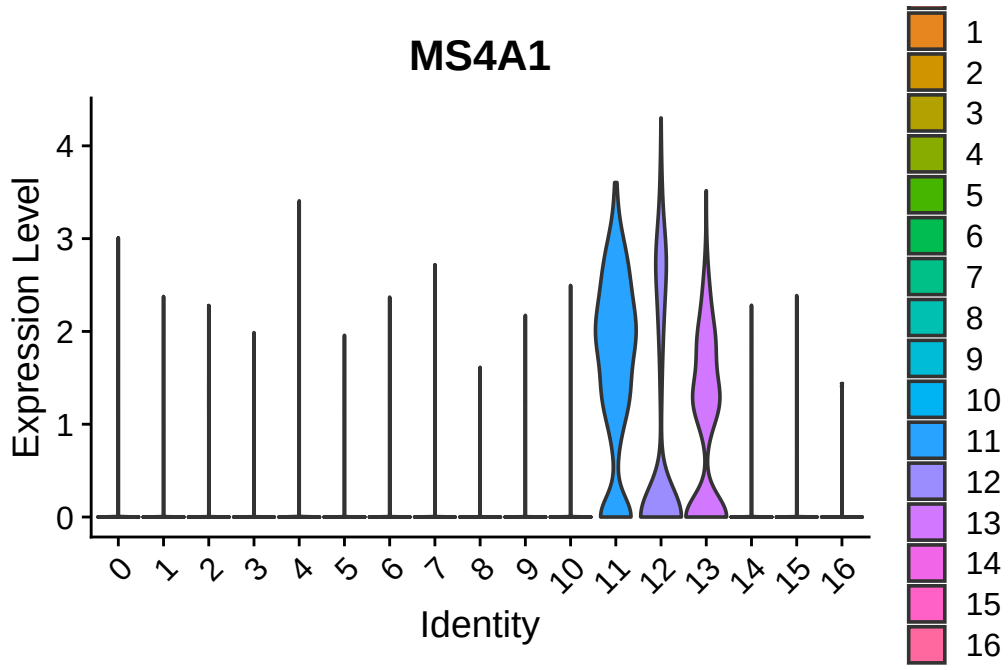
```
Warning: `aes_string()` was deprecated in ggplot2 3.0.0.
i Please use tidy evaluation idioms with `aes()`.
i See also `vignette("ggplot2-in-packages")` for more information.
i The deprecated feature was likely used in the Seurat package.
  Please report the issue at <https://github.com/satijalab/seurat/issues>.
```



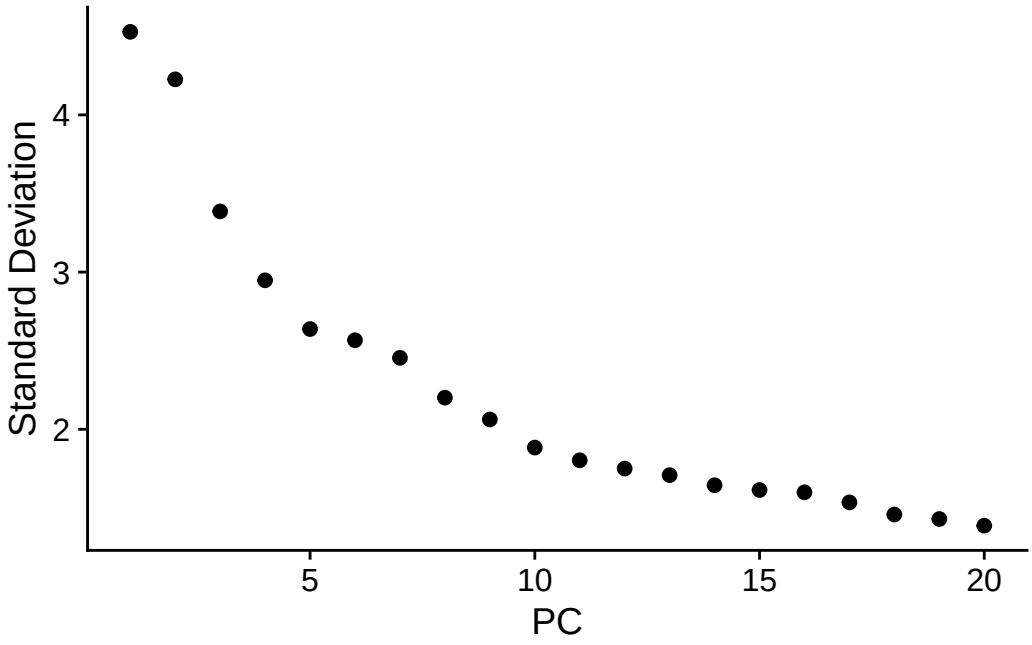

```
VlnPlot(obj1, features = c('CD8A'), pt.size = 0)  # cluster 9
```




```
VlnPlot(obj1, features = c('MS4A1'), pt.size = 0) # clusters 11,12,13
```




```
obj1.clean <- subset(obj1, subset = seurat_clusters %in% c(0:8,10,14,15))
obj1.clean$age_group <- ifelse(obj1.clean$Age < 50, "Younger", "Older")
obj1.clean <- RunPCA(obj1.clean, verbose = FALSE); ElbowPlot(obj1.clean)
```




```
obj1.clean <- harmony::RunHarmony(obj1.clean, group.by.vars = "orig.ident", assay.use = "SCT")
```

```
Transposing data matrix
```

```
Initializing state using k-means centroids initialization
```

```
Harmony 1/10
```

```
Harmony 2/10

Harmony converged after 2 iterations
```

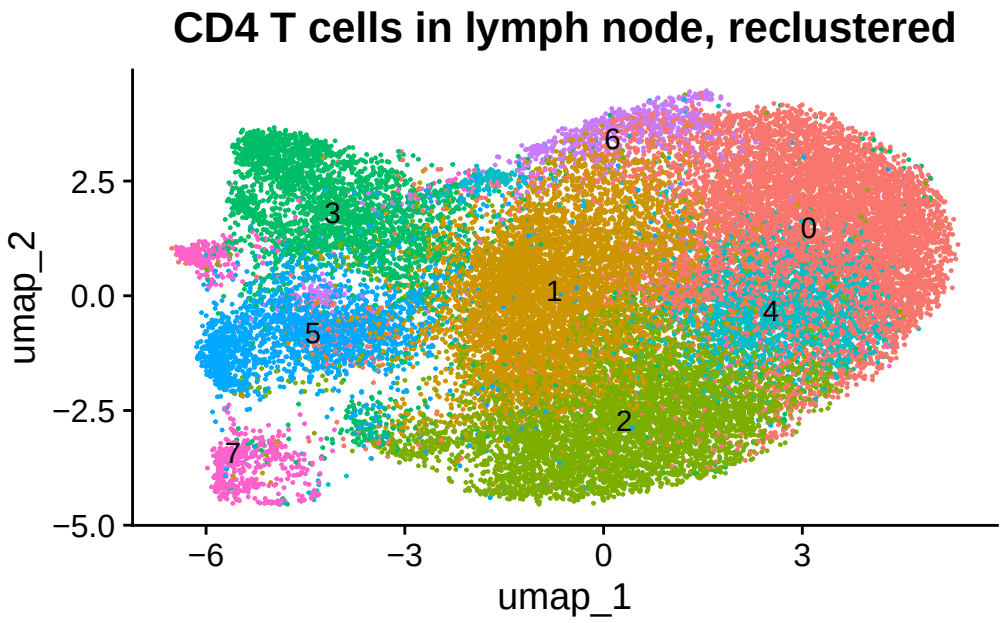
```
obj1.clean <- FindNeighbors(obj1.clean, dims = 1:10, reduction='harmony')
```

```
Computing nearest neighbor graph

Computing SNN
```

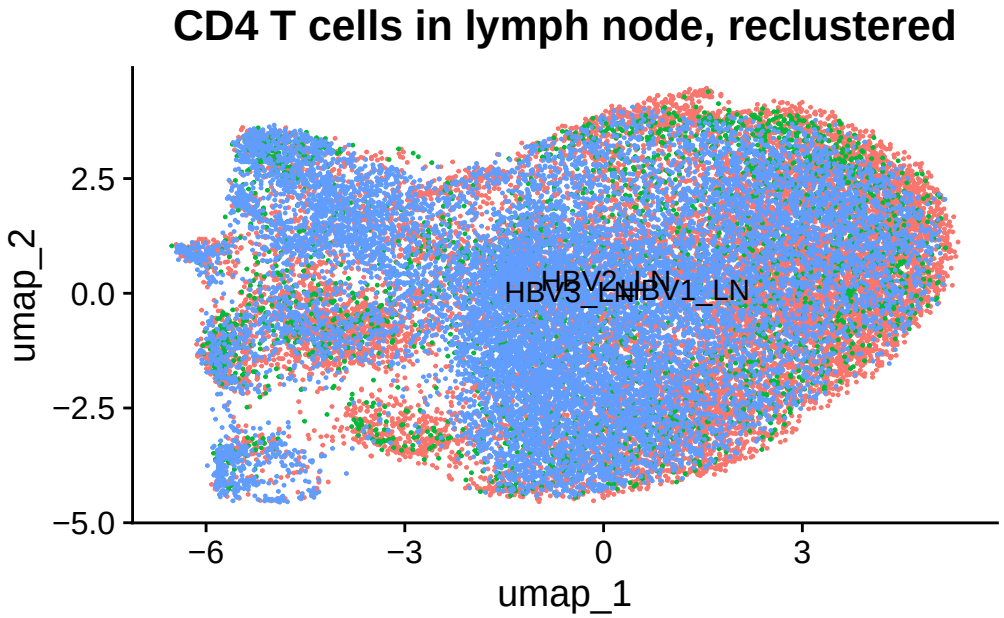
```
obj1.clean <- FindClusters(obj1.clean, resolution = 0.5, verbose = FALSE)
obj1.clean <- RunUMAP(obj1.clean, dims = 1:20, reduction='harmony')
```

```
Warning: The default method for RunUMAP has changed from calling Python UMAP via reticulate to the R-
native UWOT using the cosine metric
To use Python UMAP via reticulate, set umap.method to 'umap-learn' and metric to 'correlation'
This message will be shown once per session

19:23:45 UMAP embedding parameters a = 0.9922 b = 1.112

19:23:45 Read 29118 rows and found 20 numeric columns

19:23:45 Using Annoy for neighbor search, n_neighbors = 30

19:23:45 Building Annoy index with metric = cosine, n_trees = 50

0%   10   20   30   40   50   60   70   80   90   100%

[----|----|----|----|----|----|----|----|----|----|

**************************************************|
19:23:46 Writing NN index file to temp file C:\Users\heratr01\AppData\Local\Temp\RtmpK6wtG7\filed490682a7662
19:23:46 Searching Annoy index using 1 thread, search_k = 3000
19:23:53 Annoy recall = 100%
19:23:53 Commencing smooth kNN distance calibration using 1 thread with target n_neighbors = 30
19:23:55 Initializing from normalized Laplacian + noise (using RSpectra)
19:23:55 Commencing optimization for 200 epochs, with 1259974 positive edges
19:23:55 Using rng type: pcg
19:24:15 Optimization finished
```

```
DimPlot(obj1.clean, group.by = "seurat_clusters",label = TRUE) + NoLegend() + ggtitle("CD4 T cells in lymph node
```




```
DimPlot(obj1.clean, group.by = "orig.ident", label = TRUE) + NoLegend() + ggtitle("CD4 T cells in lymph node, re
```

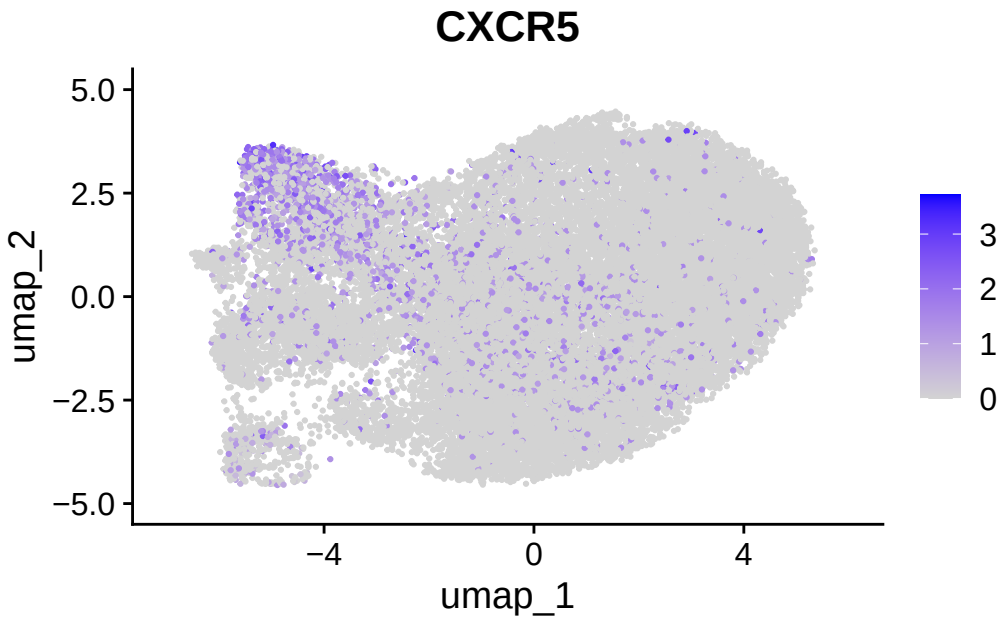



```
top.genes <- FindAllMarkers(obj1.clean, only.pos = TRUE, min.pct = 0.25,
                            logfc.threshold = 0.25)
```

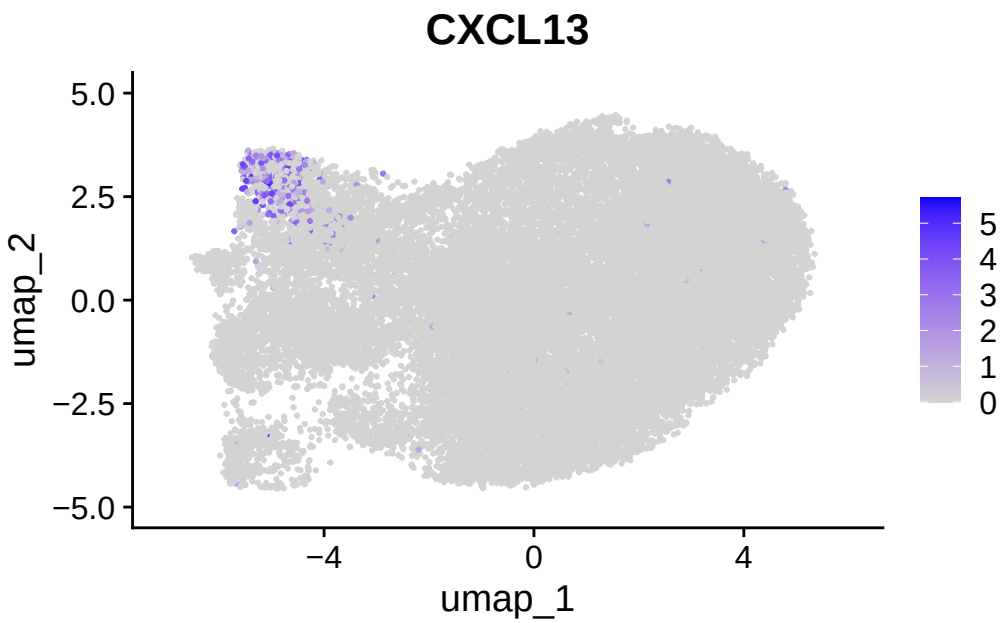
```
Calculating cluster 0
Calculating cluster 1
Calculating cluster 2
Calculating cluster 3
Calculating cluster 4
Calculating cluster 5
```

```
Calculating cluster 6
Calculating cluster 7
```

```
# write.csv(top.genes, file = 'outs/CD4-subsets-FindAllMarkers.csv')
top20 <- top.genes %>% group_by(cluster) %>% slice_max(n = 50, order_by = avg_log2FC)
FeaturePlot(obj1.clean, features = c('CXCR5'), pt.size = 0.3)
```

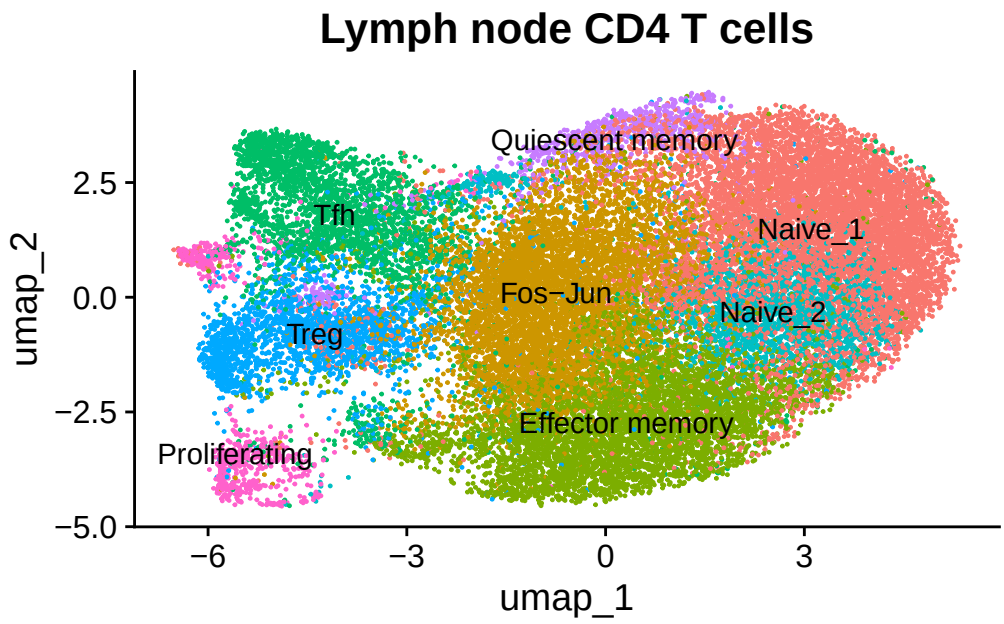



```
FeaturePlot(obj1.clean, features = c('CXCL13'), pt.size = 0.3)
```




```
# cluster annotations
# cluster 0: Naive_1 (+LEF1,+SELL)
```

```
# cluster 1: Fos-Jun activated
# cluster 2: Effector memory
# cluster 3: Tfh (+CXCR5,+BCL6)
# cluster 4: Naive_2 TIGIT memory
# cluster 5: Treg (+FOXP3,+PRDM1)
# cluster 6: Quiescent memory
# cluster 7: Proliferating (+MKI67, +BIRC5)

new.cluster.ids <- c( "0" = "Naive_1", "1" = "Fos-Jun", "2" = "Effector memory",
                      "3" = "Tfh", "4" = "Naive_2",
                      "5" = "Treg", "6" = "Quiescent memory", "7" = "Proliferating")
obj1.clean <- RenameIdents(obj1.clean, new.cluster.ids)
obj1.clean$CD4.subset.annotation <- Idents(obj1.clean)
DimPlot(obj1.clean, reduction = "umap", label = TRUE, group.by = 'CD4.subset.annotation') + NoLegend() + ggtitle
```




```
# ggsave(filename = 'Images/CD4-UMAP-subsets.pdf', height=5, width=5 )

top.genes <- FindAllMarkers(obj1.clean, only.pos = TRUE, min.pct = 0.25,
                        logfc.threshold = 0.25)
```

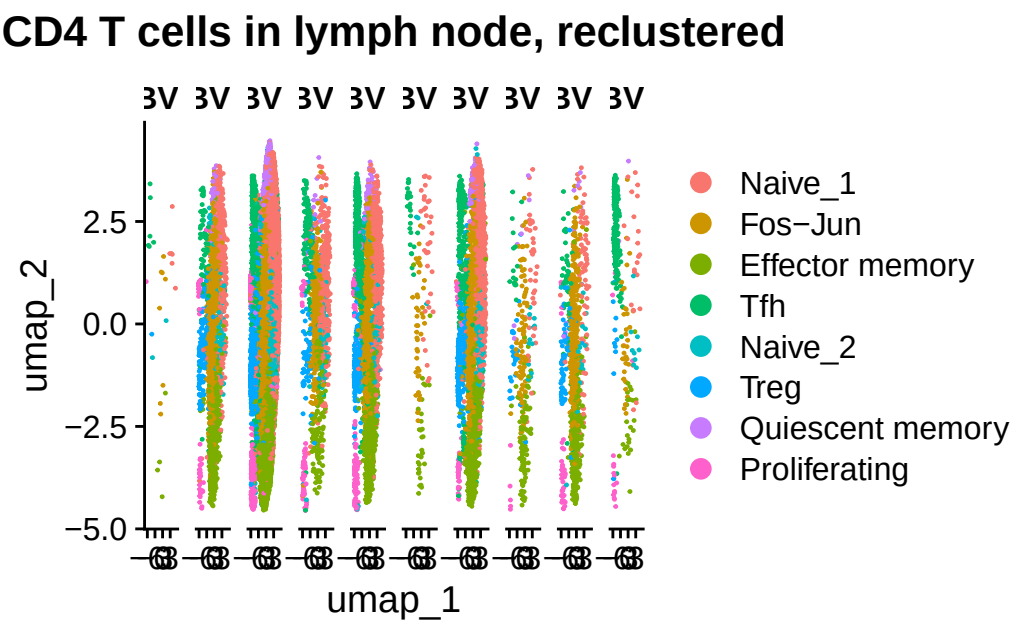
```
Calculating cluster Naive_1
Calculating cluster Fos-Jun
Calculating cluster Effector memory
Calculating cluster Tfh
Calculating cluster Naive_2
Calculating cluster Treg
Calculating cluster Quiescent memory
Calculating cluster Proliferating
```

```
# write.csv(top.genes, file = 'outs/CD4-subsets-FindAllMarkers.csv')

data.table::as.data.table(table(obj1.clean$CD4.subset.annotation, obj1.clean$subject))
```

```
                V1            V2     N
             <char>        <char> <int>
 1:          Naive_1 NLV-HBVX-0022     6
 2:          Fos-Jun NLV-HBVX-0022     8
 3:  Effector memory NLV-HBVX-0022     4
 4:              Tfh NLV-HBVX-0022     6
 5:          Naive_2 NLV-HBVX-0022     2
 6:             Treg NLV-HBVX-0022     1
 7: Quiescent memory NLV-HBVX-0022     0
 8:    Proliferating NLV-HBVX-0022     1
 9:          Naive_1 NLV-HBVX-0029   357
10:          Fos-Jun NLV-HBVX-0029   827
11:  Effector memory NLV-HBVX-0029   495
12:              Tfh NLV-HBVX-0029    78
13:          Naive_2 NLV-HBVX-0029   102
14:             Treg NLV-HBVX-0029   130
15: Quiescent memory NLV-HBVX-0029    42
16:    Proliferating NLV-HBVX-0029    46
17:          Naive_1 NLV-HBVX-0033  5604
18:          Fos-Jun NLV-HBVX-0033  2836
19:  Effector memory NLV-HBVX-0033  3215
20:              Tfh NLV-HBVX-0033  1181
21:          Naive_2 NLV-HBVX-0033  1838
22:             Treg NLV-HBVX-0033  1488
23: Quiescent memory NLV-HBVX-0033   572
24:    Proliferating NLV-HBVX-0033   361
25:          Naive_1 NLV-HBVX-0036   538
26:          Fos-Jun NLV-HBVX-0036   534
27:  Effector memory NLV-HBVX-0036   208
28:              Tfh NLV-HBVX-0036   224
29:          Naive_2 NLV-HBVX-0036    90
30:             Treg NLV-HBVX-0036    92
31: Quiescent memory NLV-HBVX-0036    10
32:    Proliferating NLV-HBVX-0036    71
33:          Naive_1 NLV-HBVX-0041   568
34:          Fos-Jun NLV-HBVX-0041  1059
35:  Effector memory NLV-HBVX-0041   796
36:              Tfh NLV-HBVX-0041   845
37:          Naive_2 NLV-HBVX-0041   146
38:             Treg NLV-HBVX-0041   210
39: Quiescent memory NLV-HBVX-0041    53
40:    Proliferating NLV-HBVX-0041   138
41:          Naive_1 NLV-HBVX-0043    34
42:          Fos-Jun NLV-HBVX-0043    44
43:  Effector memory NLV-HBVX-0043    31
44:              Tfh NLV-HBVX-0043    24
45:          Naive_2 NLV-HBVX-0043     3
46:             Treg NLV-HBVX-0043     0
47: Quiescent memory NLV-HBVX-0043     0
48:    Proliferating NLV-HBVX-0043     0
49:          Naive_1 NLV-HBVX-0058   774
50:          Fos-Jun NLV-HBVX-0058   516
51:  Effector memory NLV-HBVX-0058   453
52:              Tfh NLV-HBVX-0058   358
53:          Naive_2 NLV-HBVX-0058   110
54:             Treg NLV-HBVX-0058   276
```

```
55: Quiescent memory NLV-HBVX-0058    90
56:    Proliferating NLV-HBVX-0058    57
57:          Naive_1 NLV-HBVX-0063    37
58:          Fos-Jun NLV-HBVX-0063    90
59:  Effector memory NLV-HBVX-0063    96
60:              Tfh NLV-HBVX-0063    15
61:          Naive_2 NLV-HBVX-0063    12
62:             Treg NLV-HBVX-0063    24
63: Quiescent memory NLV-HBVX-0063     7
64:    Proliferating NLV-HBVX-0063     5
65:          Naive_1 NLV-HBVX-0069    76
66:          Fos-Jun NLV-HBVX-0069   403
67:  Effector memory NLV-HBVX-0069   272
68:              Tfh NLV-HBVX-0069    53
69:          Naive_2 NLV-HBVX-0069    23
70:             Treg NLV-HBVX-0069    70
71: Quiescent memory NLV-HBVX-0069     9
72:    Proliferating NLV-HBVX-0069    41
73:          Naive_1 NLV-HBVX-0076    24
74:          Fos-Jun NLV-HBVX-0076    35
75:  Effector memory NLV-HBVX-0076    24
76:              Tfh NLV-HBVX-0076   211
77:          Naive_2 NLV-HBVX-0076     9
78:             Treg NLV-HBVX-0076    10
79: Quiescent memory NLV-HBVX-0076     2
80:    Proliferating NLV-HBVX-0076    18
                  V1             V2     N
```

```
DimPlot(obj1.clean, split.by = "subject", group.by='CD4.subset.annotation', label = F) + ggtitle("CD4 T cells in
```




```
# ggsave(filename = 'Images/CD4-UMAP-splitSubject.pdf', height=4, width=18 )

df <- data.frame(
  subject = c( "NLV-HBVX-0022", "NLV-HBVX-0029",  "NLV-HBVX-0033", "NLV-HBVX-0036", "NLV-HBVX-0041", "NLV-HBVX-0
```

```
              "NLV-HBVX-0058", "NLV-HBVX-0063", "NLV-HBVX-0069", "NLV-HBVX-0076"),
  antibody_titer = c(350,  0.1, 27.79851,  10.25068563, 377.7495387, 135, 476.1468, 55.41045, 275, 443.3457 )
new.meta <- obj1.clean@meta.data %>% left_join(df, by = "subject")
rownames(new.meta) <- rownames(obj1.clean@meta.data)
obj1.clean@meta.data <- new.meta
obj1.clean$titer.categ <- ifelse(obj1.clean$antibody_titer < 100, "Under100", "Over100")


rm(df, obj1)
```

HBV clonotypes overlay

```
x <- obj1.clean@meta.data
x<- x %>% separate(CTaa, into = c("TRA","TRB"), sep = "_", remove=F)
x <- subset(x, TRB != "NA")
x <- x[,c("barcode","TRB")]
x <- x %>%  separate(TRB, into = c("TRB", "TRB2"), sep = ";")
```

```
Warning: Expected 2 pieces. Missing pieces filled with `NA` in 19053 rows [1, 2, 3, 4,
5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, ...].
```

```
x2 <- x[,c("barcode","TRB")]
x3 <- x[,c("barcode", "TRB2")]
x3 <- subset(x3, TRB2 != "NA")
colnames(x3) <- c("barcode","TRB")
meta4 <- rbind(x2, x3)  # now contains all of the TCRB in tonsil CD4

# matches in cd4s ln vs hep b iedb
cd4seq <- readxl::read_excel("iedb_hepatitis_b_epitopes.xlsx")
```

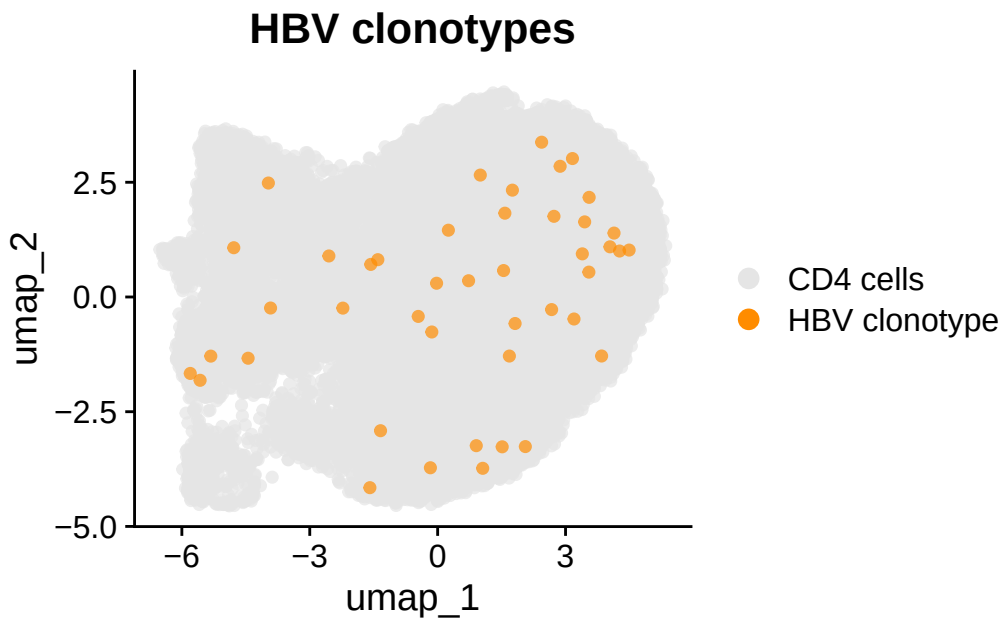
```
New names:
* `Receptor` -> `Receptor...1`
* `Receptor` -> `Receptor...2`
* `Receptor` -> `Receptor...3`
* `Receptor` -> `Receptor...4`
* `Epitope` -> `Epitope...6`
* `Epitope` -> `Epitope...7`
* `Epitope` -> `Epitope...8`
* `Epitope` -> `Epitope...9`
* `Assay` -> `Assay...10`
* `Assay` -> `Assay...11`
* `Assay` -> `Assay...12`
* `Chain 1` -> `Chain 1...13`
* `Chain 1` -> `Chain 1...14`
* `Chain 1` -> `Chain 1...15`
* `Chain 1` -> `Chain 1...16`
* `Chain 1` -> `Chain 1...17`
* `Chain 1` -> `Chain 1...18`
* `Chain 1` -> `Chain 1...19`
* `Chain 1` -> `Chain 1...20`
* `Chain 1` -> `Chain 1...21`
* `Chain 1` -> `Chain 1...22`
* `Chain 1` -> `Chain 1...23`
```

```
* `Chain 1` -> `Chain 1...24`
* `Chain 1` -> `Chain 1...25`
* `Chain 1` -> `Chain 1...26`
* `Chain 1` -> `Chain 1...27`
* `Chain 1` -> `Chain 1...28`
* `Chain 1` -> `Chain 1...29`
* `Chain 1` -> `Chain 1...30`
* `Chain 1` -> `Chain 1...31`
* `Chain 1` -> `Chain 1...32`
* `Chain 1` -> `Chain 1...33`
* `Chain 1` -> `Chain 1...34`
* `Chain 1` -> `Chain 1...35`
* `Chain 1` -> `Chain 1...36`
* `Chain 1` -> `Chain 1...37`
* `Chain 1` -> `Chain 1...38`
* `Chain 1` -> `Chain 1...39`
* `Chain 1` -> `Chain 1...40`
* `Chain 1` -> `Chain 1...41`
* `Chain 2` -> `Chain 2...42`
* `Chain 2` -> `Chain 2...43`
* `Chain 2` -> `Chain 2...44`
* `Chain 2` -> `Chain 2...45`
* `Chain 2` -> `Chain 2...46`
* `Chain 2` -> `Chain 2...47`
* `Chain 2` -> `Chain 2...48`
* `Chain 2` -> `Chain 2...49`
* `Chain 2` -> `Chain 2...50`
* `Chain 2` -> `Chain 2...51`
* `Chain 2` -> `Chain 2...52`
* `Chain 2` -> `Chain 2...53`
* `Chain 2` -> `Chain 2...54`
* `Chain 2` -> `Chain 2...55`
* `Chain 2` -> `Chain 2...56`
* `Chain 2` -> `Chain 2...57`
* `Chain 2` -> `Chain 2...58`
* `Chain 2` -> `Chain 2...59`
* `Chain 2` -> `Chain 2...60`
* `Chain 2` -> `Chain 2...61`
* `Chain 2` -> `Chain 2...62`
* `Chain 2` -> `Chain 2...63`
* `Chain 2` -> `Chain 2...64`
* `Chain 2` -> `Chain 2...65`
* `Chain 2` -> `Chain 2...66`
* `Chain 2` -> `Chain 2...67`
* `Chain 2` -> `Chain 2...68`
* `Chain 2` -> `Chain 2...69`
* `Chain 2` -> `Chain 2...70`
```

```
cd4seq <- cd4seq[,c('Chain 2...53')]  # the 53rd column has the CDR3
cd4seq <- distinct(cd4seq)  # ~500 duplicates removed

m <- merge(meta4, cd4seq, by.x="TRB", by.y="Chain 2...53")  # map one dataset to another

hbv.specific <- m$barcode

DimPlot(obj1.clean, label=F, cells.highlight = hbv.specific,
```

```
        cols.highlight = "darkorange", cols= "grey", pt.size=1.5, sizes.highlight=1.5,
        group.by = 'CD4.subset.annotation', alpha=0.7) +
  ggtitle("HBV clonotypes") +
  scale_color_manual(labels = c("CD4 cells","HBV clonotype"),
                     values = c("grey90", "darkorange"))
```

```
Scale for colour is already present.
Adding another scale for colour, which will replace the existing scale.
```



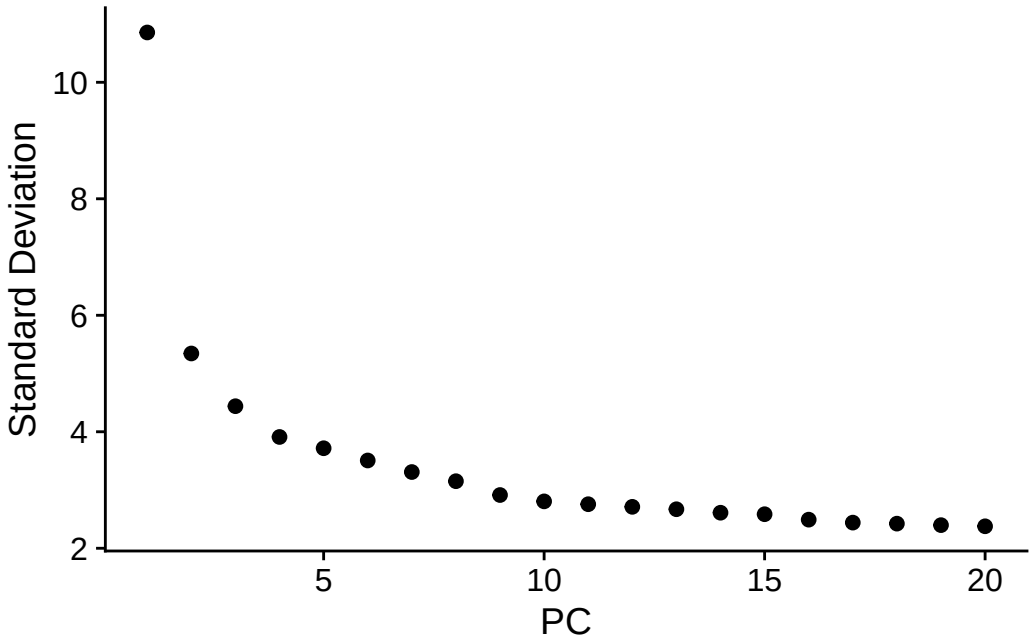

```
# ggsave(filename = 'Images/CD4-matchCDR3-HBV.pdf', height=6, width=8 )

rstatix::fisher_test(matrix(c(nrow(m),nrow(cd4seq),nrow(obj1.clean@meta.data),1e8), nrow=2))
```

```
# A tibble: 1 x 3
          n        p p.signif
*     <dbl>    <dbl> <chr>
1 100033096 6.34e-53 ****
```

```
# A tibble: 1 × 3
#          n        p p.signif
#*     <dbl>    <dbl> <chr>
#1 100006973 8.13e-96 ****

# rm(x2, x3, meta4, cd4seq, m, cellname, b)
```

Subcluster Tfh

```
tfh <- subset(obj1.clean, CD4.subset.annotation == "Tfh")
tfh <- RunPCA(tfh, verbose = FALSE); ElbowPlot(tfh)
```




```
tfh <- harmony::RunHarmony(tfh, group.by.vars = "orig.ident", assay.use = "SCT")
```

```
Transposing data matrix

Initializing state using k-means centroids initialization

Harmony 1/10

Harmony 2/10

Harmony 3/10

Harmony converged after 3 iterations
```

```
tfh <- FindNeighbors(tfh, dims = 1:10, reduction='harmony') #adjust based on elbow plot
```

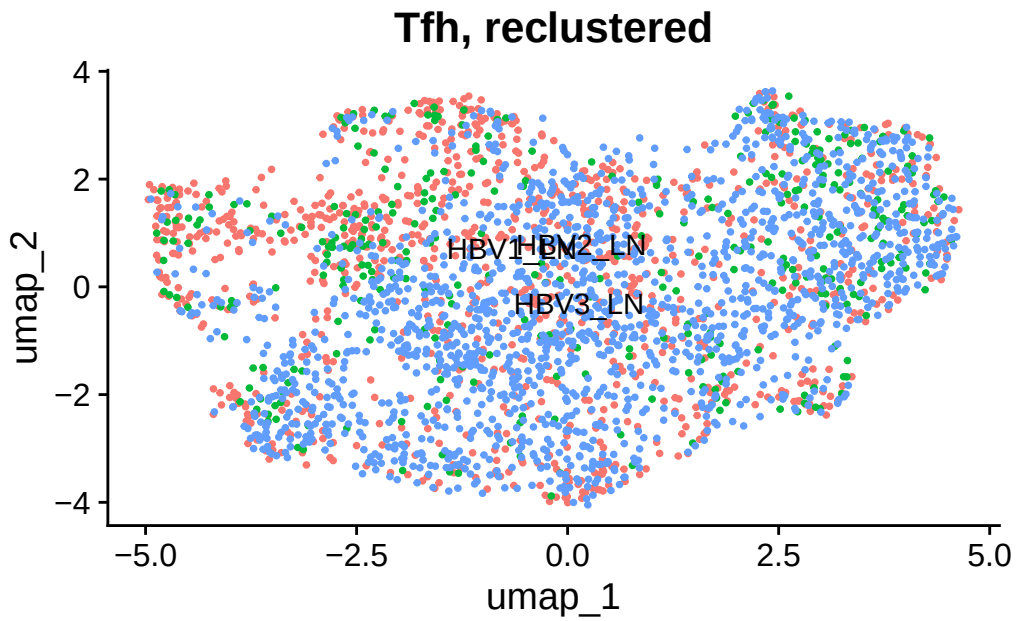
```
Computing nearest neighbor graph

Computing SNN
```

```
tfh <- FindClusters(tfh, resolution = 0.6, verbose = FALSE)
tfh <- RunUMAP(tfh, dims = 1:20, reduction='harmony') #adjust based on elbow plot
```

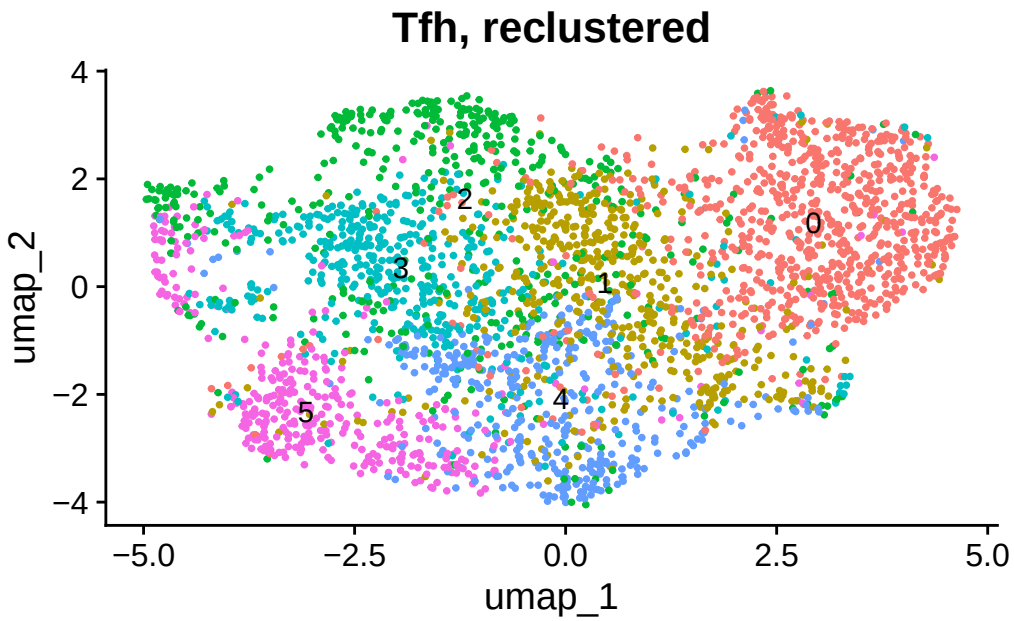
```
19:25:15 UMAP embedding parameters a = 0.9922 b = 1.112

19:25:15 Read 2995 rows and found 20 numeric columns

19:25:15 Using Annoy for neighbor search, n_neighbors = 30

19:25:15 Building Annoy index with metric = cosine, n_trees = 50
```

```
0%   10   20   30   40   50   60   70   80   90   100%

[----|----|----|----|----|----|----|----|----|----|

**************************************************|
19:25:15 Writing NN index file to temp file C:\Users\heratr01\AppData\Local\Temp\RtmpK6wtG7\filed4902b084ef4
19:25:15 Searching Annoy index using 1 thread, search_k = 3000
19:25:15 Annoy recall = 100%
19:25:16 Commencing smooth kNN distance calibration using 1 thread with target n_neighbors = 30
19:25:17 Initializing from normalized Laplacian + noise (using RSpectra)
19:25:17 Commencing optimization for 500 epochs, with 120576 positive edges
19:25:17 Using rng type: pcg
19:25:23 Optimization finished
```

```
DimPlot(tfh, group.by = "orig.ident",label = TRUE) + NoLegend() + ggtitle("Tfh, reclustered")
```



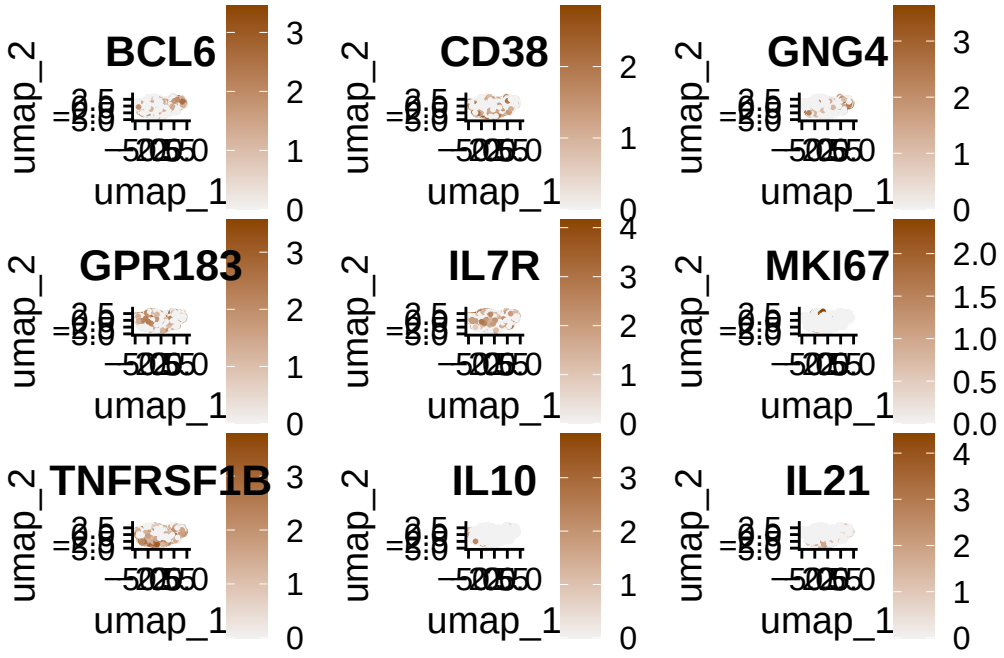

```
DimPlot(tfh, group.by = "seurat_clusters",label = TRUE) + NoLegend() + ggtitle("Tfh, reclustered")
```



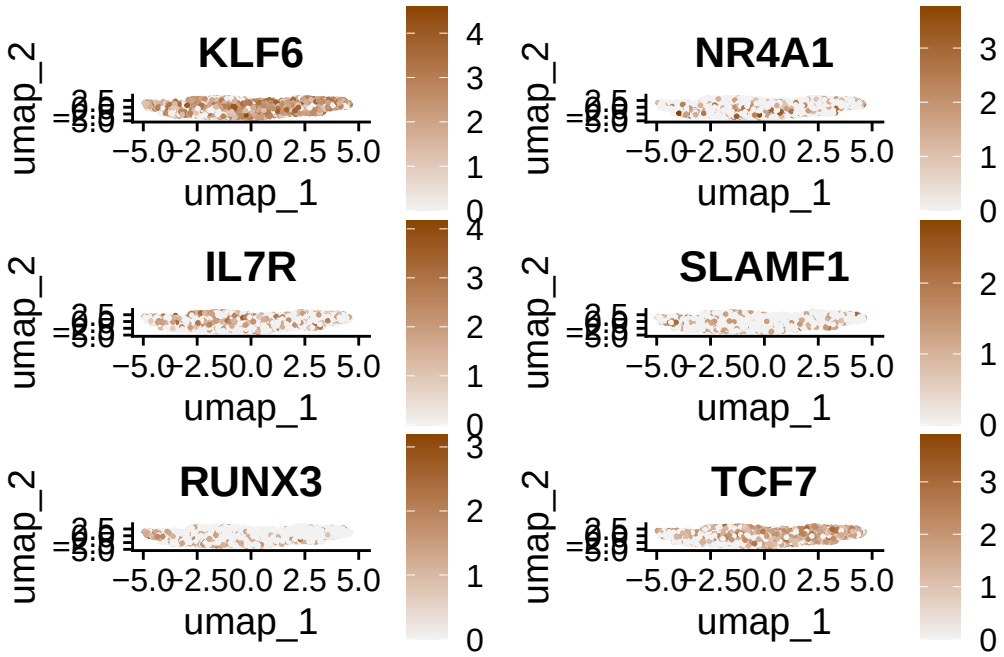

```
top20 <- FindAllMarkers(tfh, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25) %>%
  group_by(cluster) %>% slice_max(n = 50, order_by = avg_log2FC)
```

```
Calculating cluster 0
Calculating cluster 1
Calculating cluster 2
Calculating cluster 3
Calculating cluster 4
Calculating cluster 5
```

```
# cluster annotations
# cluster 0: GNG4
# cluster 1: KLF6/HSP
# cluster 2: KLF2/IL7R
# cluster 3: GPR183
# cluster 4: LAG3/VIM
# cluster 5: CD38/TNFR2
col.palette  <- viridis(n = 100, option = "viridis", direction=-1)
FeaturePlot(tfh, features = c('BCL6','CD38','GNG4','GPR183','IL7R','MKI67','TNFRSF1B',
                              'IL10','IL21'), pt.size = 0.2,
            cols=c('gray95','darkorange4') )
```

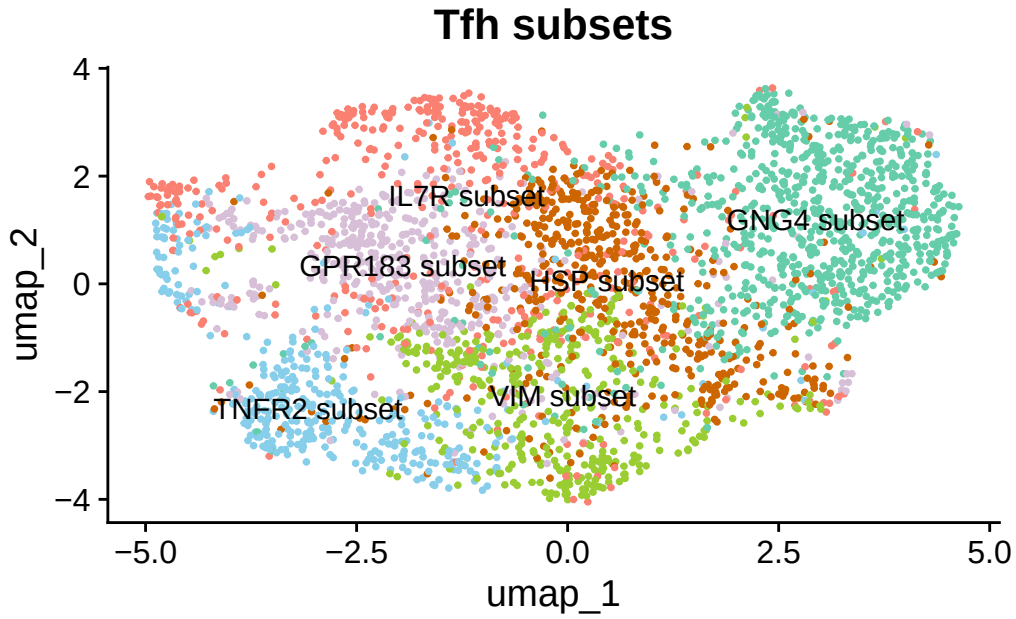



```
# ggsave(filename = 'Images/Tfh-FeaturePlot-selectedGenes.pdf',height=9, width=10)

FeaturePlot(tfh, features = c('KLF6','NR4A1','IL7R','SLAMF1','RUNX3','TCF7'),
            pt.size = 0.2, cols=c('gray95','darkorange4') )
```



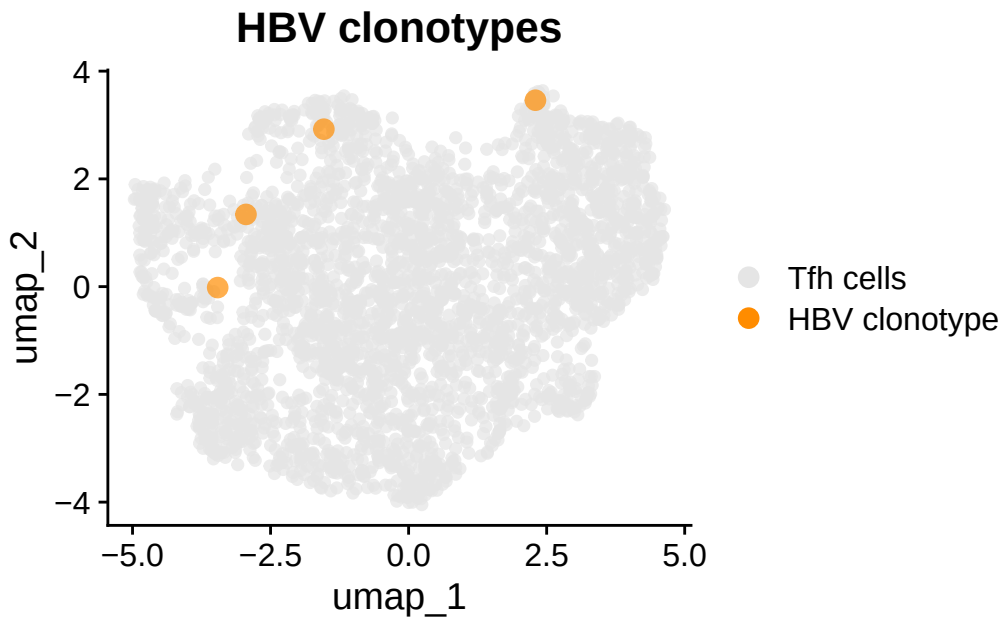

```
new.cluster.ids <- c( "0"="GNG4 subset", "1"="HSP subset", "2"="IL7R subset",
                      "3"="GPR183 subset", "4"="VIM subset", "5"="TNFR2 subset")
tfh <- RenameIdents(tfh, new.cluster.ids)
tfh$tfh.subset.annotation <- Idents(tfh)
subset.cols <- c(
  "GNG4 subset"    = "mediumaquamarine", # Mature GC-Tfh
  "HSP subset"     = "darkorange3",      # Stressed/Early (The darker orange you requested)
```

```
  "IL7R subset"    = "salmon",           # Quiescent/Memory-like
  "GPR183 subset" = "thistle",          # Transitional/Outer-follicle
  "VIM subset"     = "yellowgreen",     # Highly active/Invasive
  "TNFR2 subset"   = "skyblue"          # Early Activation/Starter
)
DimPlot(tfh, reduction = "umap", label = TRUE, group.by = 'tfh.subset.annotation',
        cols = subset.cols) + NoLegend() + ggtitle("Tfh subsets")
```




```
# ggsave(filename = 'Images/Tfh-UMAP-subsetLabels.pdf',height=4, width=4)



DimPlot(tfh, label=F, cells.highlight = hbv.specific,
        cols.highlight = "darkorange", cols= "grey", pt.size=1.5, sizes.highlight=3,
        group.by = 'CD4.subset.annotation', alpha=0.7) +
  ggtitle("HBV clonotypes") +
  scale_color_manual(labels = c("Tfh cells","HBV clonotype"),
                     values = c("grey90", "darkorange"))
```

```
Scale for colour is already present.
Adding another scale for colour, which will replace the existing scale.
```




```
# ggsave(filename = 'Images/Tfh-matchCDR3-HBV.pdf', height=6, width=8 )

rstatix::fisher_test(matrix(c(4,nrow(cd4seq),nrow(tfh@meta.data),1e8), nrow=2))
```

```
# A tibble: 1 x 3
          n          p p.signif
*     <dbl>      <dbl> <chr>
1 100006933 0.00000735 ****
```

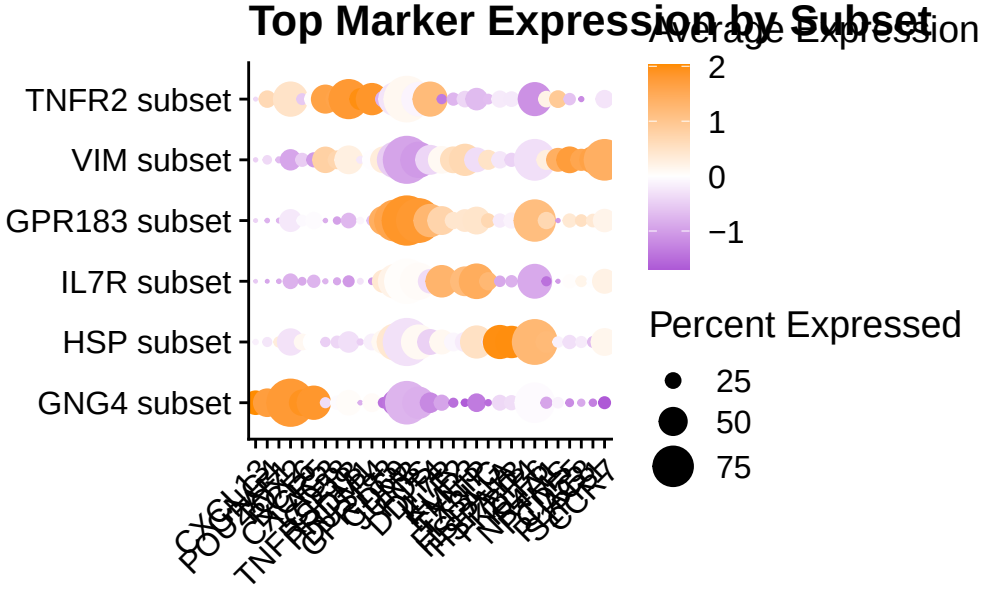
```
top.genes <- FindAllMarkers(tfh, only.pos = TRUE, min.pct = 0.25, logfc.threshold = 0.25)
```

```
Calculating cluster GNG4 subset
Calculating cluster HSP subset
Calculating cluster IL7R subset
Calculating cluster GPR183 subset
Calculating cluster VIM subset
Calculating cluster TNFR2 subset
```

```
top.fc <- top.genes %>% group_by(cluster) %>% slice_max(n = 20, order_by = avg_log2FC)
top.p <- top.genes %>% group_by(cluster) %>% slice_max(n = 20, order_by = avg_log2FC)
top.genes <- as.data.table(rbind(top.fc, top.p))
top.genes <- top.genes[avg_log2FC>1]
top.genes <- unique(top.genes, by = c("cluster", "gene"))

# write.csv(file = "outs/Tfhsubsets.FindAllMarkers.csv", x= FindAllMarkers(tfh, only.pos = TRUE, min.pct = 0.25,

genes_to_plot <- c("CXCL13", 'GNG4', 'POU2AF1', 'TOX2', 'BCL6', 'CXCR5',
                   "LAG3", "CD38", 'TNFRSF1B', 'PRDM1', 'DUSP4',
                   "GPR183", "CD69", 'JUNB', 'FOS', 'DDIT4',
                   "IL7R", "KLF2", 'EMP3', 'TXNIP', 'FLT3LG',
                   "HSPA1A", "HSPA1B", 'FKBP4', 'KLF6', 'NR4A1',
                   "PTMS", "CD55", 'PLAC8', 'S1PR1', 'CCR7')
```

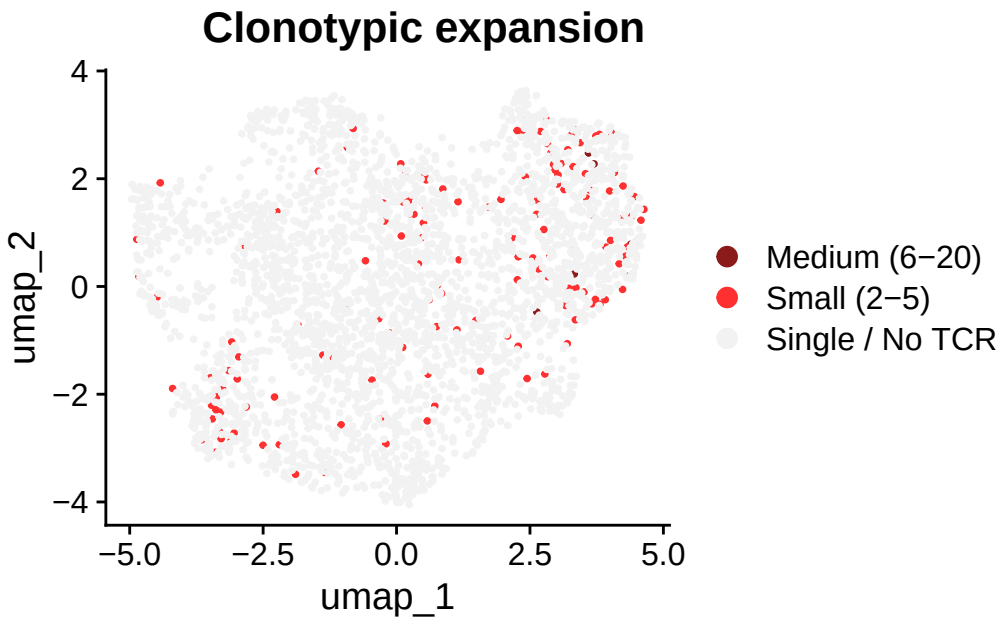
```
# GNG4: CXCL13, GNG4, POU2AF1, TOX2, BCL6, CXCR5
# TNFR2: LAG3, CD38, TNFRSF1B, PRDM1, DUSP4
# GPR183: GPR183, CD69, JUNB, FOS, DDIT4
# KLF2:  IL7R, KLF2, EMP3, TXNIP, FLT3LG
# HSP:  HSPA1A, HSPA1B, FKBP4, KLF6, NR4A1
# VIM:  PTMS, CD55, PLAC8, S1PR1, CCR7

DotPlot(tfh, features = genes_to_plot, group.by = "tfh.subset.annotation",
        dot.scale = 8) + RotatedAxis() +
  scale_color_gradient2(low = "darkorchid3", mid = "white", high = "darkorange") +
  labs(title = "Top Marker Expression by Subset", x = " ", y = " ")
```

```
Scale for colour is already present.
Adding another scale for colour, which will replace the existing scale.
```

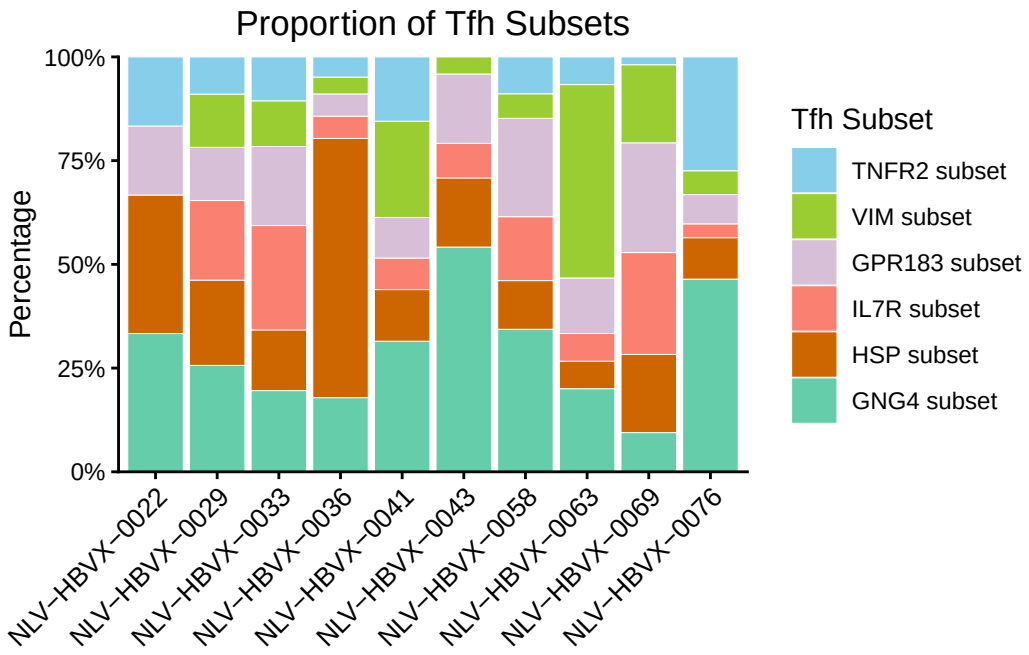



```
# ggsave(filename = 'Images/Tfh_bubbleplot_selectedGenes.pdf', height=3.5, width=10)

md <- as.data.table(tfh@meta.data, keep.rownames = "cell_id")
clone_counts <- md[!is.na(CTstrict), .(N_cells = .N), by = CTstrict]
md <- merge(md, clone_counts, by = "CTstrict", all.x = TRUE)
md[, Expansion_Status := fcase(
  is.na(CTstrict) | N_cells == 1, "Single / No TCR",
  N_cells > 1 & N_cells <= 5,    "Small (2-5)",
  N_cells > 5 & N_cells <= 20,   "Medium (6-20)",
  N_cells > 20,                  "Hyperexpanded (>20)"
)]
levels_order <- c("Single / No TCR", "Small (2-5)", "Medium (6-20)", "Hyperexpanded (>20)")
md[, Expansion_Status := factor(Expansion_Status, levels = levels_order)]

md <- md[order(Expansion_Status)]
tfh@meta.data <- data.frame(md, row.names = md$cell_id)[rownames(tfh@meta.data), ]

dark_palette <- c(
```

```
  "Single / No TCR"     = "grey95",   # Neutral for non-expanded
  "Small (2-5)"         = "firebrick1",  # Steel Blue
  "Medium (6-20)"       = "firebrick4",  # Deep Navy
  "Hyperexpanded (>20)" = "#590d22"   # Near-Black / Deep Blood Red
)

DimPlot(tfh, group.by = "Expansion_Status", reduction = "umap",
        cols = dark_palette, order = levels_order) +
  ggtitle("Clonotypic expansion") +
  theme(plot.title = element_text(hjust = 0.8) )
```




```
# ggsave(filename = 'Images/Tfh_UMAP_expandedClonotypes.pdf', height=5, width=6)


md <- as.data.table(tfh@meta.data)
bar.stats <- md[, .N, by = .(subject, tfh.subset.annotation)]
stack.order <- c("TNFR2 subset", "VIM subset", "GPR183 subset",
                 "IL7R subset", "HSP subset", "GNG4 subset")

bar.stats[, tfh.subset.annotation := factor(tfh.subset.annotation, levels = stack.order)]
ggplot(bar.stats, aes(x = subject, y = N, fill = tfh.subset.annotation)) +
  geom_bar(stat = "identity", position = "fill", color = "white", linewidth = 0.1) +
  scale_fill_manual(values = subset.cols) +
  scale_y_continuous(labels = scales::percent, expand = c(0,0)) +
  theme_classic() +
  labs(title = "Proportion of Tfh Subsets",
       x = "Participant", y = "Percentage", fill = "Tfh Subset") +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1, size = 10),
    legend.position = "right", panel.grid.major.x = element_blank(),
    plot.title = element_text(hjust = 0.5), axis.title.x = element_blank())
```




```
# ggsave(filename = 'Images/Tfh-Proportion-across-subjects.pdf', height=4, width=6)

# Calculate subset percentages per subject
subject.stats <- md[, .N, by = .(subject, tfh.subset.annotation, antibody_titer,
                                 titer.categ)]
subject.stats[, Percentage := (N / sum(N)) * 100, by = subject]
# Reshape to wide format (one row per subject) for correlation matrix
subject.stats <- dcast(subject.stats, subject + antibody_titer + titer.categ ~ tfh.subset.annotation, value.var
subject.stats
```

```
Key: <subject, antibody_titer, titer.categ>
          subject antibody_titer titer.categ GNG4 subset HSP subset IL7R subset
           <char>          <num>      <char>       <num>      <num>       <num>
 1: NLV-HBVX-0022      350.00000     Over100   33.333333  33.333333    0.000000
 2: NLV-HBVX-0029        0.10000    Under100   25.641026  20.512821   19.230769
 3: NLV-HBVX-0033       27.79851    Under100   19.559695  14.648603   25.148180
 4: NLV-HBVX-0036       10.25069    Under100   17.857143  62.500000    5.357143
 5: NLV-HBVX-0041      377.74954     Over100   31.479290  12.426036    7.573964
 6: NLV-HBVX-0043      135.00000     Over100   54.166667  16.666667    8.333333
 7: NLV-HBVX-0058      476.14680     Over100   34.357542  11.731844   15.363128
 8: NLV-HBVX-0063       55.41045    Under100   20.000000   6.666667    6.666667
 9: NLV-HBVX-0069      275.00000     Over100    9.433962  18.867925   24.528302
10: NLV-HBVX-0076      443.34570     Over100   46.445498   9.952607    3.317536
    GPR183 subset VIM subset TNFR2 subset
           <num>      <num>        <num>
 1:     16.666667   0.000000    16.666667
 2:     12.820513  12.820513     8.974359
 3:     19.051651  11.007621    10.584251
 4:      5.357143   4.017857     4.910714
 5:      9.822485  23.195266    15.502959
 6:     16.666667   4.166667     0.000000
 7:     23.743017   5.865922     8.938547
 8:     13.333333  46.666667     6.666667
 9:     26.415094  18.867925     1.886792
```

```
10:      7.109005   5.687204    27.488152
```

```
subject.stats <- md[, .N, by = .(subject, tfh.subset.annotation, antibody_titer, titer.categ)]
subject.stats[, Percentage := (N / sum(N)) * 100, by = subject]
subject.stats
```

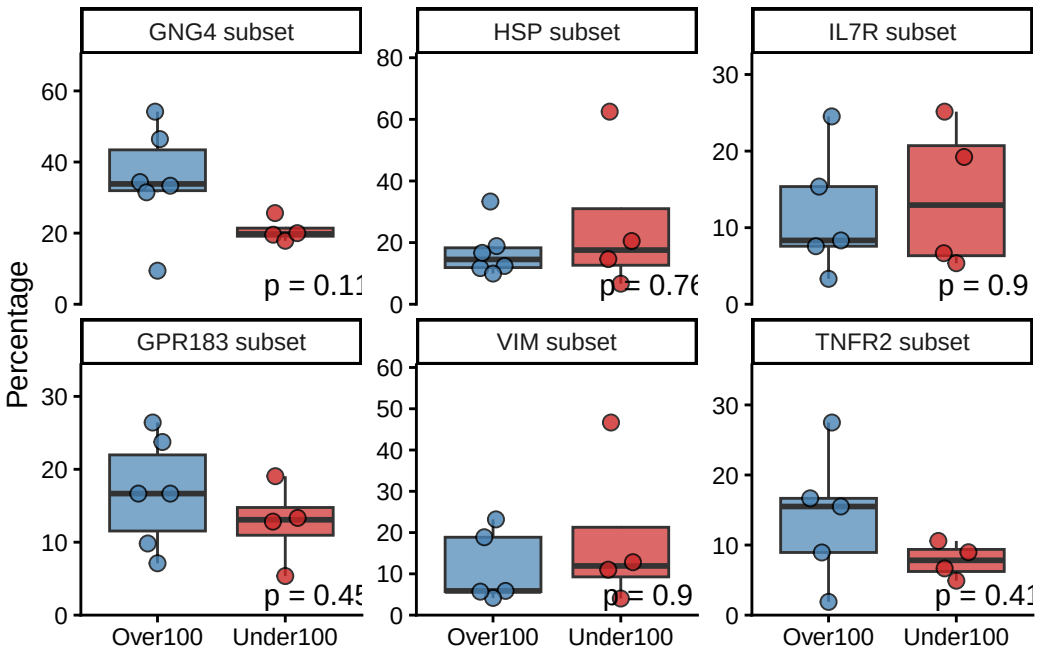
```
          subject tfh.subset.annotation antibody_titer titer.categ     N
           <char>                <fctr>          <num>      <char> <int>
 1: NLV-HBVX-0033           HSP subset       27.79851    Under100   173
 2: NLV-HBVX-0033          IL7R subset       27.79851    Under100   297
 3: NLV-HBVX-0033           VIM subset       27.79851    Under100   130
 4: NLV-HBVX-0033        GPR183 subset       27.79851    Under100   225
 5: NLV-HBVX-0033          GNG4 subset       27.79851    Under100   231
 6: NLV-HBVX-0033         TNFR2 subset       27.79851    Under100   125
 7: NLV-HBVX-0058          GNG4 subset      476.14680     Over100   123
 8: NLV-HBVX-0058         TNFR2 subset      476.14680     Over100    32
 9: NLV-HBVX-0058           HSP subset      476.14680     Over100    42
10: NLV-HBVX-0058        GPR183 subset      476.14680     Over100    85
11: NLV-HBVX-0058          IL7R subset      476.14680     Over100    55
12: NLV-HBVX-0058           VIM subset      476.14680     Over100    21
13: NLV-HBVX-0036           HSP subset       10.25069    Under100   140
14: NLV-HBVX-0041          GNG4 subset      377.74954     Over100   266
15: NLV-HBVX-0029        GPR183 subset        0.10000    Under100    10
16: NLV-HBVX-0041        GPR183 subset      377.74954     Over100    83
17: NLV-HBVX-0041           VIM subset      377.74954     Over100   196
18: NLV-HBVX-0036          IL7R subset       10.25069    Under100    12
19: NLV-HBVX-0041          IL7R subset      377.74954     Over100    64
20: NLV-HBVX-0069           HSP subset      275.00000     Over100    10
21: NLV-HBVX-0076          GNG4 subset      443.34570     Over100    98
22: NLV-HBVX-0069           VIM subset      275.00000     Over100    10
23: NLV-HBVX-0041         TNFR2 subset      377.74954     Over100   131
24: NLV-HBVX-0036          GNG4 subset       10.25069    Under100    40
25: NLV-HBVX-0069        GPR183 subset      275.00000     Over100    14
26: NLV-HBVX-0029           VIM subset        0.10000    Under100    10
27: NLV-HBVX-0041           HSP subset      377.74954     Over100   105
28: NLV-HBVX-0022        GPR183 subset      350.00000     Over100     1
29: NLV-HBVX-0076         TNFR2 subset      443.34570     Over100    58
30: NLV-HBVX-0036        GPR183 subset       10.25069    Under100    12
31: NLV-HBVX-0063           HSP subset       55.41045    Under100     1
32: NLV-HBVX-0029          IL7R subset        0.10000    Under100    15
33: NLV-HBVX-0043          GNG4 subset      135.00000     Over100    13
34: NLV-HBVX-0022         TNFR2 subset      350.00000     Over100     1
35: NLV-HBVX-0076        GPR183 subset      443.34570     Over100    15
36: NLV-HBVX-0076          IL7R subset      443.34570     Over100     7
37: NLV-HBVX-0069         TNFR2 subset      275.00000     Over100     1
38: NLV-HBVX-0029           HSP subset        0.10000    Under100    16
39: NLV-HBVX-0043           VIM subset      135.00000     Over100     1
40: NLV-HBVX-0076           HSP subset      443.34570     Over100    21
41: NLV-HBVX-0069          GNG4 subset      275.00000     Over100     5
42: NLV-HBVX-0069          IL7R subset      275.00000     Over100    13
43: NLV-HBVX-0036         TNFR2 subset       10.25069    Under100    11
44: NLV-HBVX-0063           VIM subset       55.41045    Under100     7
45: NLV-HBVX-0076           VIM subset      443.34570     Over100    12
46: NLV-HBVX-0029          GNG4 subset        0.10000    Under100    20
47: NLV-HBVX-0043           HSP subset      135.00000     Over100     4
48: NLV-HBVX-0029         TNFR2 subset        0.10000    Under100     7
```

```
49: NLV-HBVX-0063           GNG4 subset       55.41045    Under100     3
50: NLV-HBVX-0036            VIM subset       10.25069    Under100     9
51: NLV-HBVX-0043           IL7R subset      135.00000     Over100     2
52: NLV-HBVX-0063         GPR183 subset       55.41045    Under100     2
53: NLV-HBVX-0022           GNG4 subset      350.00000     Over100     2
54: NLV-HBVX-0043         GPR183 subset      135.00000     Over100     4
55: NLV-HBVX-0063           IL7R subset       55.41045    Under100     1
56: NLV-HBVX-0063          TNFR2 subset       55.41045    Under100     1
57: NLV-HBVX-0022            HSP subset      350.00000     Over100     2
          subject tfh.subset.annotation antibody_titer titer.categ     N
    Percentage
         <num>
 1:  14.648603
 2:  25.148180
 3:  11.007621
 4:  19.051651
 5:  19.559695
 6:  10.584251
 7:  34.357542
 8:   8.938547
 9:  11.731844
10:  23.743017
11:  15.363128
12:   5.865922
13:  62.500000
14:  31.479290
15:  12.820513
16:   9.822485
17:  23.195266
18:   5.357143
19:   7.573964
20:  18.867925
21:  46.445498
22:  18.867925
23:  15.502959
24:  17.857143
25:  26.415094
26:  12.820513
27:  12.426036
28:  16.666667
29:  27.488152
30:   5.357143
31:   6.666667
32:  19.230769
33:  54.166667
34:  16.666667
35:   7.109005
36:   3.317536
37:   1.886792
38:  20.512821
39:   4.166667
40:   9.952607
41:   9.433962
42:  24.528302
43:   4.910714
44:  46.666667
```

```
45:   5.687204
46:  25.641026
47:  16.666667
48:   8.974359
49:  20.000000
50:   4.017857
51:   8.333333
52:  13.333333
53:  33.333333
54:  16.666667
55:   6.666667
56:   6.666667
57:  33.333333
    Percentage
```

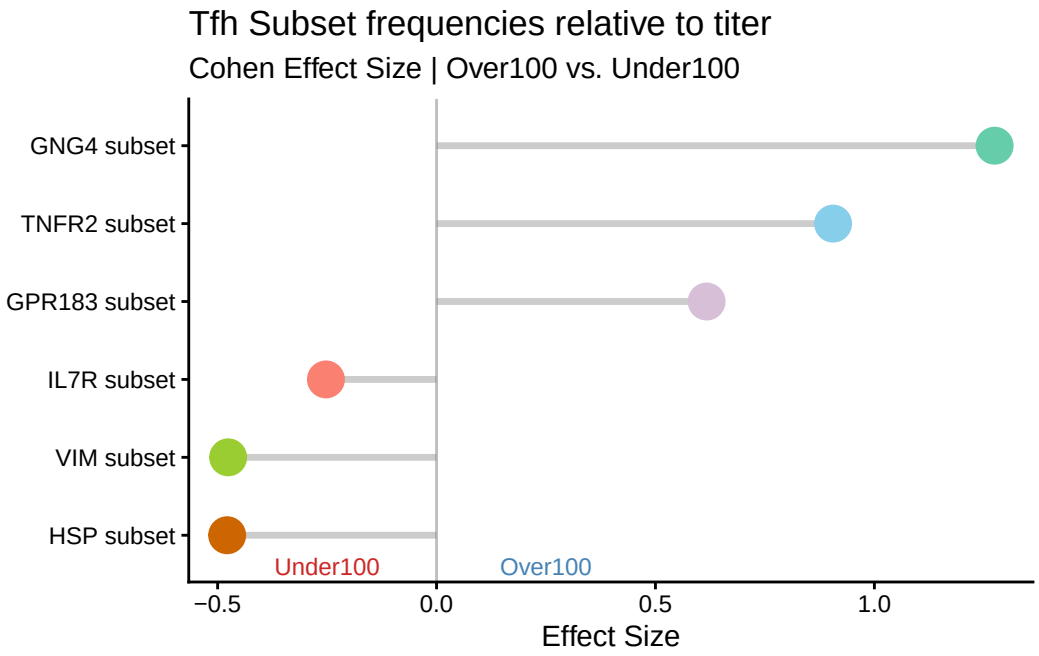
```
ggplot(subject.stats, aes(x = titer.categ, y = Percentage, fill = titer.categ)) +
  geom_boxplot(outlier.shape = NA, alpha = 0.7) +
  ggbeeswarm::geom_quasirandom(width = 0.2, size = 2.5, alpha = 0.8, shape = 21,
                               color = "black") +
  facet_wrap(~tfh.subset.annotation, scales = "free_y", ) +
  scale_fill_manual(values = c("Under100" = "firebrick3", "Over100" = "steelblue")) +
  scale_y_continuous(expand = expansion(mult = c(0, 0.3)), limits = c(0, NA)) +
  ggpubr::stat_compare_means(method = "wilcox.test",
        label = "p.format", digits = 2, label.x.npc = "right", label.y.npc = 0.05) +
  theme_classic() + theme(legend.position = "none", axis.title.x = element_blank())
```

```
Warning in ggpubr::stat_compare_means(method = "wilcox.test", label =
"p.format", : Ignoring unknown parameters: `digits`
```




```
# ggsave(filename = "Images/Tfh-subsets-vs-Abtiter.pdf", height=5, width=7)


# install.packages("coin")
```

```
eff.stats <- subject.stats[, as.data.table(rstatix::cohens_d(data = .SD, Percentage ~ titer.categ)), by = tfh.su
eff.stats[, label := paste0("r = ", round(effsize, 2), " (", magnitude, ")")]
eff.stats[, tfh.subset.annotation := reorder(tfh.subset.annotation, effsize)]

ggplot(eff.stats, aes(x = effsize, y = tfh.subset.annotation)) +
  geom_segment(aes(x = 0, xend = effsize, y = tfh.subset.annotation,
                   yend = tfh.subset.annotation), color = "grey80", linewidth = 1.2) +
  geom_point(aes(color = tfh.subset.annotation), size = 6) +
  scale_color_manual(values = subset.cols) +
  annotate("text", y = 0.6, x = 0.25, label = "Over100", color = "steelblue", size=3) +
  annotate("text", y = 0.6, x = -0.25, label = "Under100", color = "firebrick3", size=3)+
  theme_classic() +
  geom_vline(xintercept=0, colour = 'gray50', alpha=0.5)+
  labs(title = "Tfh Subset frequencies relative to titer",
       subtitle = "Cohen Effect Size | Over100 vs. Under100",
       x = "Effect Size", y = NULL) +
  theme(legend.position = "none", axis.text.y = element_text(color = "black"),
        panel.grid.minor = element_blank())
```




```
# ggsave(filename = 'Images/Tfh-EffectSize-Lollipop.pdf', height=3, width=4.5)

rm(top.genes, top.fc, top.p, genes_to_plot, clone_counts, dark_palette, levels_order, bar.stats, stack.order)
```

GNG4 clonal expansion

```
gng4.cells <- subset(tfh, subset = tfh.subset.annotation == "GNG4 subset")
Idents(gng4.cells) <- "Expansion_Status"

# use 'Single / No TCR' as the baseline to find what makes expansion unique
de_markers <- FindMarkers(
  gng4.cells, ident.1 = "Medium (6-20)", ident.2 = "Single / No TCR",
```

```
  only.pos = FALSE, min.pct = 0.1, logfc.threshold = 0.25 )

setDT(de_markers, keep.rownames = "gene")
print(de_markers, 30)
```

```
                gene        p_val avg_log2FC pct.1 pct.2    p_val_adj
              <char>        <num>      <num> <num> <num>        <num>
   1:           IQCA1 4.527702e-64  9.6974131   0.4 0.000 1.747964e-59
   2:        TRAV38-1 7.292255e-54  6.6524579   1.0 0.014 2.815248e-49
   3:       LINC01134 7.949845e-33  8.9381065   0.2 0.000 3.069117e-28
   4:       MIR3659HG 7.949845e-33  8.9145565   0.2 0.000 3.069117e-28
   5: ENSG00000232184 7.949845e-33  9.0530029   0.2 0.000 3.069117e-28
   6:           MYCN 7.949845e-33  9.2988406   0.2 0.000 3.069117e-28
   7:       LINC01102 7.949845e-33  8.9381065   0.2 0.000 3.069117e-28
   8: ENSG00000241231 7.949845e-33  9.2988406   0.2 0.000 3.069117e-28
   9:           IGIP 7.949845e-33  8.9145565   0.2 0.000 3.069117e-28
  10:       LINC01366 7.949845e-33  8.9145565   0.2 0.000 3.069117e-28
  11: ENSG00000238258 7.949845e-33  9.2101371   0.2 0.000 3.069117e-28
  12:        MARVELD1 7.949845e-33  8.9145565   0.2 0.000 3.069117e-28
  13: ENSG00000290939 7.949845e-33  8.9145565   0.2 0.000 3.069117e-28
  14: ENSG00000288767 7.949845e-33  9.2101371   0.2 0.000 3.069117e-28
  15:        IGHV3-48 7.949845e-33  9.0530029   0.2 0.000 3.069117e-28
  16:       INO80-AS1 7.949845e-33  9.2988406   0.2 0.000 3.069117e-28
  17: ENSG00000260306 7.949845e-33  9.0530029   0.2 0.000 3.069117e-28
  18:           ADAD2 7.949845e-33  8.9145565   0.2 0.000 3.069117e-28
  19:             NXN 7.949845e-33  8.9381065   0.2 0.000 3.069117e-28
  20:           NKPD1 7.949845e-33  9.2101371   0.2 0.000 3.069117e-28
  21:         LRRC75B 7.949845e-33  9.2101371   0.2 0.000 3.069117e-28
  22:           CENPI 7.949845e-33  8.9381065   0.2 0.000 3.069117e-28
  23:           PALLD 1.547092e-29  6.2129869   0.6 0.010 5.972702e-25
  24:         TGFBR3L 4.423299e-22  6.7193835   0.4 0.006 1.707659e-17
  25:        DNASE1L3 4.599035e-17  8.6678762   0.2 0.001 1.775503e-12
  26:             HGD 4.599035e-17  7.7626242   0.2 0.001 1.775503e-12
  27:          VWA5B2 4.599035e-17  7.3537209   0.2 0.001 1.775503e-12
  28:       LINC02999 4.599035e-17  7.5092493   0.2 0.001 1.775503e-12
  29:        KIAA0319 4.599035e-17  7.2235036   0.2 0.001 1.775503e-12
  30:            MYRF 4.599035e-17  7.8491058   0.2 0.001 1.775503e-12
  ---
6377:           KMT2A 9.896030e-01  0.9954836   0.4 0.448 1.000000e+00
6378:            RAE1 9.912460e-01  0.2599162   0.2 0.193 1.000000e+00
6379:            RNF6 9.912708e-01  0.2758744   0.2 0.195 1.000000e+00
6380:          SLC7A1 9.913908e-01  0.3836387   0.2 0.202 1.000000e+00
6381:          UBE3A 9.914534e-01  0.3100662   0.4 0.440 1.000000e+00
6382:          TGIF2 9.915041e-01  0.3274551   0.2 0.209 1.000000e+00
6383:          CCNT2 9.916523e-01  0.3657912   0.2 0.218 1.000000e+00
6384:          NDUFA7 9.917706e-01  0.6494006   0.2 0.227 1.000000e+00
6385:          INTS10 9.917706e-01  0.3886020   0.2 0.227 1.000000e+00
6386:          NOLC1 9.919520e-01  0.8433238   0.2 0.241 1.000000e+00
6387:         HAPSTR1 9.937109e-01  0.3040677   0.2 0.190 1.000000e+00
6388:          DMTF1 9.937471e-01  0.3060234   0.2 0.193 1.000000e+00
6389:          CPSF7 9.938671e-01  0.3831876   0.2 0.203 1.000000e+00
6390:          SFT2D1 9.938996e-01  0.4283422   0.2 0.206 1.000000e+00
6391:           BRI3 9.940373e-01  0.3668138   0.2 0.218 1.000000e+00
6392:             CS 9.941081e-01  0.2828333   0.2 0.225 1.000000e+00
6393:          ATXN7 9.942390e-01  0.7271675   0.2 0.239 1.000000e+00
6394:         PYHIN1 9.960228e-01 -0.2718795   0.4 0.371 1.000000e+00
```

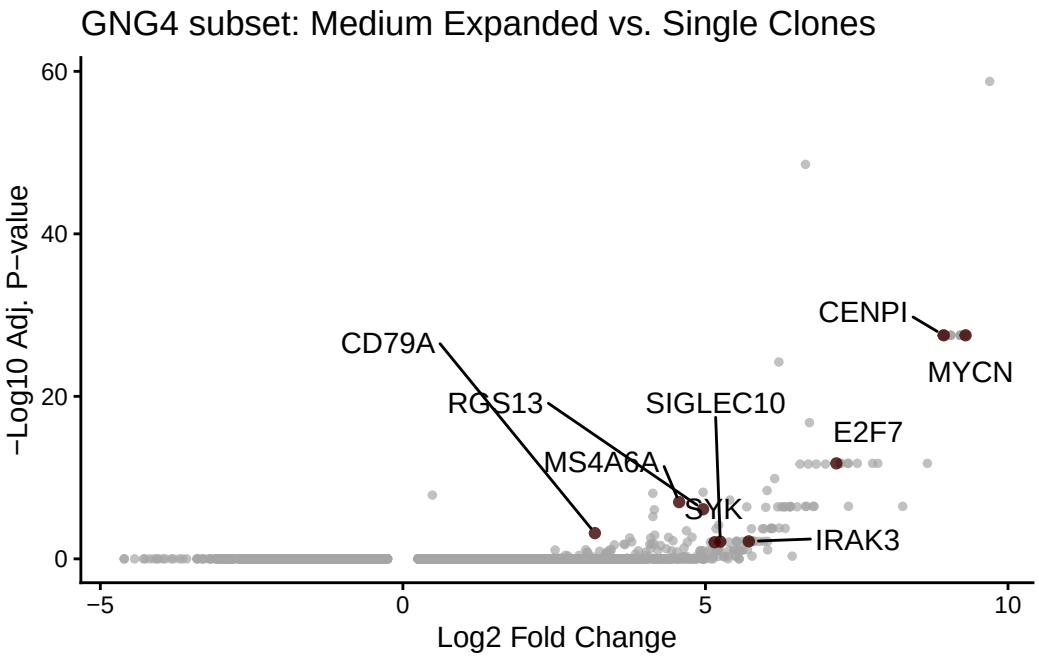
```
6395:           VPS53 9.962694e-01  0.2916297   0.2 0.196 1.000000e+00
6396:          MRPS15 9.963002e-01  0.2518552   0.2 0.200 1.000000e+00
6397:          FUNDC2 9.964136e-01  0.3176775   0.2 0.217 1.000000e+00
6398:           SPPL3 9.964565e-01  0.2899317   0.2 0.224 1.000000e+00
6399:            TANK 9.981426e-01 -0.2926496   0.6 0.482 1.000000e+00
6400:            CKS2 9.987458e-01  0.3554972   0.2 0.192 1.000000e+00
6401:           TGIF1 9.987986e-01  0.3117278   0.2 0.214 1.000000e+00
6402:            UPF2 1.000000e+00  0.3308394   0.2 0.224 1.000000e+00
6403:           CRTC3 1.000000e+00  0.4057307   0.2 0.223 1.000000e+00
6404:          RNF115 1.000000e+00  0.3662466   0.2 0.216 1.000000e+00
6405:        ANKRD13C 1.000000e+00  0.2788539   0.2 0.195 1.000000e+00
6406:          TSG101 1.000000e+00  0.2559865   0.2 0.200 1.000000e+00
```

```
de_markers[p_val_adj<0.05,gene]
```

```
  [1] "IQCA1"           "TRAV38-1"        "LINC01134"       "MIR3659HG"
  [5] "ENSG00000232184" "MYCN"            "LINC01102"       "ENSG00000241231"
  [9] "IGIP"            "LINC01366"       "ENSG00000238258" "MARVELD1"
 [13] "ENSG00000290939" "ENSG00000288767" "IGHV3-48"        "INO80-AS1"
 [17] "ENSG00000260306" "ADAD2"           "NXN"             "NKPD1"
 [21] "LRRC75B"         "CENPI"           "PALLD"           "TGFBR3L"
 [25] "DNASE1L3"        "HGD"             "VWA5B2"          "LINC02999"
 [29] "KIAA0319"        "MYRF"            "TNS2"            "E2F7"
 [33] "GOLGA6L9"        "ENSG00000248373" "ENSG00000248475" "ZFHX2"
 [37] "LKAAEAR1"        "METAP1D"         "SLC25A35"        "LIMCH1"
 [41] "TRBV7-2"         "TRBV4-2"         "STX3"            "MS4A6A"
 [45] "ITPKB-IT1"       "ENSG00000276334" "LINC02241"       "PRXL2A"
 [49] "DRAM1"           "PEX11A"          "ENSG00000286288" "ENSG00000285731"
 [53] "ENSG00000239482" "MARVELD2"        "H2AZ2-DT"        "ENSG00000229190"
 [57] "LINC00926"       "ENSG00000260367" "ENSG00000267058" "RILP"
 [61] "RGS13"           "CHRM3-AS2"       "DPY19L2P3"       "ENSG00000259865"
 [65] "ACTL10"          "LPL"             "ENSG00000228862" "ENSG00000287475"
 [69] "RASSF8-AS1"      "TCP10L"          "ENSG00000286924" "CD63-AS1"
 [73] "LCMT2"           "ZNF726"          "ZSWIM4"          "CD79A"
 [77] "ENSG00000290876" "ITFG2-AS1"       "ZSCAN26"         "CYP4V2"
 [81] "HACD1"           "ULK2"            "ENSG00000286313" "ZCCHC12"
 [85] "ENSG00000289156" "ADAMTS3"         "MGC4859"         "ENSG00000289412"
 [89] "ENSG00000289046" "IRAK3"           "KIF7"            "CYP4F29P"
 [93] "CYSLTR1"         "ENSG00000276345" "SIGLEC10"        "CYP7B1"
 [97] "KCNJ13"          "P4HA3"           "RASSF8"          "LOXL2"
[101] "SYK"             "MISP3"           "ST7"             "MSANTD7"
[105] "NMT2"            "HVCN1"           "MUTYH"           "ENSG00000291063"
[109] "DELEC1"          "TTC28"           "C1orf112"
```

```
priority_genes <- c("MYCN", "CENPI", "E2F7", "RGS13", "CD79A", "SYK", "IRAK3", "MS4A6A", "SIGLEC10")
de_markers[, color_group := "Other"]
de_markers[gene %in% priority_genes, color_group := "Expanded Signature"]
de_markers[, label_gene := ifelse(gene %in% priority_genes, gene, "")]

ggplot(de_markers, aes(x = avg_log2FC, y = -log10(p_val_adj))) +
  geom_point(data = de_markers[color_group == "Other"], color = "grey65", alpha = 0.7,
             size = 1) +
  geom_point(data = de_markers[color_group == "Expanded Signature"], color = "#3D0000",
             alpha = 0.8, size=1.5) +
  geom_text_repel(aes(label = label_gene), box.padding = 0.5, point.padding = 0.3,
                  segment.color = "black", size = 4, max.overlaps = Inf) +
```

```
  theme_classic() +
  labs(title = "GNG4 subset: Medium Expanded vs. Single Clones",
       x = "Log2 Fold Change", y = "-Log10 Adj. P-value")
```



miloR analysis

```
options(Matrix.warnDeprecatedCoerce = 0)
options(warn = 0)
##  miloR for differential abundance testing by conditions
# BiocManager::install("miloR")
library(miloR)
```

```
Loading required package: edgeR
```

```
Attaching package: 'edgeR'
```

```
The following object is masked from 'package:SingleCellExperiment':

    cpm
```

```
##  MULTI SEED APPROACH
# library(foreach); library(doParallel)
# cl <- makeCluster(detectCores() - 15, type = 'PSOCK')
# registerDoParallel(cl)
# res.daresults <- foreach( i = 1:1000) %dopar%
# {
#   library(miloR)
#   options(warn = 0)
#   print(paste("Iteration:", i))
```

```
#   set.seed(i) # 99
#   milo_obj <- Milo(Seurat::as.SingleCellExperiment(tfh))
#   milo_obj <- buildGraph(milo_obj, k = 30, d = 30)
#   milo_obj <- makeNhoods(milo_obj, prop = 0.15, k = 30, d = 30, refined = TRUE)
#   milo_obj <- calcNhoodDistance(milo_obj, d=30)
#   milo_obj <- countCells(milo_obj, meta.data = as.data.frame(SummarizedExperiment::colData(milo_obj)), sample
#   milo_obj.des.titer <- data.frame(SummarizedExperiment::colData(milo_obj))[,c("subject", "titer.categ")]
#   milo_obj.des.titer <- dplyr::distinct(milo_obj.des.titer)
#   rownames(milo_obj.des.titer) <- milo_obj.des.titer$subject
#   milo_obj.des.titer <- milo_obj.des.titer[colnames(nhoodCounts(milo_obj)), , drop=FALSE]
#   da_results <- testNhoods(milo_obj, design = ~ titer.categ, design.df = milo_obj.des.titer)
#   milo_obj <- buildNhoodGraph(milo_obj)
#   da_results <- annotateNhoods(milo_obj, da_results, coldata_col = "titer.categ")
#   da_results <- annotateNhoods(milo_obj, da_results, coldata_col = "tfh.subset.annotation")
#   da_results
# }; stopCluster(cl); rm(cl)
#
# saveRDS(res.daresults, file = "20260407_miloR_Tfh_daresults.RDS")
res.daresults <- readRDS("20260407_miloR_Tfh_daresults.RDS")


# SINGLE SEED APPROACH
set.seed(114) # 99, 111
milo_obj <- Milo(as.SingleCellExperiment(tfh))
milo_obj <- buildGraph(milo_obj, k = 30, d = 30)
```

```
Constructing kNN graph with k:30
```

```
milo_obj <- makeNhoods(milo_obj, prop = 0.10, k = 30, d = 30, refined = TRUE)
```

```
Checking valid object

Running refined sampling with reduced_dim
```

```
milo_obj <- calcNhoodDistance(milo_obj, d=30)
milo_obj <- countCells(milo_obj, meta.data = as.data.frame(colData(milo_obj)), sample = "subject")
```

```
Checking meta.data validity

Counting cells in neighbourhoods
```

```
head(nhoodCounts(milo_obj))
```

```
6 x 10 sparse Matrix of class "dgCMatrix"

  [[ suppressing 10 column names 'NLV-HBVX-0033', 'NLV-HBVX-0058', 'NLV-HBVX-0036' ... ]]


1 105  . .  . . 1 . . . .
2   5  . 3 73 3 6 6 . 1 1
3  69  6 2  6 2 1 1 . . .
4   2 76 .  1 . . . . . .
5  15  2 3 64 6 4 3 . 2 .
6   1  . 7 56 4 1 6 . 2 4
```

```
milo_obj.des.titer <- data.frame(colData(milo_obj))[,c("subject", "titer.categ")]
milo_obj.des.titer <- distinct(milo_obj.des.titer)
rownames(milo_obj.des.titer) <- milo_obj.des.titer$subject
milo_obj.des.titer <- milo_obj.des.titer[colnames(nhoodCounts(milo_obj)), , drop=FALSE] # Reorder rownames to ma
da_results <- testNhoods(milo_obj, design = ~ titer.categ, design.df = milo_obj.des.titer)
```

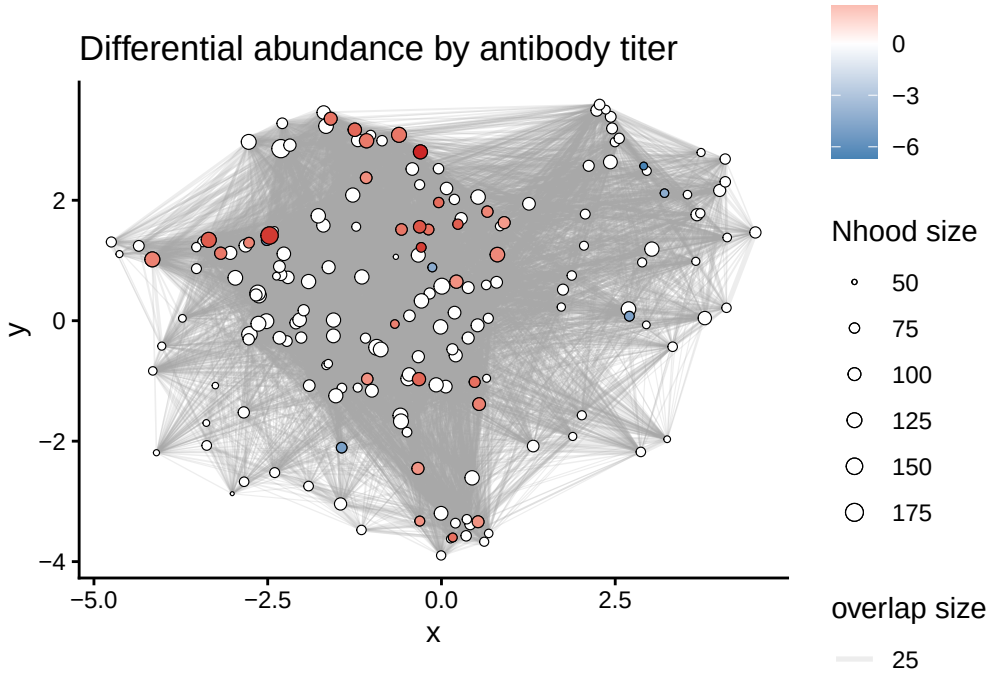
```
Using TMM normalisation

Running with model contrasts

Performing spatial FDR correction with k-distance weighting
```

```
milo_obj <- buildNhoodGraph(milo_obj)
da_results <- annotateNhoods(milo_obj, da_results, coldata_col = "titer.categ")
```

```
Converting titer.categ to factor...
```

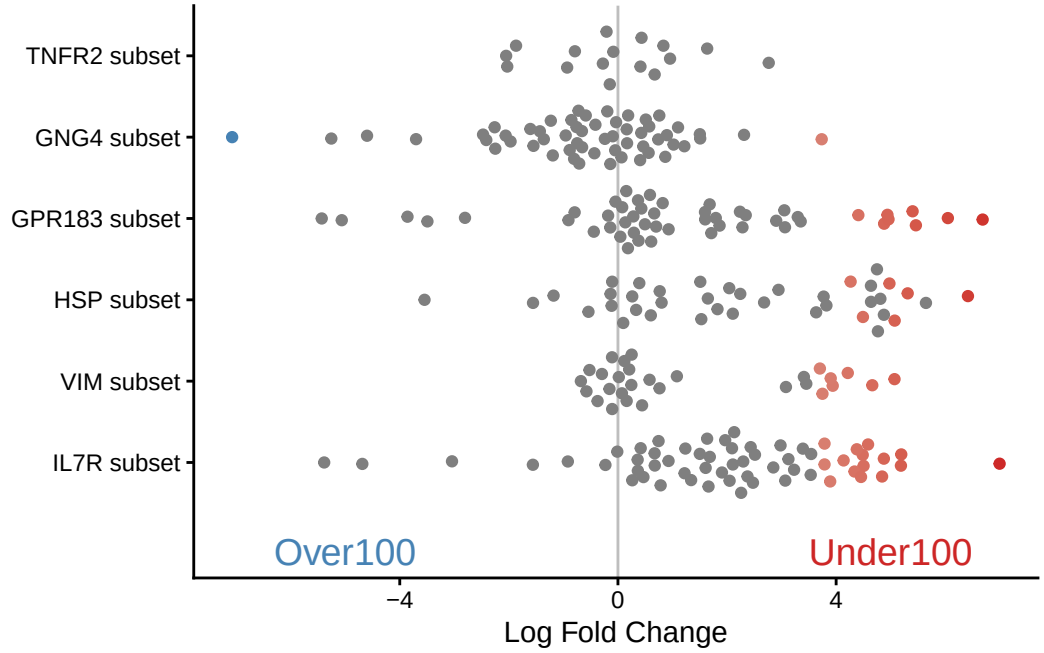
```
da_results <- annotateNhoods(milo_obj, da_results, coldata_col = "tfh.subset.annotation")

plotNhoodGraphDA(milo_obj, da_results, alpha=0.1) +
  patchwork::plot_layout(guides="collect") +
  ggtitle("Differential abundance by antibody titer") +
  scale_fill_gradient2(low = 'steelblue', mid='white', high='firebrick3') +
  theme_classic()
```

```
Adding nhood effect sizes to neighbourhood graph attributes

Scale for fill is already present.
Adding another scale for fill, which will replace the existing scale.
```



```
# ggsave(filename = "./Images/Tfh_milo_titer_network.pdf", width = 6,height=6)


plotDAbeeswarm(res.daresults[[114]], group.by='tfh.subset.annotation', alpha=0.1) +
  geom_hline(yintercept=0, color="grey50", alpha=0.5) +
  scale_color_gradient2(low = "steelblue", mid = "grey80", high = "firebrick3") +
  annotate("text", x = -0.1, y = -5, label = "Over100", color = "steelblue", size=5) +
  annotate("text", x = -0.1, y = 5, label = "Under100", color = "firebrick3", size=5)+
  theme_classic() +
  theme(axis.title.y= element_blank(), axis.text = element_text(color='black'))
```

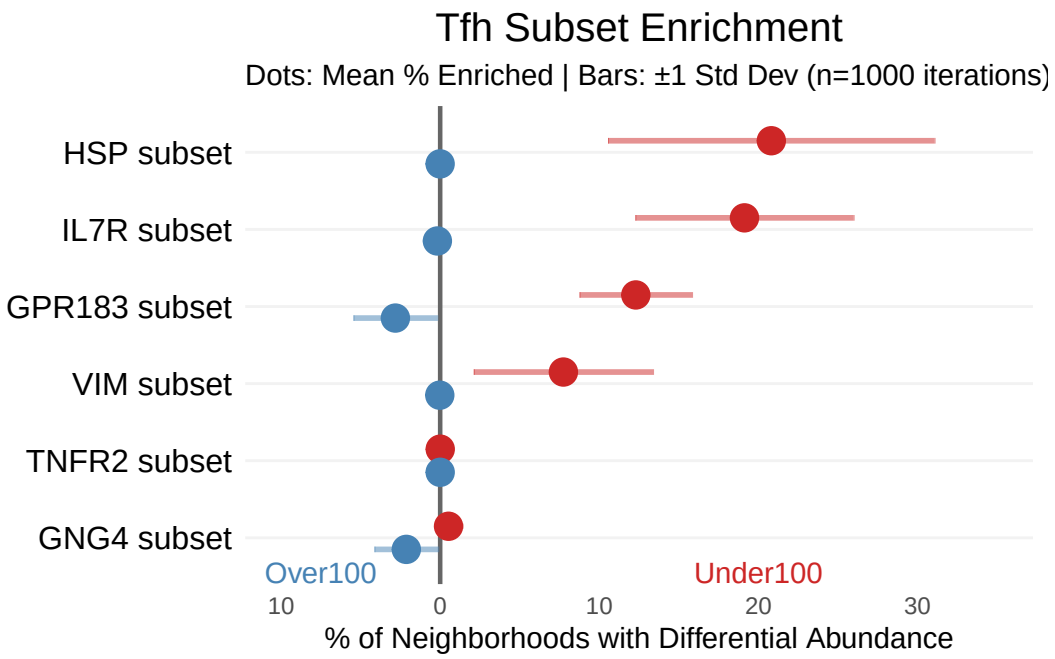
```
Converting group_by to factor...
Scale for colour is already present.
Adding another scale for colour, which will replace the existing scale.
```




```
# ggsave(filename = "./Images/Tfh_milo_titer_beeswarm.pdf", width = 4.5,height=3)


iter_stats <- rbindlist(res.daresults, idcol=T)[, .(
  Total_Nhoods = .N,
  Count_Under100 = sum(SpatialFDR < 0.1 & logFC > 0, na.rm = TRUE),
  Count_Over100  = sum(SpatialFDR < 0.1 & logFC < 0, na.rm = TRUE)
), by = .(.id, tfh.subset.annotation)]

iter_stats[, `:=`(
  Percent_Under100 = (Count_Under100 / Total_Nhoods) * 100,
  Percent_Over100  = (Count_Over100 / Total_Nhoods) * 100)  ]

summary_stats <- iter_stats[, .(
  under_mean = mean(Percent_Under100),
  over_mean  = mean(Percent_Over100),
  under_sd   = sd(Percent_Under100),
  over_sd    = sd(Percent_Over100),
```

```
  under_ci_lo = quantile(Percent_Under100, 0.025),
  under_ci_hi = quantile(Percent_Under100, 0.975),
  over_ci_lo  = quantile(Percent_Over100, 0.025),
  over_ci_hi  = quantile(Percent_Over100, 0.975)  ), by = tfh.subset.annotation]

summary_stats[, `:=`(
  x_under = under_mean,
  x_over  = -over_mean,
  x_under_sd_lo = under_mean - under_sd,
  x_under_sd_hi = under_mean + under_sd,
  x_over_sd_lo = (-over_mean) - over_sd,
  x_over_sd_hi = (-over_mean) + over_sd,
  Net_Score = under_mean - over_mean   )]

ggplot(summary_stats, aes(y = reorder(tfh.subset.annotation, Net_Score))) +
  geom_hline(yintercept = 1:nrow(summary_stats), color = "grey95", linewidth = 0.5) +
  geom_vline(xintercept = 0, color = "grey40", linewidth = 0.8) +
  geom_errorbarh(aes(xmin = x_under_sd_lo, xmax = x_under_sd_hi),
                 height = 0, color = "firebrick3", linewidth = 1, alpha = 0.5,
                 position = position_nudge(y = 0.15)) +
  geom_point(aes(x = x_under), color = "firebrick3", size = 4.5,
             position = position_nudge(y = 0.15)) +
  geom_errorbarh(aes(xmin = x_over_sd_lo, xmax = x_over_sd_hi),
                 height = 0, color = "steelblue", linewidth = 1, alpha = 0.5,
                 position = position_nudge(y = -0.15)) +
  geom_point(aes(x = x_over), color = "steelblue", size = 4.5,
             position = position_nudge(y = -0.15)) +
  scale_x_continuous(labels = abs, limits = c(-10, 35), breaks = seq(-10, 30, 10)) +
  theme_minimal() +
  labs(title = "Tfh Subset Enrichment",
       subtitle = "Dots: Mean % Enriched | Bars: ±1 Std Dev (n=1000 iterations)",
       x = "% of Neighborhoods with Differential Abundance", y = NULL) +
  theme(panel.grid.major = element_blank(), panel.grid.minor = element_blank(),
    axis.text.y = element_text(size = 12, color = "black"),
    plot.title = element_text(hjust = 0.5, size = 15) ) +
  annotate("text", x = -7.5, y = 0.55, label = "Over100",
           color = "steelblue", size = 4) +
  annotate("text", x = 20, y = 0.55, label = "Under100",
           color = "firebrick3", size = 4)
```

```
Warning: `geom_errobarh()` was deprecated in ggplot2 4.0.0.
i Please use the `orientation` argument of `geom_errorbar()` instead.

`height` was translated to `width`.
`height` was translated to `width`.
```




```
# ggsave(filename = 'Images/Tfh_milo_beeswarm_aggregate.pdf', height=3.5,width=6)


#
# da_dt <- as.data.table(da_results)
# summary_stats <- da_dt[, .(Total_Nhoods = .N,
#                            Count_Under100 = sum(SpatialFDR < 0.1 & logFC > 0,
#                                                 na.rm = TRUE),
#                            Count_Over100 = sum(SpatialFDR < 0.1 & logFC < 0,
#                                                na.rm = TRUE)),
#                        by = tfh.subset.annotation]
# summary_stats[, Percent_Under100 := (Count_Under100 / Total_Nhoods) * 100]
# summary_stats[, Percent_Over100  := (Count_Over100 / Total_Nhoods) * 100]
# summary_stats[, x_over := -Percent_Over100]
# summary_stats[, x_under := Percent_Under100]
# summary_stats[, Net_Score := Percent_Under100 - Percent_Over100]
#
# ggplot(summary_stats, aes(y = reorder(tfh.subset.annotation, Net_Score))) +
#   geom_hline(yintercept = 1:nrow(summary_stats), color = "grey90", linetype = "solid") +
#   geom_vline(xintercept = 0, color = "grey50") +
#   geom_segment(aes(x = 0, xend = x_over, yend = tfh.subset.annotation),
#                color = "blue", linewidth = 0.8) +
#   geom_point(aes(x = x_over), color = "blue", size = 4) +
#   geom_segment(aes(x = 0, xend = x_under, yend = tfh.subset.annotation),
#                color = "red", linewidth = 0.8) +
#   geom_point(aes(x = x_under), color = "red", size = 4) +
#   theme_classic() +
#   scale_x_continuous(labels = abs, limits = c(-30, 30)) +
#   labs(title = "Proportion of Neighborhoods Enriched",
#        x = "% Neighborhoods", # "← Over100 (%) | Under100 (%) →",
#        y = NULL) +
#   theme(axis.text.y = element_text(size = 11), plot.title = element_text(hjust = 0.5),
#         legend.position = "none") +
#   annotate("text", x = -20, y = 0.6, label = "Over100", color = "blue", size = 4) +
```

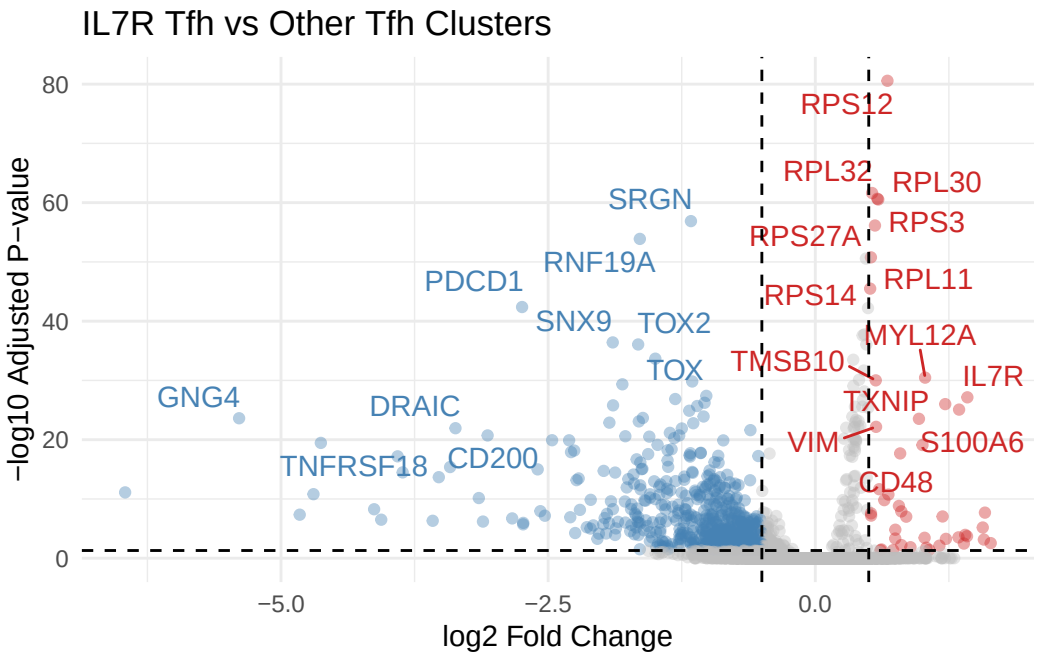
```
#   annotate("text", x = 20, y = 0.6, label = "Under100", color = "red", size = 4)




# rm(da_dt, summary_stats, iter_stats, da_results, milo_obj, milo_obj.des.titer)



# *****************  no significant or interpretable results
# da_results <- groupNhoods(milo_obj, da_results, max.lfc.delta = 2)
# plotDAbeeswarm(da_results, "NhoodGroup")
# hvg <- scran::getTopHVGs(scran::modelGeneVar(milo_obj), n=2000)
# nhood_markers <- findNhoodGroupMarkers(milo_obj, da_results, subset.row = hvg, aggregate.samples = TRUE,
#                                        sample_col = "titer.categ")
# ggplot(nhood_markers, aes(logFC_4,-log10(adj.P.Val_4 ))) + geom_point(alpha=0.5, size=0.5) + geom_hline(yinter
#
# ************************************************************
```

Differential expression by subset

```
deg <- FindMarkers(tfh, ident.1 = "IL7R subset", min.pct = 0.1, logfc.threshold = 0)
deg$diffexpressed <- "Not Significant"
deg$diffexpressed[deg$avg_log2FC > 0.5 & deg$p_val_adj < 0.05] <- "Up"
deg$diffexpressed[deg$avg_log2FC < -0.5 & deg$p_val_adj < 0.05] <- "Down"
deg$gene_label <- NA
top_genes <- rownames(deg[deg$diffexpressed != "Not Significant", ])[1:60]
deg$gene_label[rownames(deg) %in% top_genes] <- rownames(deg)[rownames(deg) %in% top_genes]
ggplot(data = deg, aes(x = avg_log2FC, y = -log10(p_val_adj), col = diffexpressed, label = gene_label)) +
  geom_point(alpha = 0.4, size = 1.5) + theme_minimal() +
  ggrepel::geom_text_repel() +
  scale_color_manual(values = c("steelblue", "grey", "firebrick3")) +
  geom_vline(xintercept = c(-0.5, 0.5), col = "black", linetype = "dashed") +
  geom_hline(yintercept = -log10(0.05), col = "black", linetype = "dashed") +
  labs(title = "IL7R Tfh vs Other Tfh Clusters", x = "log2 Fold Change",
       y = "-log10 Adjusted P-value") +
  theme(legend.title = element_blank(), legend.position = 'none')
```

```
Warning: Removed 5352 rows containing missing values or values outside the scale range
(`geom_text_repel()`).
```

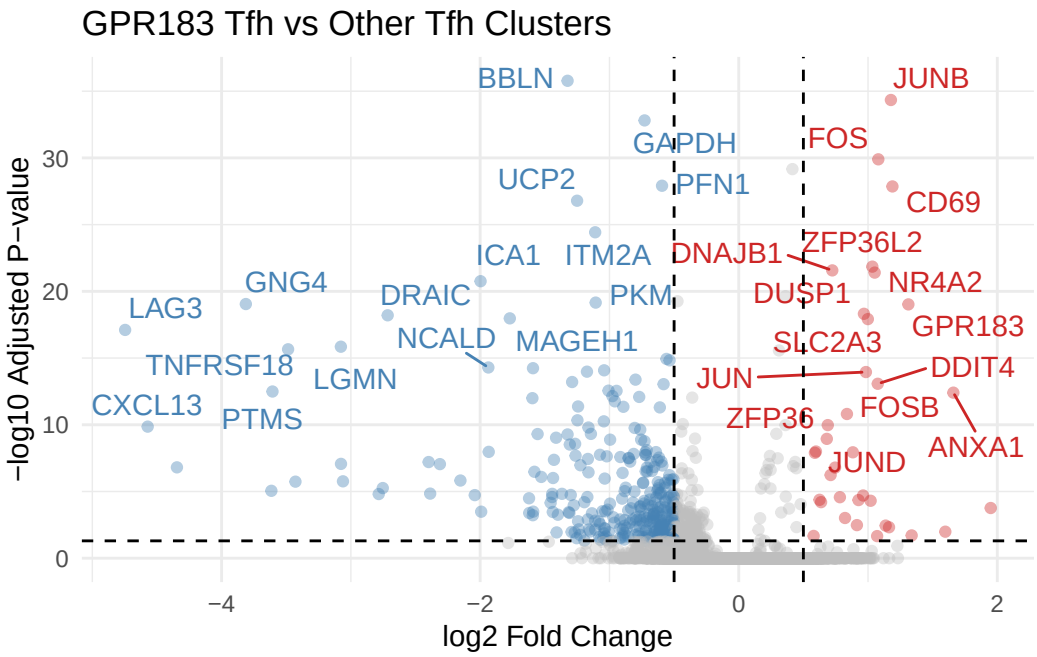
```
Warning: ggrepel: 36 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```




```
# ggsave(filename = 'Images/Tfh-Volcano-IL7RvsOthers.pdf', height=4, width=5)


deg <- FindMarkers(tfh, ident.1 = "GPR183 subset", min.pct = 0.1, logfc.threshold = 0)
deg$diffexpressed <- "Not Significant"
deg$diffexpressed[deg$avg_log2FC > 0.5 & deg$p_val_adj < 0.05] <- "Up"
deg$diffexpressed[deg$avg_log2FC < -0.5 & deg$p_val_adj < 0.05] <- "Down"
deg$gene_label <- NA
top_genes <- rownames(deg[deg$diffexpressed != "Not Significant", ])[1:60]
deg$gene_label[rownames(deg) %in% top_genes] <- rownames(deg)[rownames(deg) %in% top_genes]
ggplot(data = deg, aes(x = avg_log2FC, y = -log10(p_val_adj), col = diffexpressed, label = gene_label)) +
  geom_point(alpha = 0.4, size = 1.5) + theme_minimal() +
  ggrepel::geom_text_repel() +
  scale_color_manual(values = c("steelblue", "grey", "firebrick3")) +
  geom_vline(xintercept = c(-0.5, 0.5), col = "black", linetype = "dashed") +
  geom_hline(yintercept = -log10(0.05), col = "black", linetype = "dashed") +
  labs(title = "GPR183 Tfh vs Other Tfh Clusters", x = "log2 Fold Change",
       y = "-log10 Adjusted P-value") +
  theme(legend.title = element_blank(), legend.position = 'none')
```

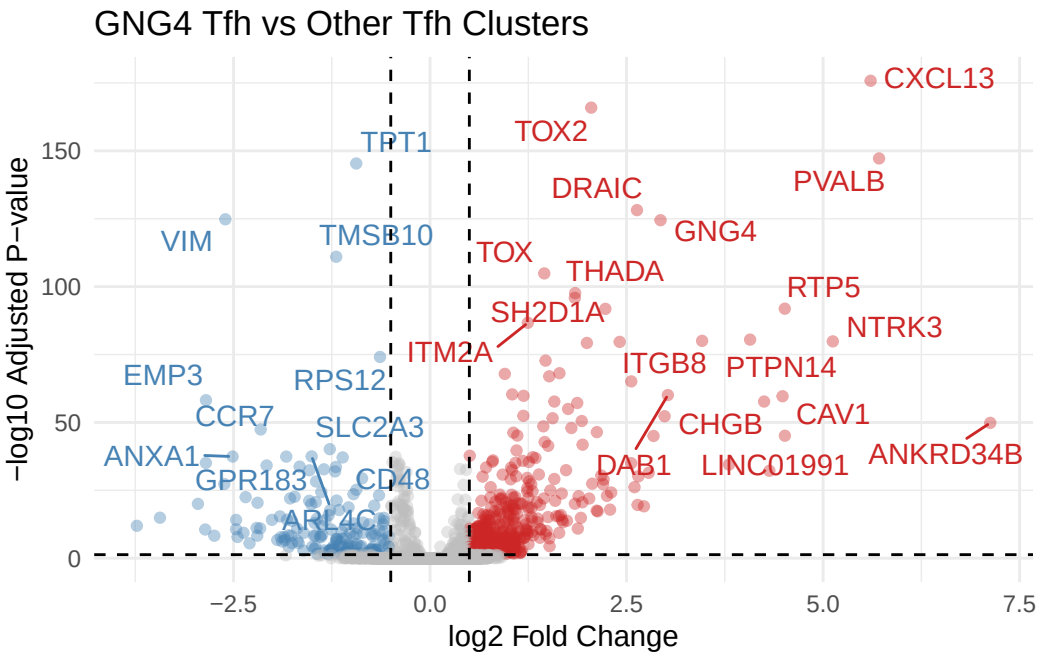
```
Warning: Removed 5315 rows containing missing values or values outside the scale range
(`geom_text_repel()`).
```

```
Warning: ggrepel: 29 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



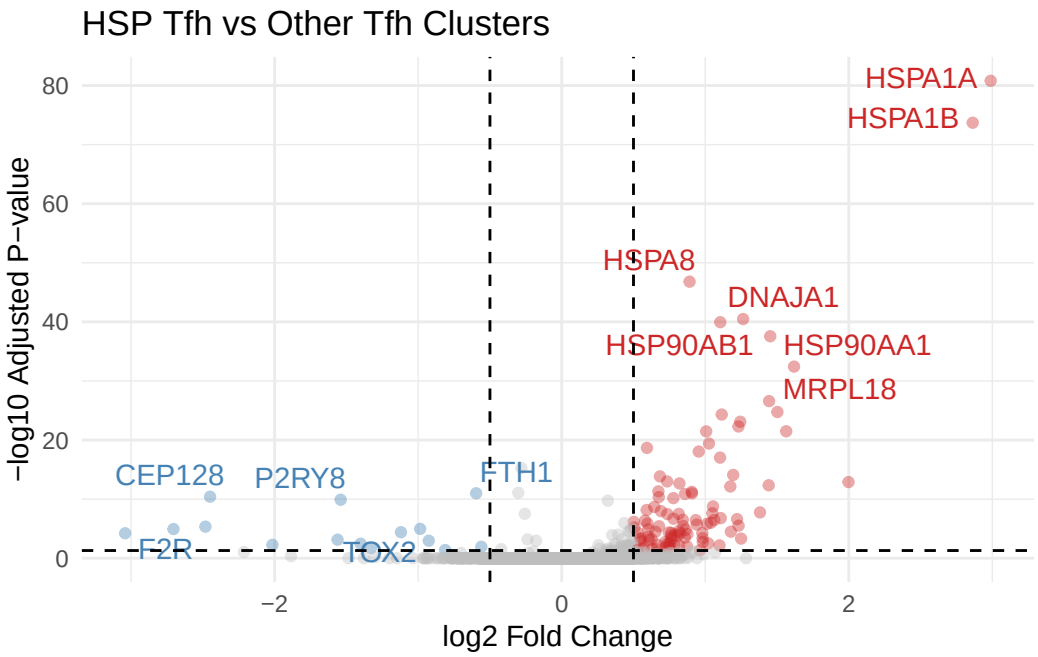

```
# ggsave(filename = 'Images/Tfh-Volcano-GPR183vsOthers.pdf', height=4, width=5)

deg <- FindMarkers(tfh, ident.1 = "GNG4 subset", min.pct = 0.1, logfc.threshold = 0)
deg$diffexpressed <- "Not Significant"
deg$diffexpressed[deg$avg_log2FC > 0.5 & deg$p_val_adj < 0.05] <- "Up"
deg$diffexpressed[deg$avg_log2FC < -0.5 & deg$p_val_adj < 0.05] <- "Down"
deg$gene_label <- NA
top_genes <- rownames(deg[deg$diffexpressed != "Not Significant", ])[1:60]
deg$gene_label[rownames(deg) %in% top_genes] <- rownames(deg)[rownames(deg) %in% top_genes]
ggplot(data = deg, aes(x = avg_log2FC, y = -log10(p_val_adj), col = diffexpressed, label = gene_label)) +
  geom_point(alpha = 0.4, size = 1.5) + theme_minimal() +
  ggrepel::geom_text_repel() +
  scale_color_manual(values = c("steelblue", "grey", "firebrick3")) +
  geom_vline(xintercept = c(-0.5, 0.5), col = "black", linetype = "dashed") +
  geom_hline(yintercept = -log10(0.05), col = "black", linetype = "dashed") +
  labs(title = "GNG4 Tfh vs Other Tfh Clusters", x = "log2 Fold Change",
       y = "-log10 Adjusted P-value") +
  theme(legend.title = element_blank(), legend.position = 'none')
```

```
Warning: Removed 5761 rows containing missing values or values outside the scale range
(`geom_text_repel()`).

Warning: ggrepel: 31 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



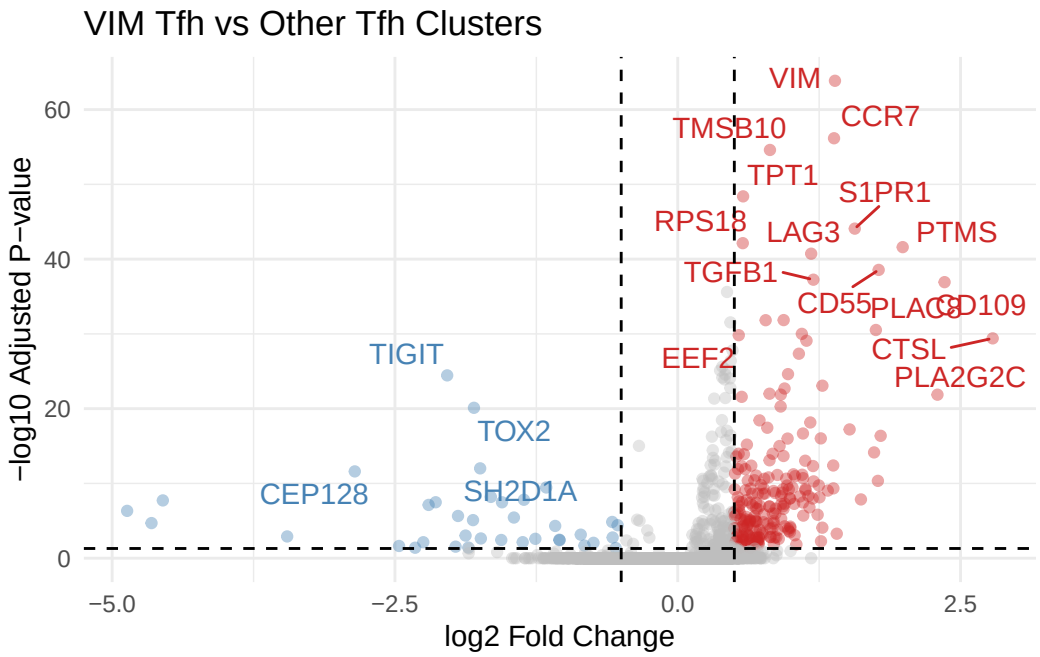

```
# ggsave(filename = 'Images/Tfh-Volcano-GNG4vsOthers.pdf', height=4, width=5)

deg <- FindMarkers(tfh, ident.1 = "HSP subset", min.pct = 0.1, logfc.threshold = 0)
deg$diffexpressed <- "Not Significant"
deg$diffexpressed[deg$avg_log2FC > 0.5 & deg$p_val_adj < 0.05] <- "Up"
deg$diffexpressed[deg$avg_log2FC < -0.5 & deg$p_val_adj < 0.05] <- "Down"
deg$gene_label <- NA
top_genes <- rownames(deg[deg$diffexpressed != "Not Significant", ])[1:60]
deg$gene_label[rownames(deg) %in% top_genes] <- rownames(deg)[rownames(deg) %in% top_genes]
ggplot(data = deg, aes(x = avg_log2FC, y = -log10(p_val_adj), col = diffexpressed, label = gene_label)) +
  geom_point(alpha = 0.4, size = 1.5) + theme_minimal() +
  ggrepel::geom_text_repel() +
  scale_color_manual(values = c("steelblue", "grey", "firebrick3")) +
  geom_vline(xintercept = c(-0.5, 0.5), col = "black", linetype = "dashed") +
  geom_hline(yintercept = -log10(0.05), col = "black", linetype = "dashed") +
  labs(title = "HSP Tfh vs Other Tfh Clusters", x = "log2 Fold Change",
       y = "-log10 Adjusted P-value") +
  theme(legend.title = element_blank(), legend.position = 'none')
```

```
Warning: Removed 5722 rows containing missing values or values outside the scale range
(`geom_text_repel()`).

Warning: ggrepel: 48 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



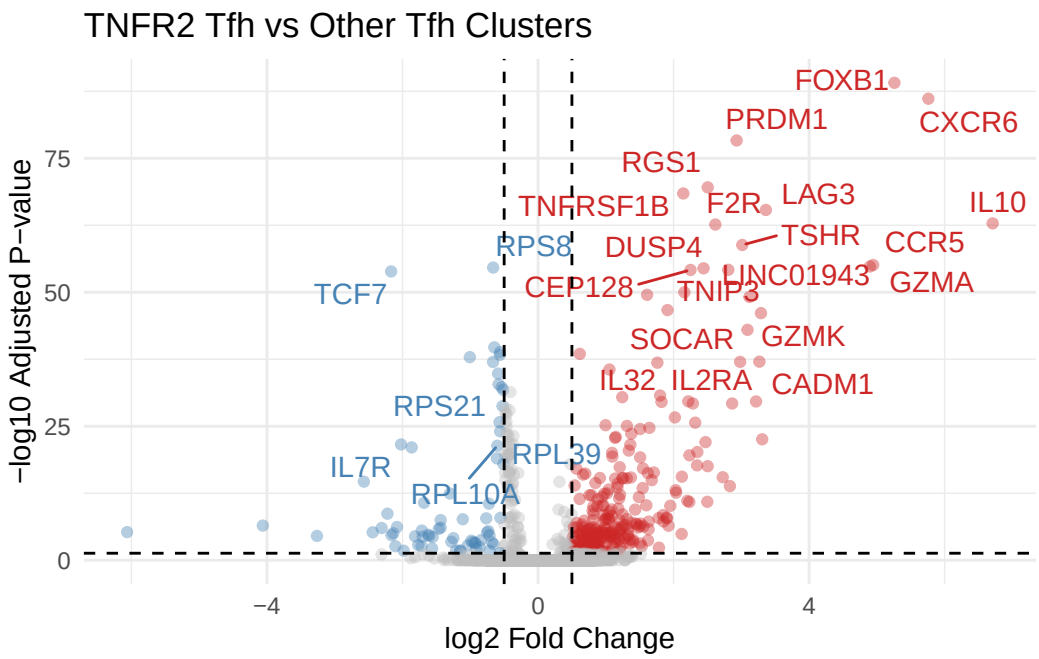

```
# ggsave(filename = 'Images/Tfh-Volcano-HSPvsOthers.pdf', height=4, width=5)

deg <- FindMarkers(tfh, ident.1 = "VIM subset", min.pct = 0.1, logfc.threshold = 0)
deg$diffexpressed <- "Not Significant"
deg$diffexpressed[deg$avg_log2FC > 0.5 & deg$p_val_adj < 0.05] <- "Up"
deg$diffexpressed[deg$avg_log2FC < -0.5 & deg$p_val_adj < 0.05] <- "Down"
deg$gene_label <- NA
top_genes <- rownames(deg[deg$diffexpressed != "Not Significant", ])[1:60]
deg$gene_label[rownames(deg) %in% top_genes] <- rownames(deg)[rownames(deg) %in% top_genes]
ggplot(data = deg, aes(x = avg_log2FC, y = -log10(p_val_adj), col = diffexpressed, label = gene_label)) +
  geom_point(alpha = 0.4, size = 1.5) + theme_minimal() +
  ggrepel::geom_text_repel() +
  scale_color_manual(values = c("steelblue", "grey", "firebrick3")) +
  geom_vline(xintercept = c(-0.5, 0.5), col = "black", linetype = "dashed") +
  geom_hline(yintercept = -log10(0.05), col = "black", linetype = "dashed") +
  labs(title = "VIM Tfh vs Other Tfh Clusters", x = "log2 Fold Change",
       y = "-log10 Adjusted P-value") +
  theme(legend.title = element_blank(), legend.position = 'none')
```

```
Warning: Removed 5592 rows containing missing values or values outside the scale range
(`geom_text_repel()`).

Warning: ggrepel: 41 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```




```
# ggsave(filename = 'Images/Tfh-Volcano-VIMvsOthers.pdf', height=4, width=5)

deg <- FindMarkers(tfh, ident.1 = "TNFR2 subset", min.pct = 0.1, logfc.threshold = 0)
deg$diffexpressed <- "Not Significant"
deg$diffexpressed[deg$avg_log2FC > 0.5 & deg$p_val_adj < 0.05] <- "Up"
deg$diffexpressed[deg$avg_log2FC < -0.5 & deg$p_val_adj < 0.05] <- "Down"
deg$gene_label <- NA
top_genes <- rownames(deg[deg$diffexpressed != "Not Significant", ])[1:60]
deg$gene_label[rownames(deg) %in% top_genes] <- rownames(deg)[rownames(deg) %in% top_genes]
ggplot(data = deg, aes(x = avg_log2FC, y = -log10(p_val_adj), col = diffexpressed, label = gene_label)) +
  geom_point(alpha = 0.4, size = 1.5) + theme_minimal() +
  ggrepel::geom_text_repel() +
  scale_color_manual(values = c("steelblue", "grey", "firebrick3")) +
  geom_vline(xintercept = c(-0.5, 0.5), col = "black", linetype = "dashed") +
  geom_hline(yintercept = -log10(0.05), col = "black", linetype = "dashed") +
  labs(title = "TNFR2 Tfh vs Other Tfh Clusters", x = "log2 Fold Change",
       y = "-log10 Adjusted P-value") +
  theme(legend.title = element_blank(), legend.position = 'none')
```

```
Warning: Removed 5605 rows containing missing values or values outside the scale range
(`geom_text_repel()`).

Warning: ggrepel: 34 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```




```
# ggsave(filename = 'Images/Tfh-Volcano-TNFR2vsOthers.pdf', height=4, width=5)


rm(deg, top_genes)

# ******************  no significant or interpretable results
# #  devtools::install_github('cole-trapnell-lab/monocle3')
# # had to download grr from tarball and manual install
# # remotes::install_github('satijalab/seurat-wrappers')
# library(monocle3)
# library(SeuratWrappers)
# cds <- as.cell_data_set(tfh)
# cds <- cluster_cells(cds, reduction_method = "UMAP")
# cds <- learn_graph(cds)
# cell_ids <- which(colData(cds)[, "tfh.subset.annotation"] == "GNG4 subset")
# closest_vertex <- cds@principal_graph_aux[["UMAP"]]$pr_graph_cell_proj_closest_vertex
# root_pr_nodes <- igraph::V(principal_graph(cds)[["UMAP"]])$name[as.numeric(names(which.max(table(closest_verte
# cds <- order_cells(cds, reduction_method = "UMAP", root_pr_nodes = root_pr_nodes)
# plot_cells(cds, color_cells_by = "pseudotime")
# ************************************************************
```

Hotspot for Tfh - calculate

```
md <- tfh@meta.data
tfh$predicted.celltype.l1 <- tfh$Comments <- tfh$Randomized <- tfh$V2weightkg <- tfh$V3weightkg <- tfh$Immunizat
  tfh$Notes <- tfh$scDblFinder.class <-   tfh$HBsAb2 <- tfh$HBsAb2date <-  NULL   # empty columns

# devtools::install_github("YosefLab/VISION")
library(VISION)
# library(reticulate); py_install("hotspotsc")
options(mc.cores=1) # Use 1 core
vis <- Vision(tfh, signatures = "h.all.v2026.1.Hs.symbols.gmt")
```

```
Importing counts from obj[["RNA"]]@counts ...

Normalizing to counts per 10,000...

Importing Meta Data from obj@meta.data ...

Importing latent space from Embeddings(obj, "pca") using first 50 components

Loading data from h.all.v2026.1.Hs.symbols.gmt ...


Using 18084/38606 genes detected in 0.10% of cells for signature analysis.

See the `sig_gene_threshold` input to change this behavior.

Dropping 'CTstrict' from meta data as it is of type 'character' and has more than 20 unique values.  If you want

Dropping 'cell_id' from meta data as it is of type 'character' and has more than 20 unique values.  If you want

Dropping 'barcode' from meta data as it is of type 'character' and has more than 20 unique values.  If you want

Dropping 'HTO_classification' from meta data as it is of type 'character' and has more than 20 unique values.  I

Dropping 'CTgene' from meta data as it is of type 'character' and has more than 20 unique values.  If you want t

Dropping 'CTnt' from meta data as it is of type 'character' and has more than 20 unique values.  If you want to

Dropping 'CTaa' from meta data as it is of type 'character' and has more than 20 unique values.  If you want to

Adding Visualization: Seurat_pca

Adding Visualization: Seurat_umap

Adding Visualization: Seurat_integrated_dr

Adding Visualization: Seurat_ref.umap

Adding Visualization: Seurat_harmony
```

```
vis <- analyze(vis)
```

```
Beginning Analysis

Clustering cells...

Using latent space to cluster cells...

completed

Projecting data into 2 dimensions...
```

```
  Running method 1/1: tSNE30 ...



Evaluating signature scores on cells...

Evaluating signature-gene importance...

Creating 5 background signature groups with the following parameters:

  sigSize sigBalance
1      23  1.0000000
2      53  1.0000000
3      92  1.0000000
4     162  1.0000000
5     243  0.5704967

  signatures per group: 3000

Computing KNN Cell Graph in the Latent Space...

Evaluating local consistency of signatures in latent space...

Clustering signatures...

Computing differential signature tests...

Computing correlations between signatures and latent space components...

Analysis Complete!
```

```
vis <- runHotspot(vis, model = "danb")
```

```
    Applying Threshold filter...removing genes detected in less than 150 cells

      Genes Retained: 7632

    Applying Fano filter...removing genes with Fano < 2.0 MAD in each of 30 bins

      Genes Retained: 1267

Computing pair-wise local correlation on 1000 features...
Computing scores for 19 modules...

Computing Module-Signature Enrichment

Generating Module Signature Overlaps...

Evaluating module scores on cells...
```

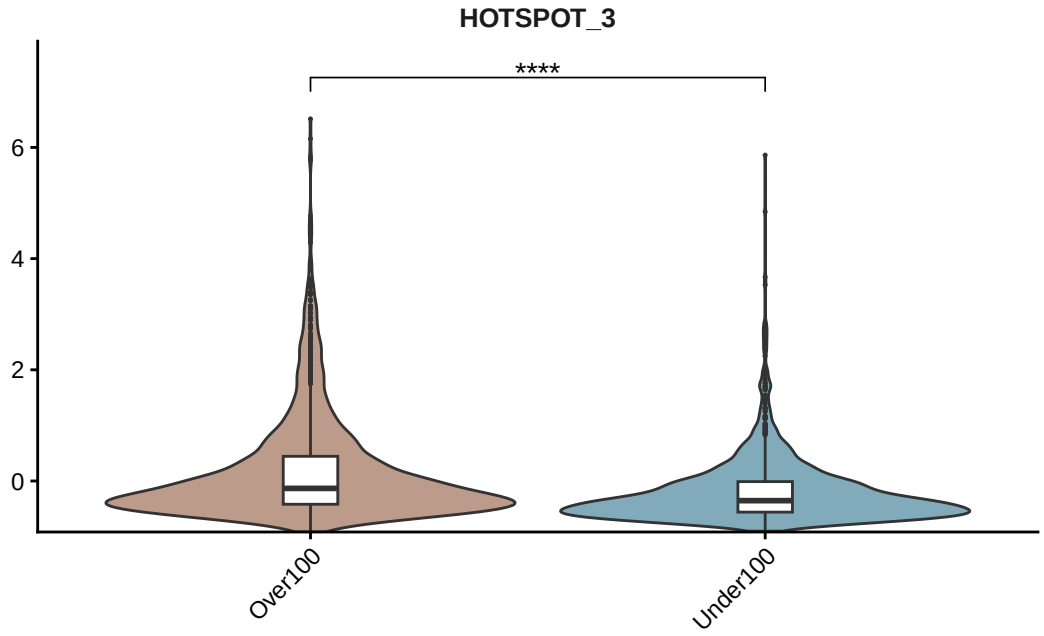
```
Evaluating signature-gene importance...

Computing differential modules and overlaps tests...

Creating 5 background signature groups with the following parameters:

  sigSize sigBalance
1       2          1
2       5          1
3      24          1
4      45          1
5      96          1

  signatures per group: 3000

Computing KNN Cell Graph in the Latent Space...

Evaluating local consistency of modules in latent space...
```

```r
# viewResults(vis)


# saveRDS(vis, "tfh_vision_hotspot_results.rds")
vis <- readRDS("tfh_vision_hotspot_results.rds")


tfh <- AddMetaData(tfh, metadata = as.data.frame( vis@ModuleHotspotScores ))

# saveRDS(tfh, "20260407_tfh_subclustered.RDS")
tfh <- readRDS("20260407_tfh_subclustered.RDS")
```

HOTSPOT for Tfh - analysis and plots

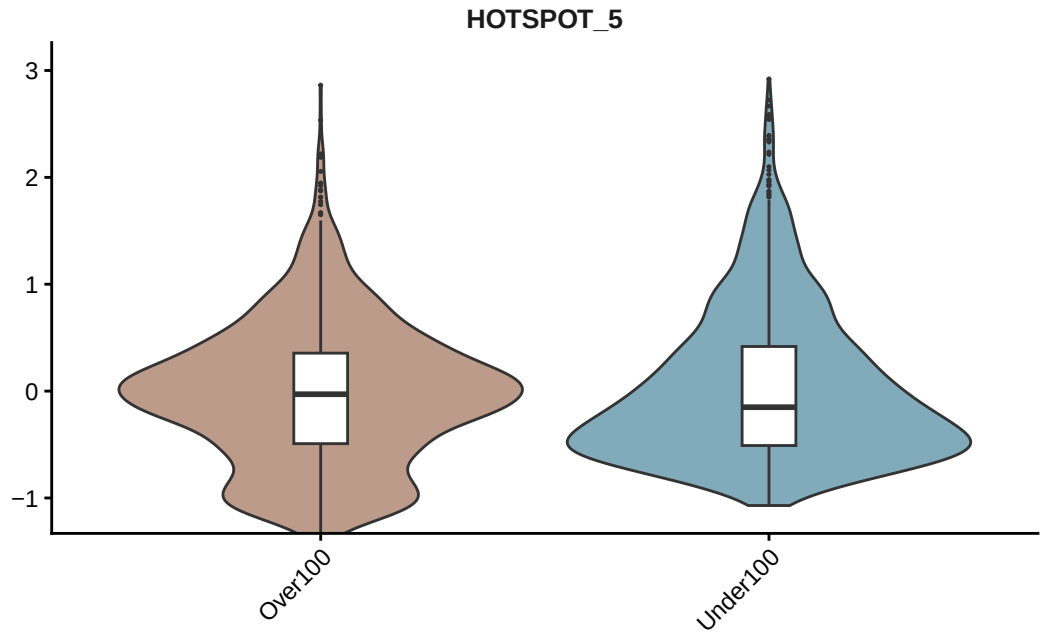
```r
VlnPlot2(tfh, features = "HOTSPOT_3", group.by = "titer.categ", pt=F, stat.method = "wilcox.test") + theme(plot.
```

```
The 'I want hue' color presets were generated from: https://medialab.github.io/iwanthue/
This message is shown once per session
```




```
VlnPlot2(tfh, features = "HOTSPOT_5", group.by = "titer.categ", pt=F, stat.method = "wilcox.test") + theme(plot.
```

```
No statistical significance.
```



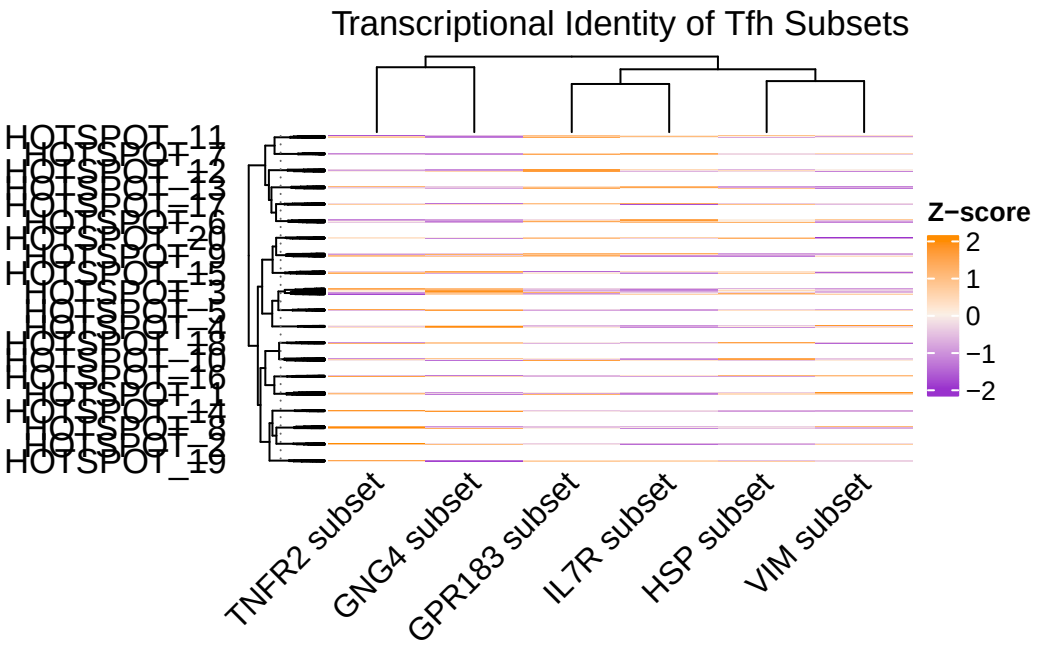

```
wilcox.test(HOTSPOT_4 ~ titer.categ, data = tfh@meta.data)
```

```
    Wilcoxon rank sum test with continuity correction

data:  HOTSPOT_4 by titer.categ
W = 839684, p-value < 2.2e-16
alternative hypothesis: true location shift is not equal to 0
```

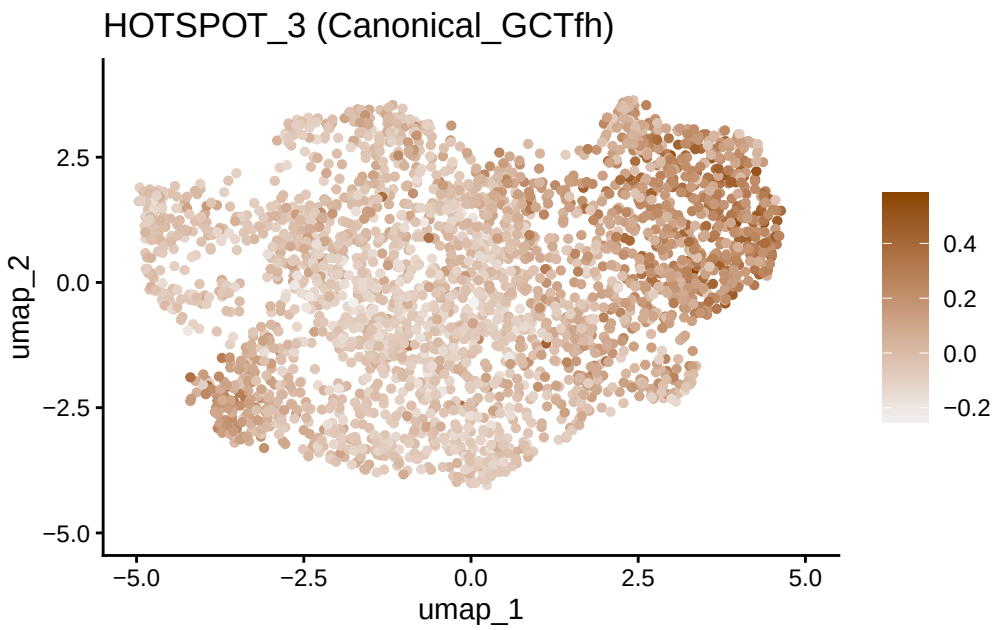
```
hotspot_4_genes <- as.data.frame(names(vis@modData[["HOTSPOT_4"]]@sigDict))

hotspot_names <- grep("^HOTSPOT_", names(vis@modData), value = TRUE)
hotspot_list <- lapply(hotspot_names, function(mod_name) {
  names(vis@modData[[mod_name]]@sigDict)  }) # Extract the gene lists for every module
names(hotspot_list) <- hotspot_names # Name the lists

# Pad the shorter modules with empty spaces to make them fit in a table
padded_modules <- lapply(hotspot_list, function(genes) {
  length(genes) <- max(sapply(hotspot_list, length)) # maximum # genes in any module
  genes[is.na(genes)] <- ""
  return(genes)
})
hotspot_export_df <- as.data.frame(padded_modules) # Bind all together into dataframe
# write.csv(hotspot_export_df, file = "HBV_Study_Hotspot_Modules_Complete.csv",
#           row.names = FALSE, quote = FALSE)

genes.modules <-  hotspot_export_df %>% unlist
genes.modules <- genes.modules[which(genes.modules != "")]
Idents(tfh) <- "tfh.subset.annotation"
DefaultAssay(tfh) <- "RNA"
genes.modules <- lapply(as.list(padded_modules), function(x) {
  x[x != "" & !is.na(x)] })  # For each column, remove the empty strings ("")
existing_genes <- intersect(unlist(genes.modules), rownames(tfh[["RNA"]]))
avg_exp <- AverageExpression(tfh, features = existing_genes,
                             group.by = "tfh.subset.annotation",
                             assay = "RNA", slot = "data")$RNA
```

```
As of Seurat v5, we recommend using AggregateExpression to perform pseudo-bulk analysis.
This message is displayed once per session.
```

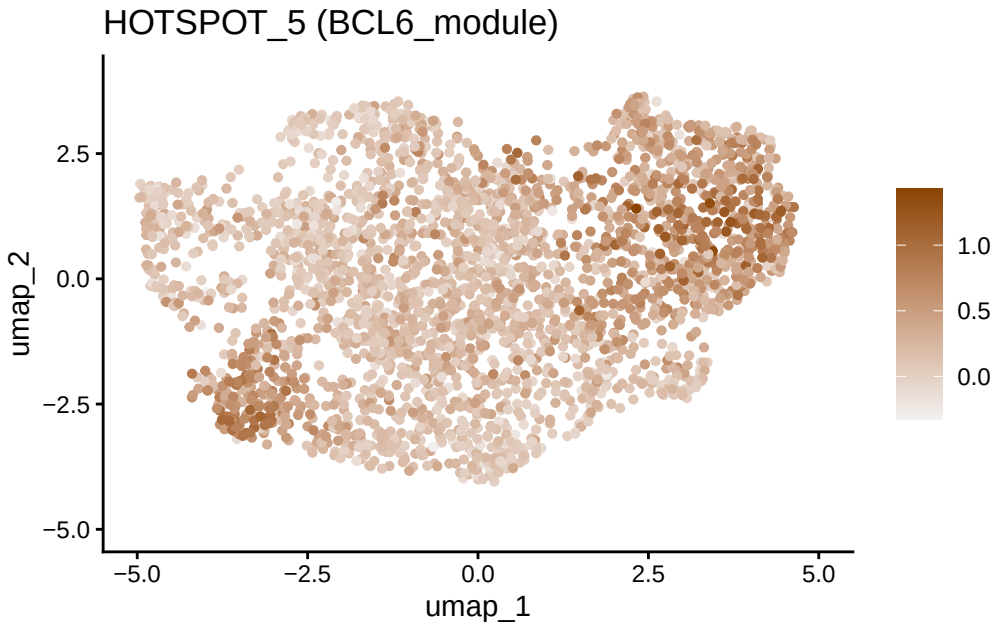
```
avg_exp <- t(scale(t(avg_exp)))
gene_groups <- rep(names(genes.modules), times = sapply(genes.modules, length))
names(gene_groups) <- unlist(genes.modules)
gene_groups <- gene_groups[existing_genes]
row_split_factor <- factor(gene_groups, levels = names(genes.modules))
col_fun <- circlize::colorRamp2(c(-2, 0, 2), c("darkorchid3", "linen", "darkorange"))
ComplexHeatmap::Heatmap(
  matrix = as.matrix(avg_exp),
  col = col_fun,
  name = "Z-score",
  cluster_columns = TRUE,
  cluster_rows = TRUE,
  row_split = row_split_factor,
  row_title_rot = 0,
  column_names_rot = 45,
  show_row_names = FALSE,    # Hides the overlapping gene names
  column_title = "Transcriptional Identity of Tfh Subsets",
  row_gap = unit(2, "mm")
)
```



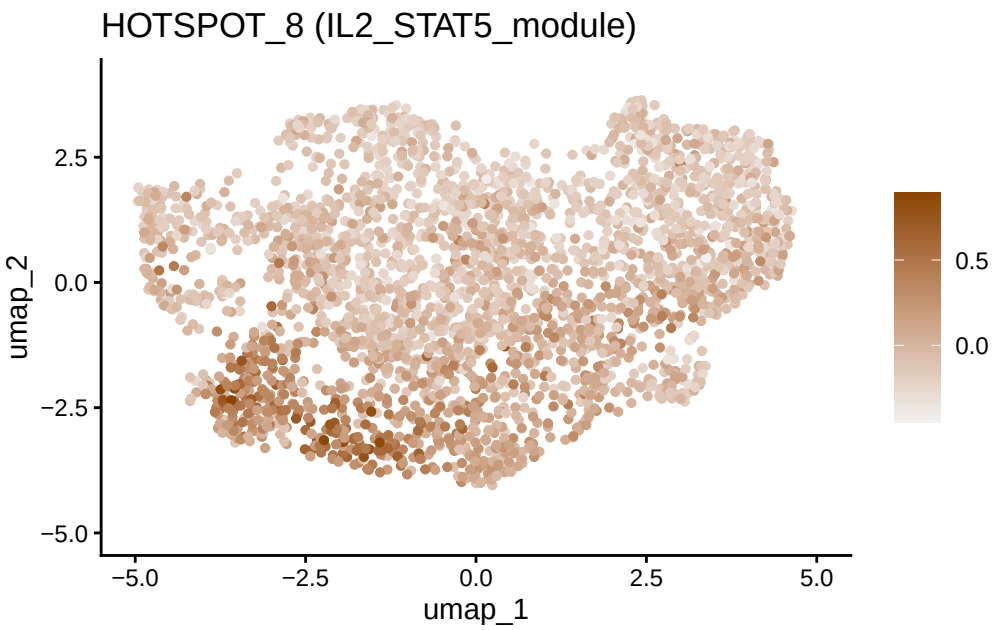

```r
tfh <- AddModuleScore(object = tfh,
                      features = list(
                        Canonical_GCTfh = genes.modules[["HOTSPOT_3"]],
                        BCL6_module = genes.modules[["HOTSPOT_5"]],
                        IL2_STAT5_module = genes.modules[["HOTSPOT_8"]]),
                      name = "HS_Score_") # prefix to the metadata columns
colnames(tfh@meta.data)[grep("HS_Score_1", colnames(tfh@meta.data))] <- "Canonical_GCTfh"
colnames(tfh@meta.data)[grep("HS_Score_2", colnames(tfh@meta.data))] <- "BCL6_module"
colnames(tfh@meta.data)[grep("HS_Score_3", colnames(tfh@meta.data))] <- "IL2_STAT5_module"

FeaturePlot(tfh, features = "Canonical_GCTfh", pt.size =1,
            cols =c('gray95','darkorange4')) +
  labs(title = "HOTSPOT_3 (Canonical_GCTfh)") +  theme_classic()
```




```
# ggsave(filename = 'Images/Tfh-UMAP-hotspot-3.pdf', height=4, width=5)

FeaturePlot(tfh, features = "BCL6_module", pt.size =1,
            cols =c('gray95','darkorange4')) +
  labs(title = "HOTSPOT_5 (BCL6_module)") +  theme_classic()
```

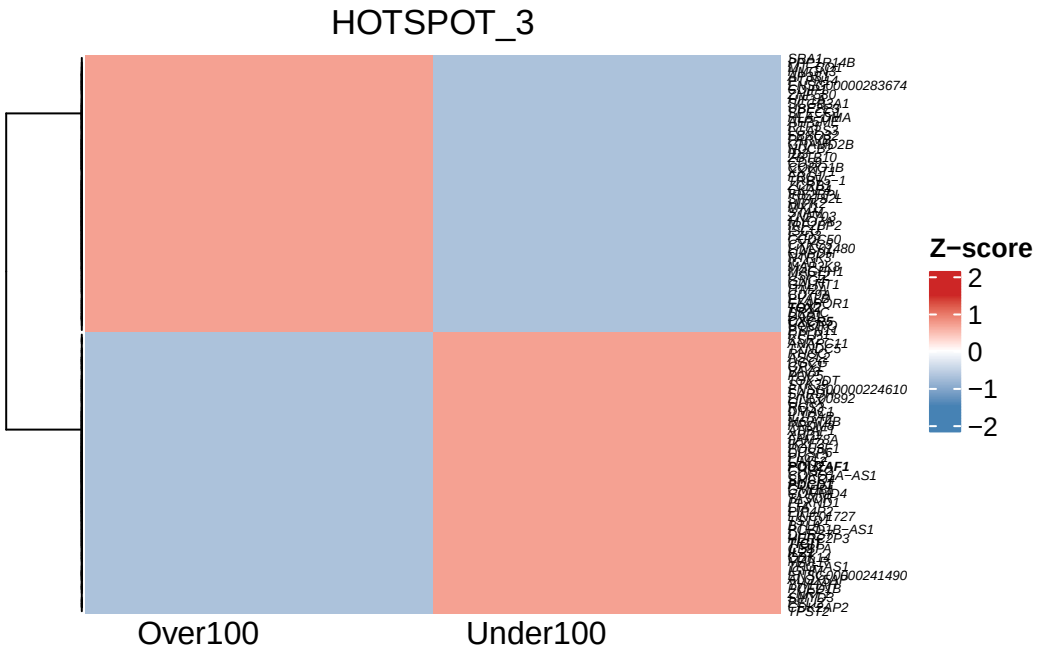



```
# ggsave(filename = 'Images/Tfh-UMAP-hotspot-5.pdf', height=4, width=5)

FeaturePlot(tfh, features = "IL2_STAT5_module", pt.size =1,
            cols =c('gray95','darkorange4')) +
  labs(title = "HOTSPOT_8 (IL2_STAT5_module)")+  theme_classic()
```

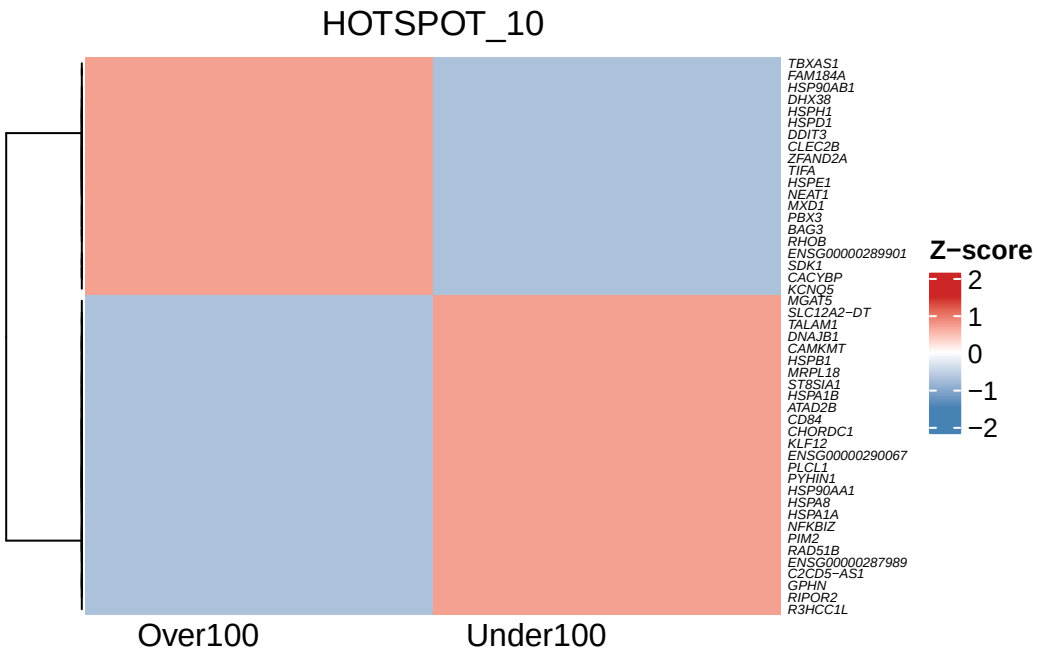



```
# ggsave(filename = 'Images/Tfh-UMAP-hotspot-8.pdf', height=4, width=5)


# saveRDS(tfh, "20260319_tfh_subclustered.RDS")
# tfh <- readRDS("20260319_tfh_subclustered.RDS")


plot.hotspot.genes <- function(object, module.name, gene.mod, emph.genes=NULL,
                               metadata.feature)
{
  module_genes <- gene.mod[[module.name]]
  avg_exp <- AverageExpression(tfh, features = module_genes, group.by = metadata.feature,
                                    assay = "RNA", slot = "data")$RNA
  existing_target_genes <- intersect(module_genes, rownames(avg_exp))
  avg_exp_subset <- avg_exp[existing_target_genes, ]
  avg_exp <- t(scale(t(avg_exp_subset)))
  col_fun = colorRamp2(c(-1.5, 0, 1.5), c("steelblue", "white", "firebrick3"))
  genes_in_heatmap <- rownames(avg_exp)
  font_faces <- rep("italic", length(genes_in_heatmap))
  bold_genes <- emph.genes
  bold_indices <- which(genes_in_heatmap %in% bold_genes)
  font_faces[bold_indices] <- "bold.italic"

  a <- ComplexHeatmap::Heatmap(avg_exp,
          name = "Z-score",
          col = col_fun,
          cluster_columns = FALSE,
          row_names_gp = grid::gpar(fontsize = 5, fontface=font_faces ),
          column_title = module.name,
          column_names_rot = 0)
    return(a)
}

plot.hotspot.genes(object = tfh, module.name = 'HOTSPOT_3', gene.mod = genes.modules,
```

```
                metadata.feature ='titer.categ',
                emph.genes = c("PDCD1", "TIGIT", "CXCR5", "IL21",
                               "TOX2","FAS", "POU2AF1" ))
```



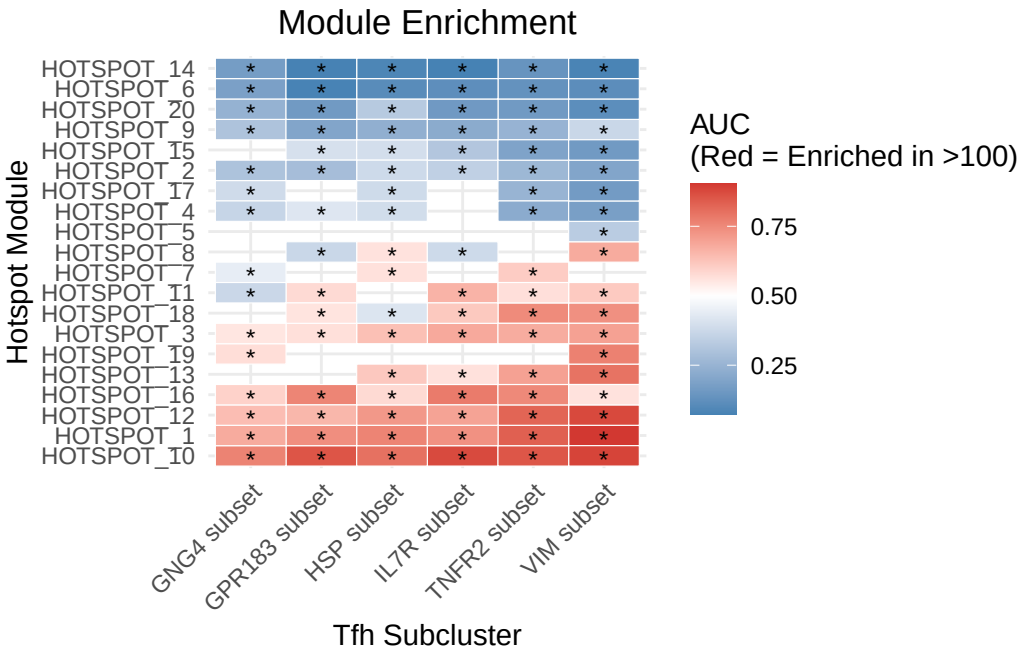

```
plot.hotspot.genes(object = tfh, module.name = 'HOTSPOT_10', gene.mod = genes.modules,
                metadata.feature ='titer.categ')
```




```
md <- tfh@meta.data
clusters <- unique(md$tfh.subset.annotation)
results_list <- list()
modules <- grep("^HOTSPOT_", colnames(md), value = TRUE)
```

```r
for (clust in clusters) {
  meta_clust <- md[md$tfh.subset.annotation == clust, ]   # Isolate cells per cluster
  # Skip the cluster if it doesn't have enough cells in both titer groups to test
  if(sum(meta_clust$titer.categ == "Over100") < 5 | sum(meta_clust$titer.categ == "Under100") < 5) next
  for (mod in modules) {
    scores_high <- meta_clust[meta_clust$titer.categ == "Over100", mod]
    scores_low  <- meta_clust[meta_clust$titer.categ == "Under100", mod]
    test_result <- wilcox.test(scores_high, scores_low, exact = FALSE) # Wilcoxon test
    W <- test_result$statistic      # Calculate AUC
    auc_value <- unname(W / (length(scores_high) * length(scores_low)))
    results_list[[paste(clust, mod)]] <- data.frame(
      Cluster = clust,
      Module = mod,
      AUC = auc_value,
      P_val = test_result$p.value
    )
  }
}

#
# Method details
#
# Step 2: The Inner Loop (The Module Test)
#
# for (mod in modules) { ... }
#
# Inside each cluster, the code now loops through all 18 Hotspot modules.
#
# Group Splitting: It separates the Hotspot scores for that specific module into two buckets: scores_high (Respo
#
# The Wilcoxon Test: wilcox.test(scores_high, scores_low) This is a non-parametric test (it doesn't assume your
#
# Step 3: The AUC Calculation (Effect Size)
#
# auc_value <- unname(W / (length(scores_high) * length(scores_low)))
#
# This is the most critical part of your "Success vs. Failure" story.
#
# W: This is the "U-statistic" from the Wilcoxon test-it's essentially a count of how many times a cell from the
#
# The Denominator: By dividing by the product of the group sizes, we turn that count into a probability (0 to 1)
#
# AUC = 0.5: Means the module is identical in both groups (random chance).
#
# AUC = 0.9: Means there is a 90% probability that a responder cell has a higher score than a non-responder cell



rm(scores_high); rm(scores_low); rm(test_result); rm(auc_value); rm(W)
enrichment_df <- bind_rows(results_list)
enrichment_df$FDR <- p.adjust(enrichment_df$P_val, method = "fdr")
enrichment_df <- enrichment_df %>% filter(FDR < 0.05) %>% arrange(-AUC)
enrichment_df$Significance <- ifelse(enrichment_df$FDR < 0.05, "*", "")

setDT(enrichment_df)
avg.hotspot <- enrichment_df[, .(AvgAUC = mean(AUC)), by = Module]
```

```
avg.hotspot <- avg.hotspot[order(AvgAUC, decreasing=T)]
enrichment_df[, Module := factor(Module, levels = avg.hotspot$Module)]  # sort the results

ggplot(enrichment_df, aes(x = Cluster, y = Module, fill = AUC)) +
  geom_tile(color = "white") +
  geom_text(aes(label = Significance), color = "black", size = 3, vjust = 0.7) +
  scale_fill_gradient2(low = "steelblue", mid = "white",
                       high = "firebrick3", midpoint = 0.5,
                       name = "AUC\n(Red = Enriched in >100)") +
  theme_minimal() +
  labs(title = "Module Enrichment",
       x = "Tfh Subcluster",
       y = "Hotspot Module") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        plot.title = element_text(hjust = 0.5))
```




```
rm(padded_modules, meta_clust, avg.hotspot, hotspot_list, hotspot_4_genes,
   hotspot_export_df, avg_exp, results_list, new.meta)
```

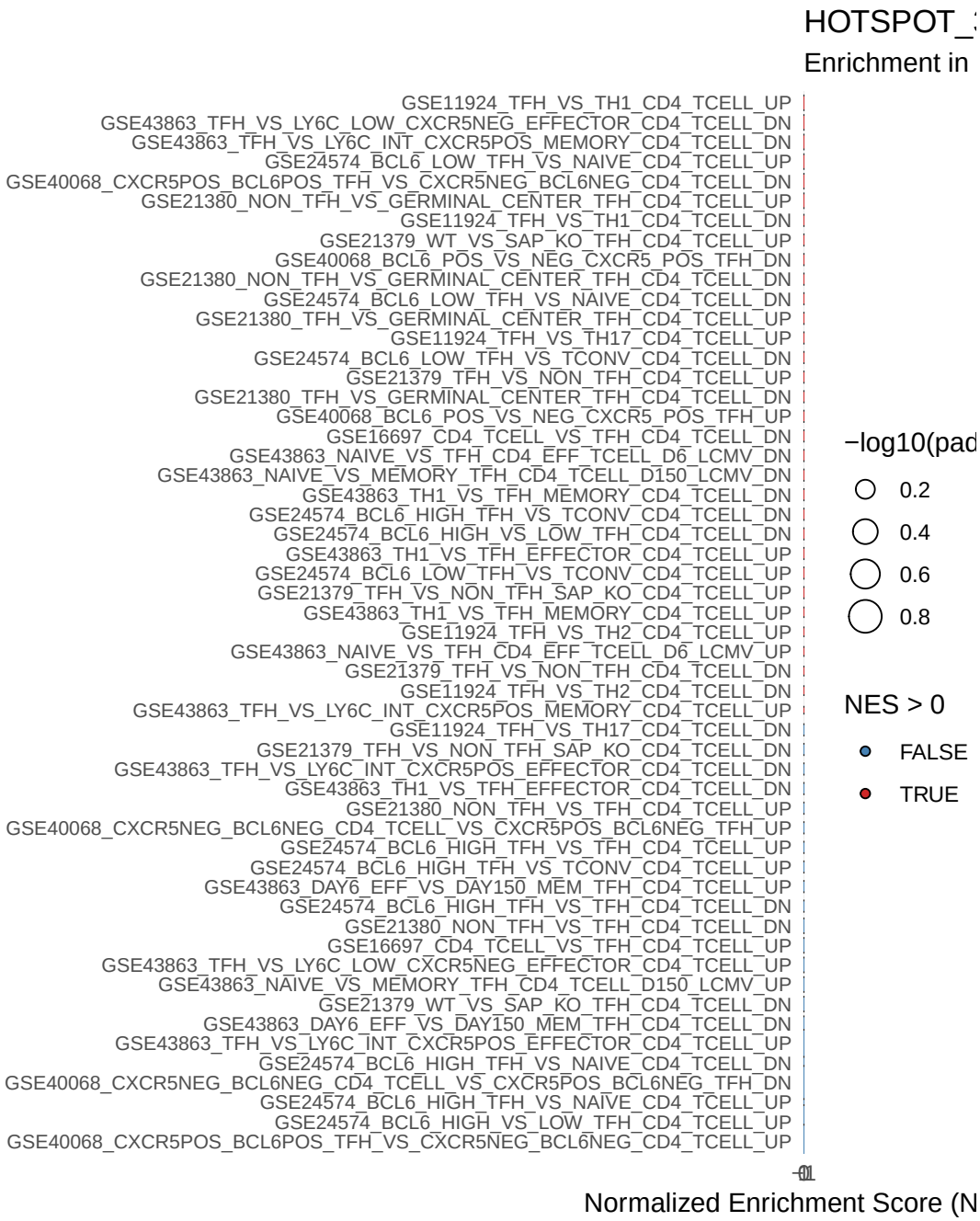
So this tells me that the Hotspots are not uniform. Hotspots are identifying co-correlated genes but this is across the full Tfh spectrum. There must be another layer to consider, because miloR suggested differential abundance for sub-groups in narrowly-defined regions of the UMAP projection.

Although Hotspot_10 is the most interesting for relevance to high antibody titer, not all of these subsets are performing the same function so hard to say what this means. These enrichments could reflect global effects but may not offer insight into what really distinguishes strong vs weak antibody producers. Probably we need to focus in on the specific interesting subset.

FGSEA for the hotspot modules

```
library(fgsea)
# top <- FindMarkers(tfh, ident.1 = "GNG4 subset",
```

```
#                               group.by = "tfh.subset.annotation", assay = "RNA",
#                               slot = "data", logfc.threshold = 0.5)
#   setDT(top, keep.rownames = "gene")
#   min_p <- min(top[p_val > 0, p_val])
#   top[p_val == 0, p_val := min_p]
#   top[, rank_score := avg_log2FC * -log10(p_val)]
#   ranks <- setNames(top$rank_score, top$gene)
#   ranks <- sort(ranks, decreasing = TRUE)
#   pathways <- tfh_msig # genes.modules
#   fgseaRes <- fgsea(pathways = pathways, stats = ranks, minSize = 10,maxSize = 500)
#   significant_modules <- fgseaRes[padj < 0.05][order(NES, decreasing = TRUE)]
# plotEnrichment(comp_pathways, ranks) +
#   labs(title = "GNG4 Identity: Comparison to Canonical Tfh",
#        subtitle = paste0("NES: ", round(fgseaRes[pathway == "GSE21379_TFH_VS_NON_TFH_UP", NES], 2)))
#

#
# tfh_msig <- msigdbr::msigdbr(species = "human", collection = "C7") %>%
#   filter(grepl("TFH", gs_name)) %>% split(x = .$gene_symbol, f = .$gs_name)
# saveRDS(tfh_msig, "20260318_MSigDB_C7_grepTfh.RDS")
tfh_msig <- readRDS("20260318_MSigDB_C7_grepTfh.RDS")

top <- FindMarkers(tfh, ident.1 = "GNG4 subset",
                               group.by = "tfh.subset.annotation", assay = "RNA",
                               slot = "data", logfc.threshold = 0.5)
setDT(top, keep.rownames = "gene")
min_p <- min(top[p_val > 0, p_val])
top[p_val == 0, p_val := min_p]
top[, rank_score := avg_log2FC * -log10(p_val)]
ranks <- setNames(top$rank_score, top$gene)
ranks <- sort(ranks, decreasing = TRUE)
selected <- tfh_msig[c(
  "GSE21379_GC_VS_NON_GC_TFH_CD4_T_CELL_UP",
  "GSE10325_CD4_T_CELL_VS_B_CELL_UP", # Tfh-like help genes
  "GSE21379_MAFF_VS_CTRL_TFH_CD4_T_CELL_UP" # Tfh maturation
)]
comp_pathways <- tfh_msig # c(list(HOTSPOT_3 = genes.modules[["HOTSPOT_3"]]), selected)
comp_results <- fgsea(pathways = comp_pathways, stats = ranks, minSize = 10, maxSize = 500)
ggplot(comp_results, aes(x = reorder(pathway, NES), y = NES, fill = NES > 0)) +
  geom_point(aes(size=-log10(padj)), color = "black", pch=21) +
  coord_flip() + scale_fill_manual(values = c("steelblue", "firebrick3")) +
  theme_minimal() +
  labs(title = "HOTSPOT_3 vs. Canonical Tfh Biology",
       subtitle = "Enrichment in Over100 vs. Under100",
       x = "Signature", y = "Normalized Enrichment Score (NES)") +
  theme(axis.text.y = element_text(size = 8))
```




```
rm(comp_pathways, ranks, top, min_p, selected)
```

Again, I think this tells me that the Hotspots are not uniform. I see a lot of results that are in-line with expectations. For example, top hit for NES is Tfh-vs-Th1-UP for HOTSPOT_3. But then the next hit is Tfh-vs-LY6Cloweffector_DOWN which does not make sense (same for the next few hits). But anyway these enrichments are weak and the -log10(Padj) do not exceed 1.3 (= Padj<0.05). So this result may not be reliable.

Clonal sharing in Tfh subsets

```
clonotype_df <- tfh@meta.data %>%
  select(tfh.subset.annotation, CTstrict) %>% # Use your clonotype column name
```

```
  filter(!is.na(CTstrict))

# 2. Identify clones present in more than one cluster
shared_clones <- clonotype_df %>% group_by(CTstrict) %>%
  filter(n_distinct(tfh.subset.annotation) > 1) %>% ungroup()

# 3. Create the adjacency matrix (Sharing counts)
# This counts how many times Cluster A and Cluster B share the same clone
sharing_matrix <- table(shared_clones$tfh.subset.annotation, shared_clones$CTstrict)
adjacency_matrix <- sharing_matrix %*% t(sharing_matrix)
diag(adjacency_matrix) <- 0 # Remove self-sharing to focus on cross-cluster links
grid_cols = c("TNFR2 subset"   = "skyblue",           # Closest to #80B1D3
              "GNG4 subset"   = "mediumaquamarine",  # Closest to #8DD3C7
              "GPR183 subset" = "thistle",           # Closest to #BEBADA
              "IL7R subset"   = "salmon",            # Closest to #FB8072
              "HSP subset"   = "sandybrown",        # Closest to #FDB462
              "VIM subset"   = "yellowgreen"        # Closest to #B3DE69
              )
circos.clear()
pdf(file = 'Images/Tfh-Circos-subsets-clonalSharing.pdf', width = 7, height = 7)
circos.par(canvas.xlim = c(-1.5, 1.5),
           canvas.ylim = c(-1.5, 1.5),
           gap.after = c(rep(2, length(grid_cols)-1), 10)) # Small gap for readability

# Draw the Chord Diagram but SUPPRESS the default text and axis ticks
# (annotationTrack = "grid" draws only the colored rings)
chordDiagram(adjacency_matrix,
             grid.col = grid_cols,
             transparency = 0.4,
             annotationTrack = "grid", # Only the ring, no default text
             preAllocateTracks = list(track.height = 0.1)) # Allocate outer space

# Add labels manually outside the grids using low-level functions
circos.track(track.index = 1,
             panel.fun = function(x, y)  {
    # Get info for current sector (the Tfh subset)
    xlim = get.cell.meta.data("xlim")
    xcenter = get.cell.meta.data("xcenter")
    sector.name = get.cell.meta.data("sector.index")
    clean_name <- gsub(" subset", "", sector.name)
    circos.text(
      xcenter,
      get.cell.meta.data("ylim")[1] + mm_y(2), # Small offset to push text away
      labels = clean_name,
      facing = "clockwise",
      niceFacing = TRUE,
      adj = c(0, 0.5), # Anchor at the start of the text
      cex = 0.8       # Font size
      )
    # circos.text(xcenter, 1, # Position at top center of outer track
    #             sector.name,
    #             facing = "clockwise",
    #             niceFacing = TRUE, # makes text readable
    #             adj = c(0, 0.5), # Anchor text at start of text, vertically centered
    #             cex = 0.8) # Shrink font slightly
    },
```

```
  bg.border = NA) # No border for text track
dev.off()
```

```
pdf
  2
```

```
circos.clear()

print(adjacency_matrix)
```

```
               GNG4 subset HSP subset IL7R subset GPR183 subset VIM subset
  GNG4 subset             0         13           2             0          1
  HSP subset             13          0           2             0          0
  IL7R subset             2          2           0             1          1
  GPR183 subset           0          0           1             0          1
  VIM subset              1          0           1             1          0
  TNFR2 subset           19          4           2             0          0

               TNFR2 subset
  GNG4 subset            19
  HSP subset              4
  IL7R subset             2
  GPR183 subset           0
  VIM subset              0
  TNFR2 subset            0
```

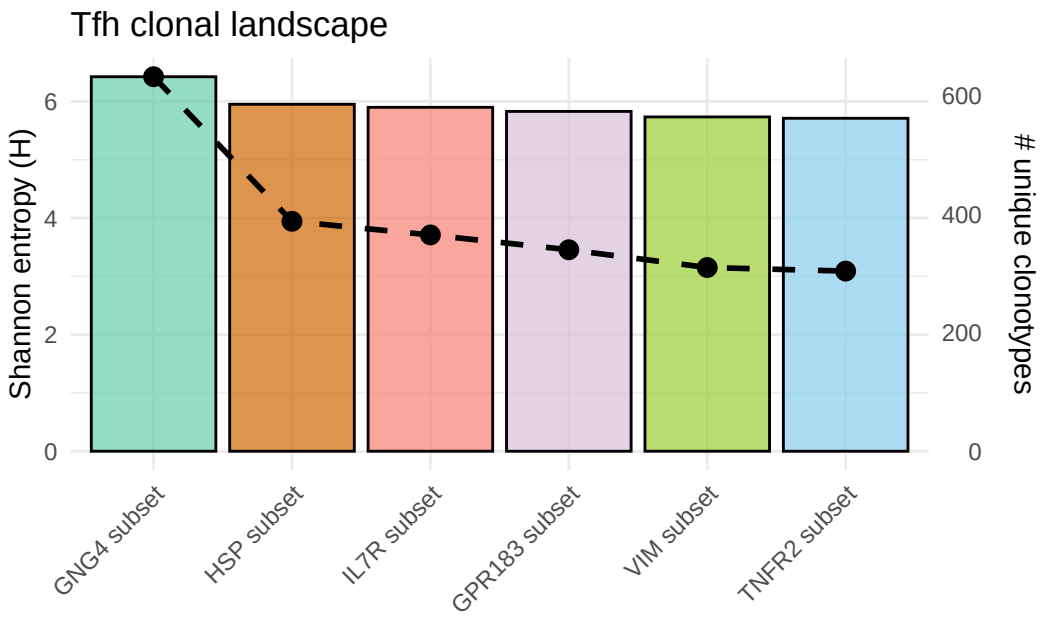
```
rm(adjacency_matrix, shared_clones, clonotype_df, sharing_matrix)
```

Assessing clonal diversity

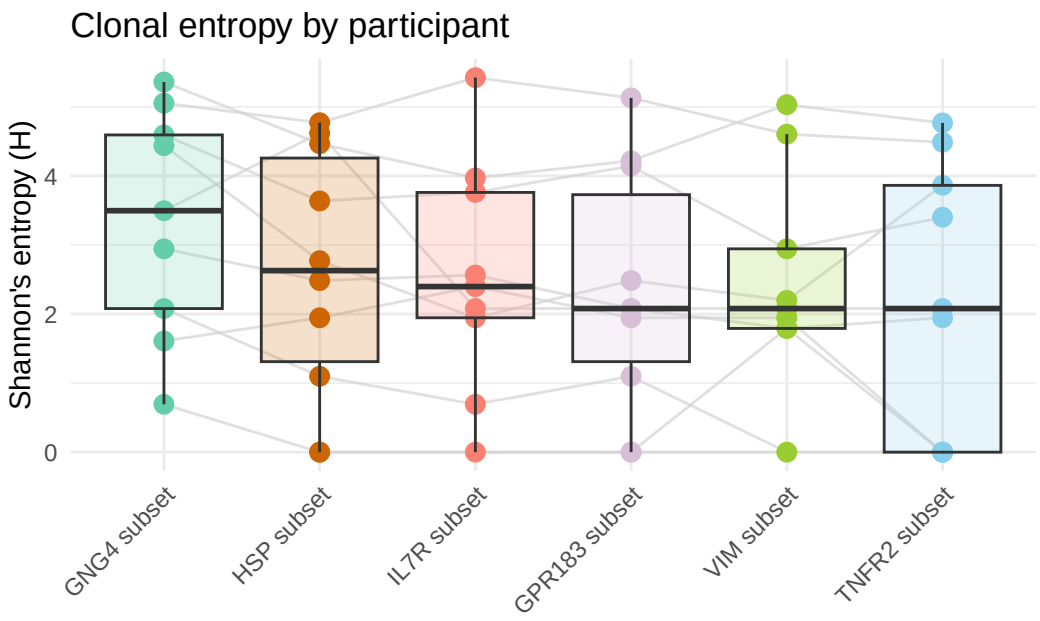
```
md <- as.data.table(tfh@meta.data)
entropy_results <- md[!is.na(CTstrict), .N, by = .(tfh.subset.annotation, CTstrict)][,
  .(
    Unique_Clonotypes = .N,
    Shannon_H = -sum((N/sum(N)) * log(N/sum(N)))
  ),
  by = tfh.subset.annotation]

setorder(entropy_results, -Shannon_H)
scale_factor <- max(entropy_results$Unique_Clonotypes) / max(entropy_results$Shannon_H)
ggplot(entropy_results, aes(x = reorder(tfh.subset.annotation, -Shannon_H))) +
  geom_bar(aes(y = Shannon_H, fill = tfh.subset.annotation),
           stat = "identity", color = "black", alpha = 0.7) +
  # We divide by the scale_factor to bring the numbers down to the Entropy range
  geom_path(aes(y = Unique_Clonotypes / scale_factor, group = 1),
            color = "black", size = 1, linetype = "dashed") +
  geom_point(aes(y = Unique_Clonotypes / scale_factor),
             size = 3, color = "black") +
  scale_y_continuous(name = "Shannon entropy (H)",
    sec.axis = sec_axis(~ . * scale_factor, name = "# unique clonotypes")) +
  scale_fill_manual(values = subset.cols) +
```

```
  theme_minimal() +
  labs(title = "Tfh clonal landscape", x="") +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none",
    axis.title.y.right = element_text(color = "black", margin = margin(l = 10))  )
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```




```
# ggsave(filename = 'Images/Tfh-subsets-Entropy-perSubset.pdf', height=3.5, width=2.5)


entropy_results <- md[!is.na(CTstrict), .N, by = .(subject, tfh.subset.annotation, CTstrict)][, .(
  Shannon_H = -sum((N/sum(N)) * log(N/sum(N)))
    ), by = .(subject, tfh.subset.annotation)]

ggplot(entropy_results, aes(x = tfh.subset.annotation, y = Shannon_H)) +
  geom_line(aes(group = subject), color = "lightgrey", size = 0.5, alpha = 0.7) +
  geom_point(aes(color = tfh.subset.annotation), size = 3) +
  geom_boxplot(aes(fill = tfh.subset.annotation), alpha = 0.2, outlier.shape = NA) +
  scale_color_manual(values = subset.cols) +
  scale_fill_manual(values = subset.cols) +
  theme_minimal() +
  labs(title = "Clonal entropy by participant", x="", y = "Shannon's entropy (H)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1),
        legend.position = "none")
```




```
ggsave(filename = 'Images/Tfh-subsets-Entropy-perparticipant.pdf', height=3, width=4)
rm(entropy_results, scale_factor)
```

Slingshot trajectory analysis

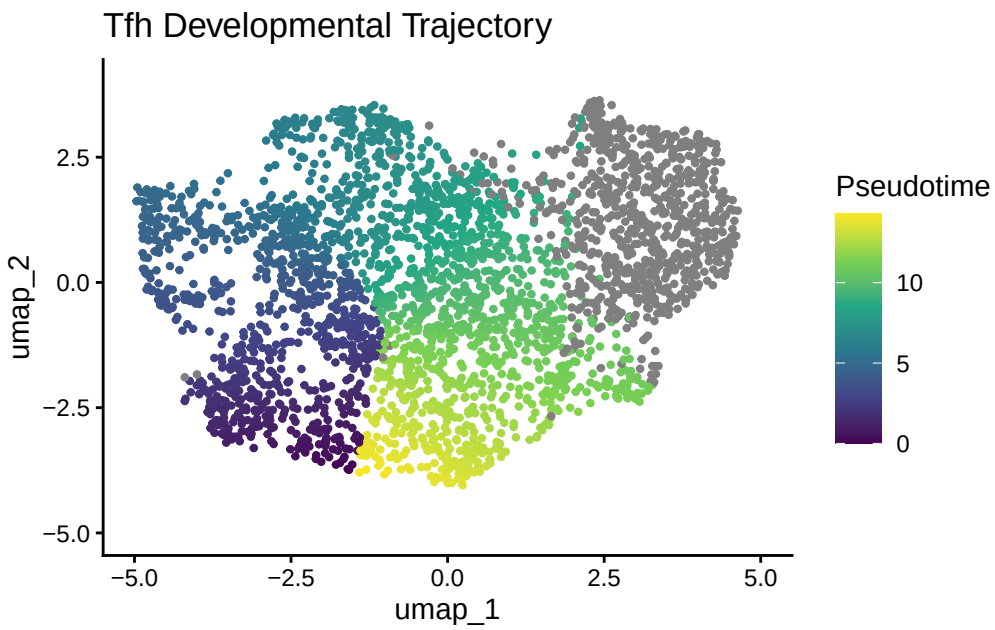
```
library(slingshot)
```

```
Loading required package: princurve
```

```
Loading required package: TrajectoryUtils
```

```
sce <- as.SingleCellExperiment(tfh)
sce <- slingshot(sce, clusterLabels = 'tfh.subset.annotation',
                 reducedDim = 'UMAP')
pt_matrix <- slingPseudotime(sce)
tfh$Pseudotime_Lin1 <- pt_matrix[, 1]
FeaturePlot(tfh, features = "Pseudotime_Lin1", pt.size = 0.7) +
  scale_color_viridis_c(option = "viridis", name = "Pseudotime") +
  labs(title = "Tfh Developmental Trajectory") +
  theme_classic()
```

```
Scale for colour is already present.
Adding another scale for colour, which will replace the existing scale.
```



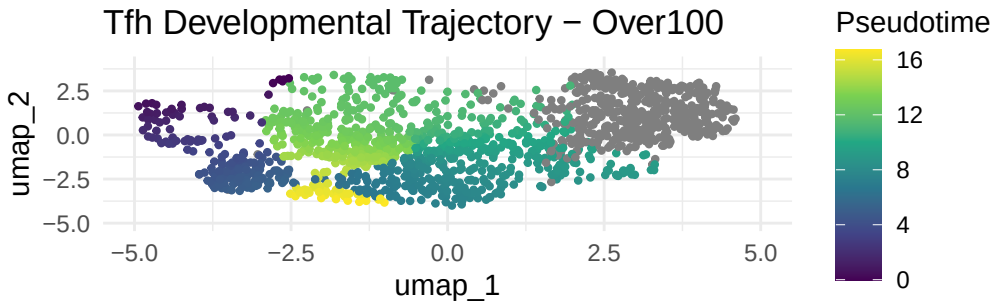

```
# ggsave(filename = 'Images/Tfh-UMAP-Slingshot-Pseudotime.pdf', height=4, width=5)

tfh.over100 <- subset(tfh, titer.categ=='Over100' )
tfh.under100 <- subset(tfh, titer.categ=='Under100' )
sce.over100 <- as.SingleCellExperiment(tfh.over100)
sce.under100 <- as.SingleCellExperiment(tfh.under100)
sce.over100 <- slingshot(sce.over100, clusterLabels = 'tfh.subset.annotation',
                         reducedDim = 'UMAP')
sce.under100 <- slingshot(sce.under100, clusterLabels = 'tfh.subset.annotation',
                         reducedDim = 'UMAP')
tfh.over100$Pseudotime_Lin1 <- slingPseudotime(sce.over100)[, 1]
tfh.under100$Pseudotime_Lin1 <- slingPseudotime(sce.under100)[, 1]
x<-FeaturePlot(tfh.over100, features = "Pseudotime_Lin1", pt.size = 0.7) +
  scale_color_viridis_c(option = "viridis", name = "Pseudotime") +
  labs(title = "Tfh Developmental Trajectory - Over100") + theme_minimal()
```

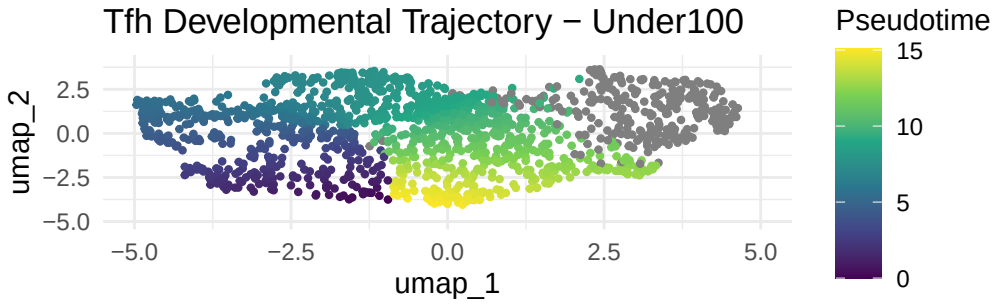
```
Scale for colour is already present.
Adding another scale for colour, which will replace the existing scale.
```

```
y<-FeaturePlot(tfh.under100, features = "Pseudotime_Lin1", pt.size = 0.7) +
  scale_color_viridis_c(option = "viridis", name = "Pseudotime") +
  labs(title = "Tfh Developmental Trajectory - Under100") + theme_minimal()
```

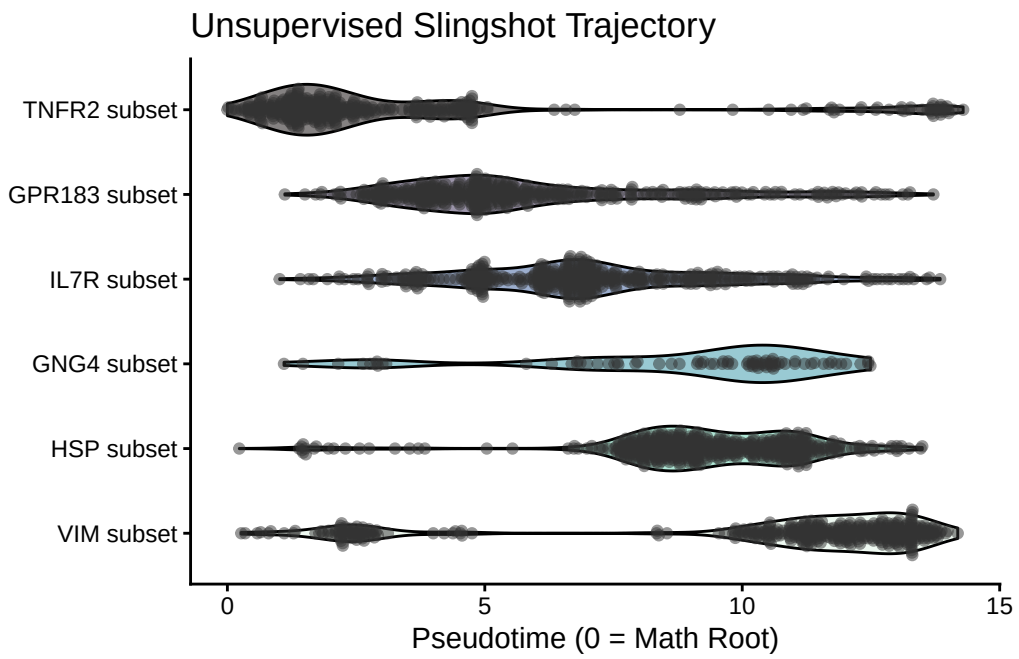
```
Scale for colour is already present.
Adding another scale for colour, which will replace the existing scale.
```

```
x+y
```

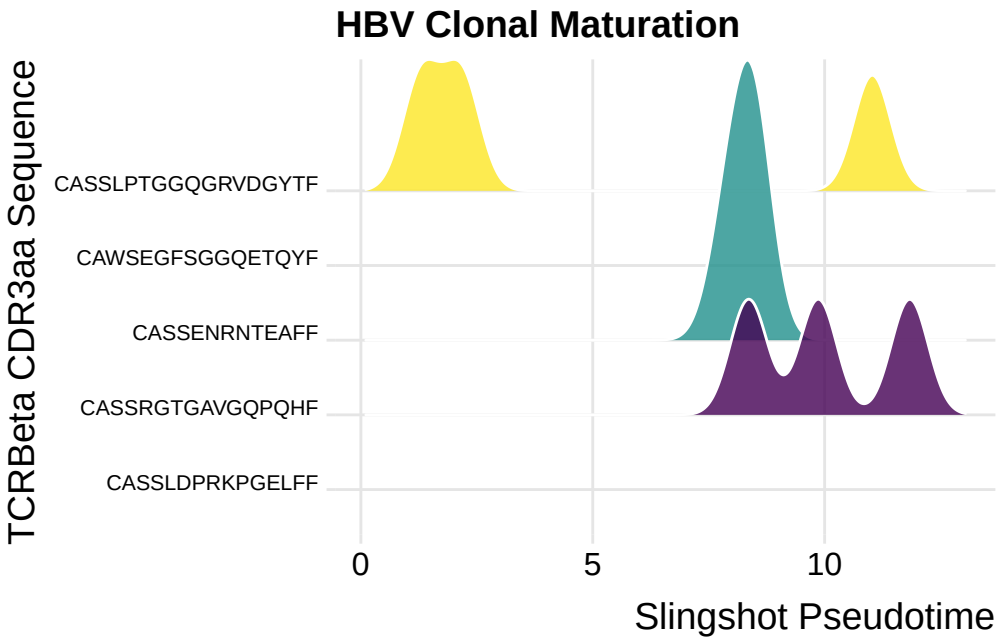





```r
rm(sce.under100, sce.over100, x, y)


md <- as.data.table(tfh@meta.data)
root_audit <- md[!is.na(Pseudotime_Lin1), .(
  Mean_PT = mean(Pseudotime_Lin1),
  Median_PT = median(Pseudotime_Lin1)
), by = tfh.subset.annotation][order(Mean_PT)]

invisible(
  md[, tfh.subset.annotation := factor(tfh.subset.annotation,
                                       levels = rev(root_audit$tfh.subset.annotation))]
)

ggplot(md[!is.na(Pseudotime_Lin1)], aes(x = Pseudotime_Lin1,
                                        y = tfh.subset.annotation,
                                        fill = tfh.subset.annotation)) +
  geom_violin(alpha = 0.5, color = "black", width = 0.6) +
  ggbeeswarm::geom_beeswarm(cex = 0.5, alpha = 0.5, color = "grey20", corral='wrap') +
  scale_fill_viridis_d(option = "mako", direction = -1) +
  theme_classic() +
  labs(title = "Unsupervised Slingshot Trajectory",
       x = "Pseudotime (0 = Math Root)",
       y = " ") +
  theme(legend.position = "none")
```




```
# ggsave(filename = 'Images/Tfh-Slingshot-Pseudotime-subsets.pdf', height=3, width=5)

invisible(md[, TCR_beta := sub(".*_", "", CTaa)])
top_clones <- md[!is.na(CTstrict), .N, by = .(CTstrict, TCR_beta)][order(-N)][1:10]  # filter out NAs so we are
plot_md <- md[CTstrict %in% top_clones$CTstrict & !is.na(Pseudotime_Lin1)]  # Keep top clones AND cells assigned
invisible(
  plot_md[, TCR_beta := factor(TCR_beta, levels = top_clones[order(N)]$TCR_beta)]  # This ensures the Y-axis fol
)
ggplot(plot_md, aes(x = Pseudotime_Lin1, y = TCR_beta, fill = TCR_beta)) +
  ggridges::geom_density_ridges(bandwidth = 0.4, scale = 2.5, alpha = 0.8,
                                color = "white") +
  scale_fill_viridis_d(option = "viridis") +
  ggridges::theme_ridges() +
  labs(title = "HBV Clonal Maturation", x = "Slingshot Pseudotime",
       y = "TCRBeta CDR3aa Sequence") +
  theme(legend.position = "none", axis.text.y = element_text(size = 8))
```




```
# rm(sce, root_audit, plot_md, top_clones, pt_matrix)

# saveRDS(tfh, "20260407_tfh_slingshot.RDS")
tfh <- readRDS("20260407_tfh_slingshot.RDS")
```

CytoTRACE2 calculation

```
library(CytoTRACE2)
tfh <- cytotrace2(tfh, is_seurat=T, slot_type = "counts",
                  species='human', batch_size = 10000)

res.cytotrace <- cytotrace2(tfh, is_seurat=T, slot_type = "counts",
                  species='human', parallelize_models = F)
plots <- plotData(cytotrace2_result = res.cytotrace,
                  annotation = res.cytotrace@meta.data, is_seurat = T)
plots$CytoTRACE2_Relative_UMAP
```

CytoTRACE2 analysis and plots

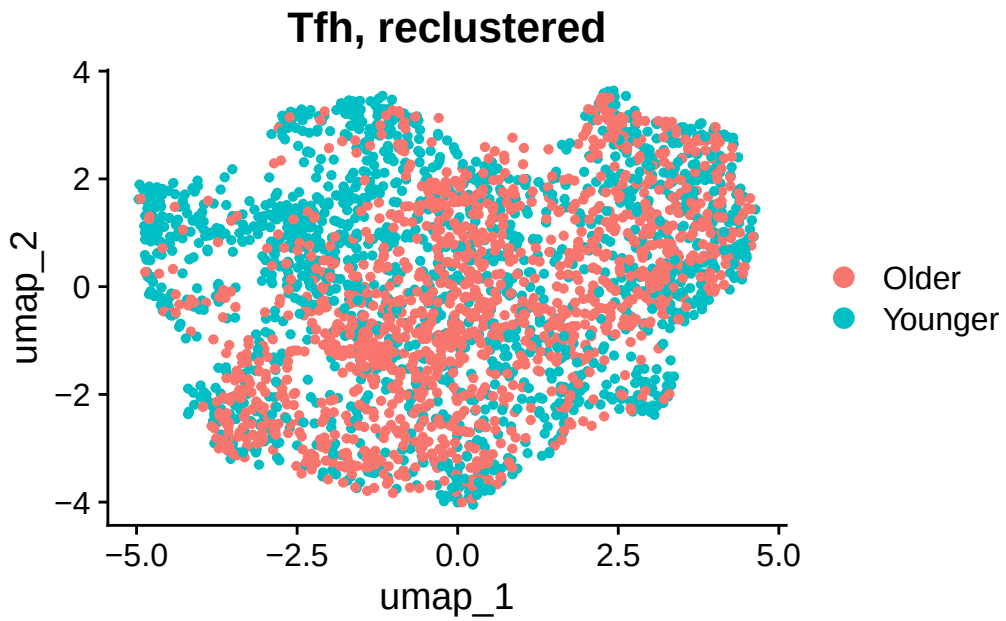
```
md <- as.data.table(res.cytotrace@meta.data)
summary_md <- md[, .(Mean_Score = mean(CytoTRACE2_Score, na.rm = TRUE),
  SEM = sd(CytoTRACE2_Score, na.rm = TRUE) / sqrt(.N)), by = tfh.subset.annotation]
ggplot(summary_md, aes(x = reorder(tfh.subset.annotation, -Mean_Score),
                       y = Mean_Score, fill = Mean_Score)) +
  geom_bar(stat = "identity", color = "black", width = 0.7) +
  geom_errorbar(aes(ymin = Mean_Score - SEM, ymax = Mean_Score + SEM), width = 0.2) +
  scale_fill_viridis_c(option = "magma", direction = -1) +
  theme_classic() +
  labs(title = "Average Developmental Potential by Subset",
       x = "Tfh Subset",
```

```
       y = "Mean CytoTRACE2 Score (High = Primitive)") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

ggplot(md, aes(x = nCount_RNA, y = nFeature_RNA, color = tfh.subset.annotation)) +
    geom_point(alpha = 0.3, size = 0.5) +
    geom_smooth(method = "lm", se = FALSE) +
    scale_x_log10() + scale_y_log10() +
    theme_minimal() +
    labs(title = "Transcriptional Diversity vs. Sequencing Depth",
         x = "Total RNA Counts (log10)",
         y = "Unique Genes Detected (log10)")

ggplot(md[!is.na(Pseudotime_Lin1)], aes(x = Pseudotime_Lin1, y = nFeature_RNA, color = tfh.subset.annotation)) +
  geom_point(alpha = 0.4, size = 1) +
  # Using a GAM (Generalized Additive Model) to see the non-linear trend
  geom_smooth(method = "gam", color = "black", linewidth = 1.2) +
  scale_color_brewer(palette = "Set1") +
  theme_classic() +
  labs(title = "Trajectory Audit: Pseudotime vs. Transcriptional Diversity",
       subtitle = "Is the mathematical timeline driven by gene counts?",
       x = "Slingshot Pseudotime (Maturation)",
       y = "nFeature_RNA (Unique Genes Detected)") +
  theme(legend.title = element_blank())



rm(summary_md, res.cytotrace, plots)
```

From this, it seems like Cytotrace is unable to find enough meaningful differences to properly call differentiation, as the result will depend on how transcriptionally diverse the subset is. The presumption is that differentiation -> less transcriptional diversity but whether that's true for Tfh is unknown. Likely these subsets are relatively similar. The plots for # features detected vs slingshot pseudotime is interesting and worth coming back to later.

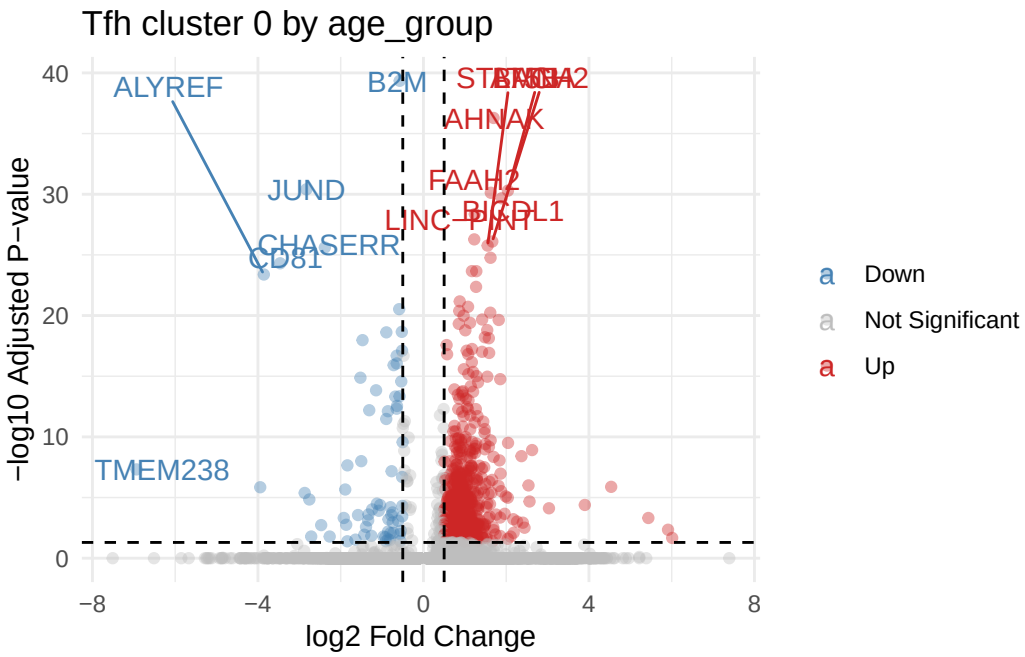
Differential expression by aging

```
DimPlot(tfh, group.by = "age_group",label = F, pt.size = 1) + ggtitle("Tfh, reclustered")
```




```
cluster0 <- subset(tfh, idents = "GNG4 subset")
Idents(cluster0) <- "age_group"
deg <- FindMarkers(cluster0, ident.1 = "Older", ident.2 = "Younger", test.use = "wilcox")
deg$gene_label <- rownames(deg)
deg$diffexpressed <- "Not Significant"
deg$diffexpressed[deg$avg_log2FC > 0.5 & deg$p_val_adj < 0.05] <- "Up"
deg$diffexpressed[deg$avg_log2FC < -0.5 & deg$p_val_adj < 0.05] <- "Down"
top_genes <- rownames(deg[deg$diffexpressed != "Not Significant", ])[1:90]
deg$gene_label[rownames(deg) %in% top_genes] <- rownames(deg)[rownames(deg) %in% top_genes]
ggplot(data = deg, aes(x = avg_log2FC, y = -log10(p_val_adj), col = diffexpressed, label = gene_label)) +
  geom_point(alpha = 0.4, size = 1.5) + theme_minimal() +
  ggrepel::geom_text_repel() +
  scale_color_manual(values = c("steelblue", "grey", "firebrick3")) +
  geom_vline(xintercept = c(-0.5, 0.5), col = "black", linetype = "dashed") +
  geom_hline(yintercept = -log10(0.05), col = "black", linetype = "dashed") +
  labs(title = "Tfh cluster 0 by age_group", x = "log2 Fold Change",y = "-log10 Adjusted P-value") +
  theme(legend.title = element_blank())
```

```
Warning: ggrepel: 11412 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```

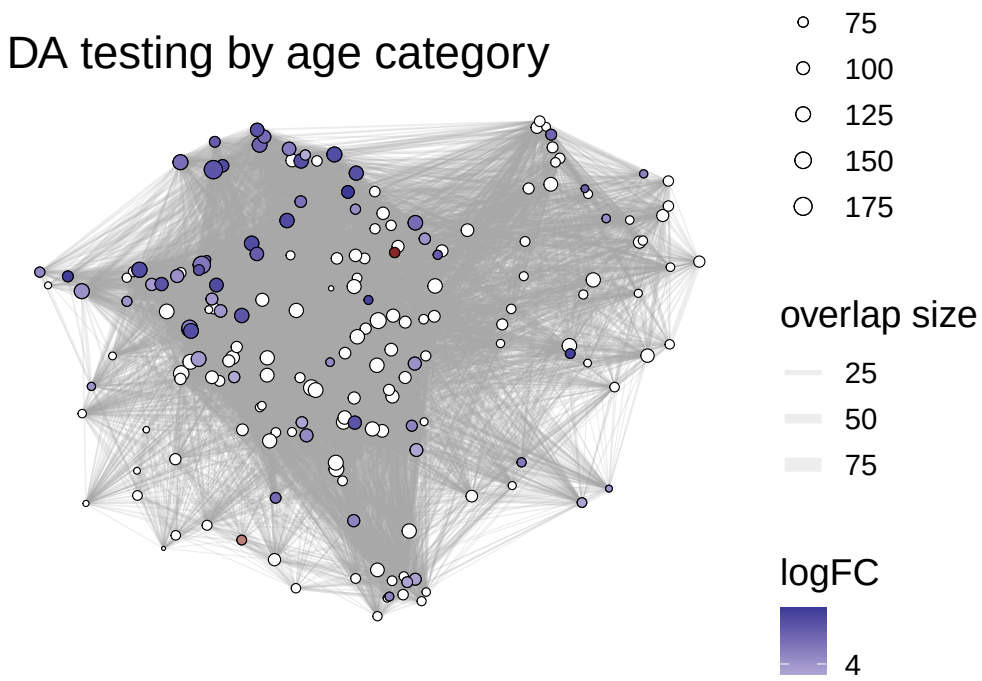



```
# milo_obj <- readRDS(file="20260321_miloR_tfh.RDS")

milo_obj.des.age <- data.frame(colData(milo_obj))[,c("subject", "age_group")]
milo_obj.des.age <- distinct(milo_obj.des.age)
rownames(milo_obj.des.age) <- milo_obj.des.age$subject
milo_obj.des.age <- milo_obj.des.age[colnames(nhoodCounts(milo_obj)), , drop=FALSE] ## Reorder rownames to match
da_results <- testNhoods(milo_obj, design = ~ age_group, design.df = milo_obj.des.age)
```

```
Using TMM normalisation

Running with model contrasts

Performing spatial FDR correction with k-distance weighting
```

```
milo_obj <- buildNhoodGraph(milo_obj)
plotNhoodGraphDA(milo_obj, da_results, alpha=0.05) +
  patchwork::plot_layout(guides="collect") + ggtitle("DA testing by age category")
```

```
Adding nhood effect sizes to neighbourhood graph attributes
```

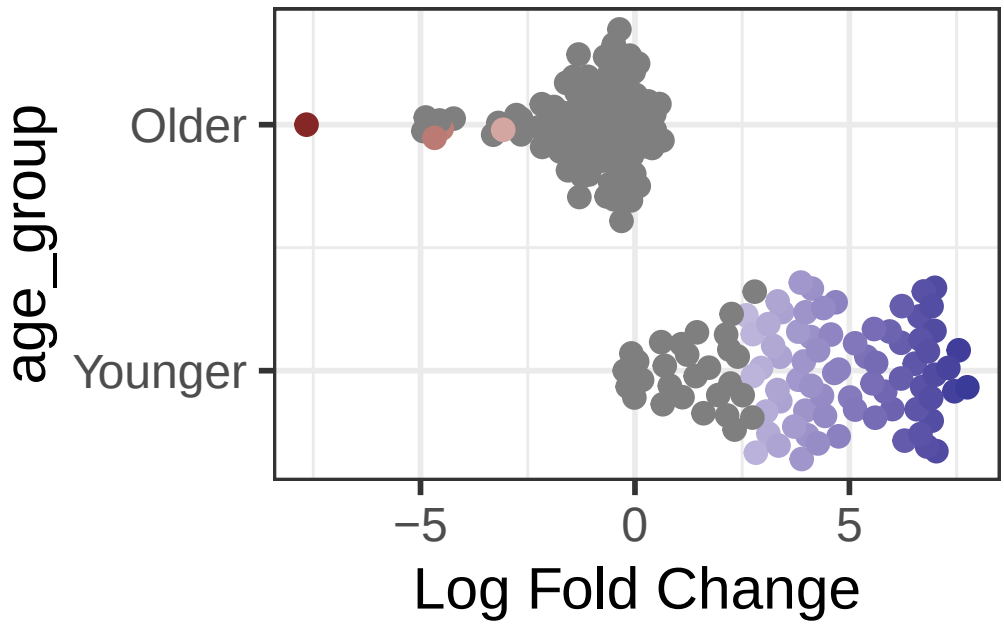



```
da_results <- annotateNhoods(milo_obj, da_results, coldata_col = "age_group")
```

```
Converting age_group to factor...
```

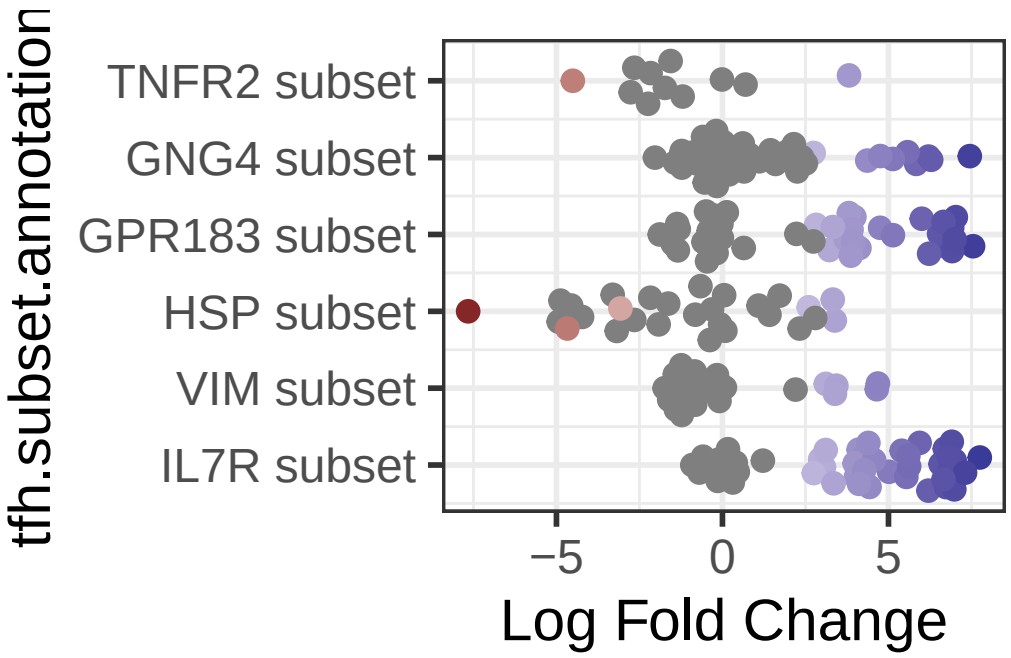
```
da_results <- annotateNhoods(milo_obj, da_results, coldata_col = "tfh.subset.annotation")
plotDAbeeswarm(da_results, group.by='age_group', alpha=0.1)
```

```
Converting group_by to factor...
```



```
plotDAbeeswarm(da_results, group.by='tfh.subset.annotation', alpha=0.1)
```

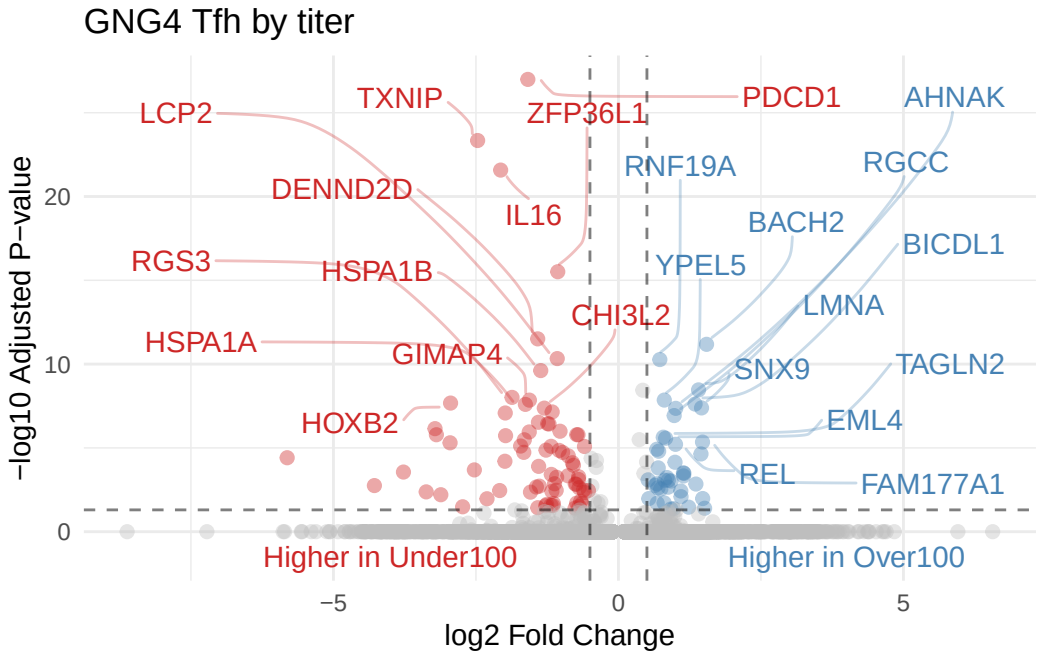
```
Converting group_by to factor...
```



Differential expression by titer

```
cluster.sub <- subset(tfh, idents = "GNG4 subset")
Idents(cluster.sub) <- "titer.categ"
deg <- FindMarkers(cluster.sub, ident.1 = "Over100", ident.2 = "Under100", test.use = "wilcox")
deg$gene_label <- rownames(deg)
deg$diffexpressed <- "Not Significant"
deg$diffexpressed[deg$avg_log2FC > 0.5 & deg$p_val_adj < 0.05] <- "Higher in Over100"
deg$diffexpressed[deg$avg_log2FC < -0.5 & deg$p_val_adj < 0.05] <- "Higher in Under100"
# top_genes <- rownames(deg[deg$diffexpressed != "Not Significant", ])[1:90]
# deg$gene_label[rownames(deg) %in% top_genes] <- rownames(deg)[rownames(deg) %in% top_genes]
setDT(deg)
top_up <- deg[diffexpressed == "Higher in Over100"][order(p_val_adj)][1:12, gene_label]
top_down <- deg[diffexpressed == "Higher in Under100"][order(p_val_adj)][1:12, gene_label]
deg$gene_label <- ifelse(deg$gene %in% c(top_up, top_down), deg$gene_label, "")

a <- ggplot(data = deg, aes(x = avg_log2FC, y = -log10(p_val_adj), col = diffexpressed,
                      label = gene_label)) +
  geom_point(alpha = 0.4, size = 2) + theme_minimal() +
  # ggrepel::geom_text_repel(max.overlaps = 20, box.padding = 0.5) +
  ggrepel::geom_text_repel(
    max.overlaps = Inf,       # Ensure our 30 genes show up
    force = 12,                # Strength of push between labels
    box.padding = 0.8,        # Space around the text
    point.padding = 0.5,      # Space between the point and the start of the line
    segment.curvature = -0.1, # Adds a subtle curve to the lines
    segment.ncp = 3,          # Smoother curve
```

```
    segment.angle = 50,      # Angle of the callout
    segment.alpha = 0.3
  ) +
  scale_color_manual(values = c("Higher in Over100" = "steelblue",
                                "Not Significant" = "grey",
                                "Higher in Under100" = "firebrick3")) +
  geom_vline(xintercept = c(-0.5, 0.5), col = "black", linetype = "dashed", alpha=0.5) +
  geom_hline(yintercept = -log10(0.05), col = "black", linetype = "dashed", alpha=0.5) +
  labs(title="GNG4 Tfh by titer", x="log2 Fold Change", y="-log10 Adjusted P-value") +
  coord_cartesian(clip = "off") +
  annotate("text", x = 4, y = -1.5, label = "Higher in Over100", color = "steelblue") +
  annotate("text", x = -4, y = -1.5, label = "Higher in Under100", color = "firebrick3") +
  theme(legend.position = 'none'); a
```



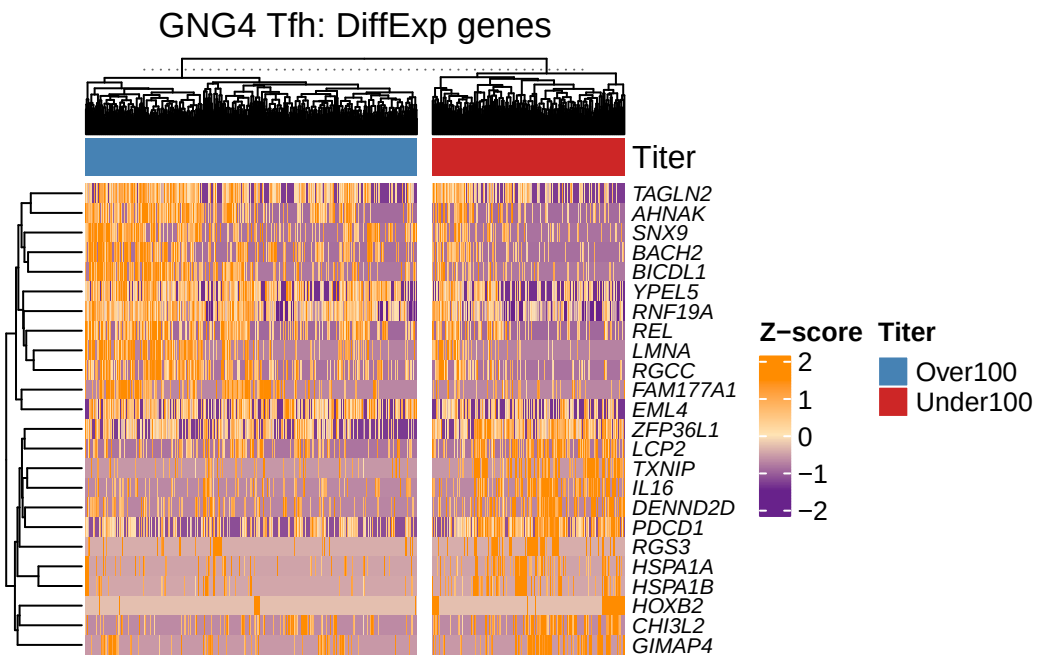

```
# ggsave(a, filename = 'Images/Tfh-Volcano-GNG4-Hi-v-Lo.pdf', height=5, width=6)


# genes_to_plot <- c("PDCD1", 'TXNIP', 'IL16', 'ZFP36L1', 'DENND2D',
#                    "BACH2", "RNF19A", 'LMNA', 'YPEL5', 'BICDL1',
#                    "CD40LG", "JUN", 'IL21', 'CDK14', 'REL',
#                    "IL4", "ZEB1", 'JUNB', 'DUSP6', 'CD44'
#                   )
genes_under <- deg[diffexpressed == "Higher in Under100" & gene_label != "", gene_label]
genes_over  <- deg[diffexpressed == "Higher in Over100" & gene_label != "", gene_label]
genes_to_plot <- c(genes_under[1:15], genes_over[1:15])
genes_to_plot <- intersect(na.omit(genes_to_plot), rownames(cluster.sub))

# genes_to_plot <- deg[which(deg$diffexpressed == "Higher in Under100"), 'gene_label'][1:15]
# genes_to_plot <- c(genes_to_plot, deg[which(deg$diffexpressed == "Higher in Over100"), 'gene_label'][1:15])

cluster.sub <- ScaleData(cluster.sub, features = genes_to_plot)
```

```
Centering and scaling data matrix
```

```
mat <- as.matrix(GetAssayData(cluster.sub, slot = "scale.data")[genes_to_plot, ])
```

```
Warning: The `slot` argument of `GetAssayData()` is deprecated as of SeuratObject 5.0.0.
i Please use the `layer` argument instead.
```

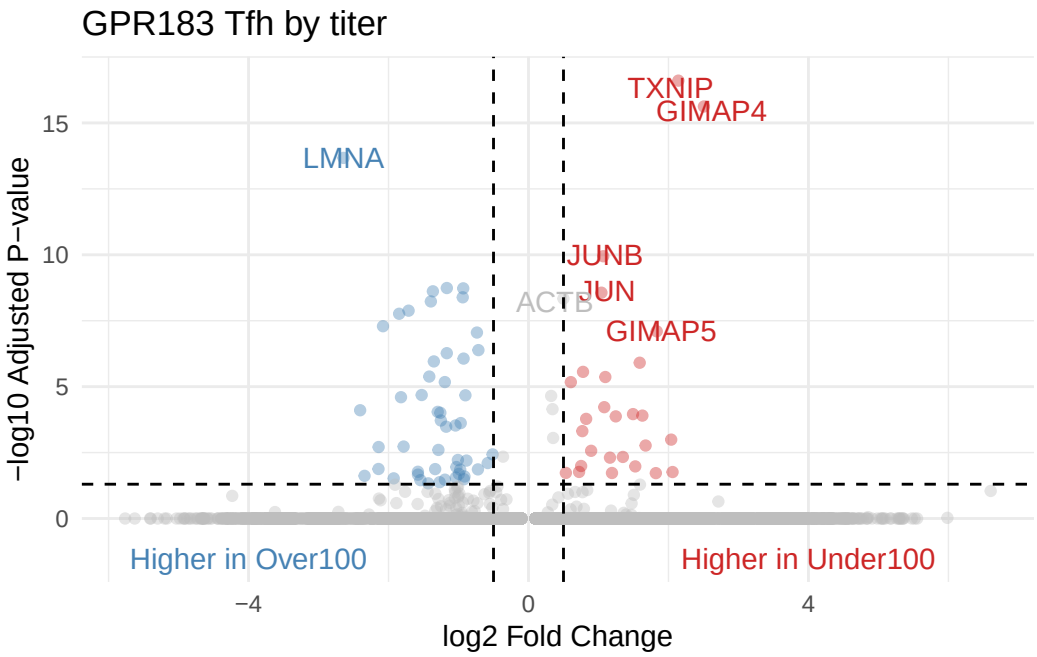
```
col_annot <- ComplexHeatmap::HeatmapAnnotation(
  Titer = cluster.sub$titer.categ,
  col = list(Titer = c("Under100" = "firebrick3", "Over100" = "steelblue")),
  show_legend = TRUE )

# pdf(file = 'Images/GNG4-diffexp-Hi-v-Lo-Ab.pdf', height=4.5,width=5)
ComplexHeatmap::Heatmap(mat,
  name = "Z-score",
  top_annotation = col_annot,
  column_split = cluster.sub$titer.categ, # Groups cells by titer
  cluster_column_slices = TRUE,           # Clusters cells WITHIN the groups
  cluster_columns = TRUE,                # Enables the internal clustering
  column_gap = unit(2, "mm"),             # Adds a visual gap between blocks
  cluster_rows = TRUE,                    # Keep gene clustering
  show_row_dend = TRUE,
  col = colorRamp2(c(-1.5, 0, 1.5), c("darkorchid4", "moccasin", "darkorange")),
  row_names_gp = grid::gpar(fontsize = 8, fontface = "italic"),
  show_column_names = FALSE,
  column_title = "GNG4 Tfh: DiffExp genes",
  row_names_side = "right",
  row_dend_width = unit(10, "mm")

)
```




```
# dev.off()


cluster.sub <- subset(tfh, idents = "GPR183 subset")
```

```r
Idents(cluster.sub) <- "titer.categ"
deg <- FindMarkers(cluster.sub, ident.1 = "Under100", ident.2 = "Over100", test.use = "wilcox")
deg$gene_label <- rownames(deg)
deg$diffexpressed <- "Not Significant"
deg$diffexpressed[deg$avg_log2FC > 0.5 & deg$p_val_adj < 0.05] <- "Higher in Under100"
deg$diffexpressed[deg$avg_log2FC < -0.5 & deg$p_val_adj < 0.05] <- "Higher in Over100"
top_genes <- rownames(deg[deg$diffexpressed != "Not Significant", ])[1:90]
deg$gene_label[rownames(deg) %in% top_genes] <- rownames(deg)[rownames(deg) %in% top_genes]
ggplot(data = deg, aes(x = avg_log2FC, y = -log10(p_val_adj), col = diffexpressed, label = gene_label)) +
  geom_point(alpha = 0.4, size = 1.5) + theme_minimal() +
  ggrepel::geom_text_repel(max.overlaps = 12, box.padding = 0.5) +
  scale_color_manual(values = c("Higher in Over100" = "steelblue",
                                "Not Significant" = "grey",
                                "Higher in Under100" = "firebrick3")) +
  geom_vline(xintercept = c(-0.5, 0.5), col = "black", linetype = "dashed") +
  geom_hline(yintercept = -log10(0.05), col = "black", linetype = "dashed") +
  labs(title = "GPR183 Tfh by titer", x = "log2 Fold Change",
       y = "-log10 Adjusted P-value") +
  coord_cartesian(clip = "off") +
  annotate("text", x = -4, y = -1.5,
           label = "Higher in Over100", color = "steelblue") +
  annotate("text", x = 4, y = -1.5,
           label = "Higher in Under100", color = "firebrick3") +
  theme(legend.position = 'none')
```

```
Warning: ggrepel: 11057 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



Tfh subset pathway analysis

```
# devtools::install_version("crossmatch", version = "1.3.1", repos = "http://cran.us.r-project.org")
# devtools::install_version("multicross", version = "2.1.0", repos = "http://cran.us.r-project.org")
# devtools::install_github("jackbibby1/SCPA")

gng4.cells <- subset(tfh, subset = tfh.subset.annotation == "GNG4 subset")
over100_mat <- as.matrix( GetAssayData(gng4.cells, assay = "SCT", slot = "data")[, gng4.cells$titer.categ == "Ov
under100_mat <- as.matrix( GetAssayData(gng4.cells, assay = "SCT", slot = "data")[, gng4.cells$titer.categ == "U
pathways <- msigdbr::msigdbr(species = "Homo sapiens", category = "H") %>%
  SCPA::format_pathways()
```

```
Warning: The `category` argument of `msigdbr()` is deprecated as of msigdbr 10.0.0.
i Please use the `collection` argument instead.
```

```
scpa_results <- SCPA::compare_pathways( samples = list(over100_mat, under100_mat),
                                  pathways = pathways)
```

```
Using single core processing. Specify 'parallel = TRUE' and `cores = x` arguments for parallel processing

Cell numbers in population 1 = 507

Cell numbers in population 2 = 294

- If greater than 500 cells, these populations will be downsampled

All 50 pathways passed the min/max genes threshold

Calculating pathway fold changes...

Performing a two-sample analysis with SCPA...


  |
  |                                                      |   0%
  |
  |=                                                     |   2%
  |
  |==                                                    |   4%
  |
  |===                                                   |   6%
  |
  |====                                                  |   8%
  |
  |=====                                                 |  10%
  |
  |======                                                |  12%
  |
  |=======                                               |  14%
  |
  |========                                              |  16%
  |
  |=========                                             |  18%
  |
```

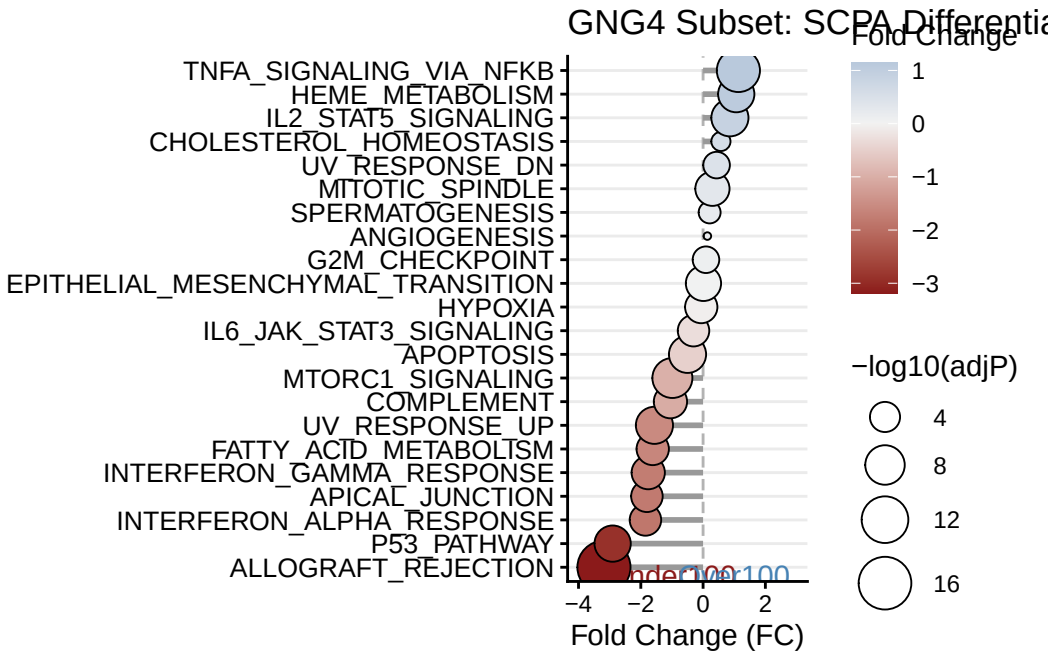
```
|==========                                        |  20%
|
|===========                                       |  22%
|
|============                                      |  24%
|
|=============                                     |  26%
|
|==============                                    |  28%
|
|===============                                   |  30%
|
|================                                  |  32%
|
|=================                                 |  34%
|
|==================                                |  36%
|
|===================                               |  38%
|
|====================                              |  40%
|
|=====================                             |  42%
|
|======================                            |  44%
|
|=======================                           |  46%
|
|========================                          |  48%
|
|=========================                         |  50%
|
|==========================                        |  52%
|
|===========================                       |  54%
|
|============================                      |  56%
|
|=============================                     |  58%
|
|==============================                    |  60%
|
|===============================                   |  62%
|
|================================                  |  64%
|
|=================================                 |  66%
|
|==================================                |  68%
|
|===================================               |  70%
|
|====================================              |  72%
|
|=====================================             |  74%
|
```

```
  |=======================================             |  76%
  |
  |========================================            |  78%
  |
  |=========================================           |  80%
  |
  |==========================================          |  82%
  |
  |===========================================         |  84%
  |
  |============================================        |  86%
  |
  |=============================================       |  88%
  |
  |==============================================      |  90%
  |
  |===============================================     |  92%
  |
  |================================================    |  94%
  |
  |=================================================   |  96%
  |
  |================================================== |  98%
  |
  |==================================================| 100%
```

```r
scpa_results <- scpa_results %>%  mutate(adjPvalBH = p.adjust(Pval, method = "BH"),
                                         logP = -log10(adjPvalBH)) %>%
  arrange(desc(qval))

setDT(scpa_results)
scpa_results[, direction := ifelse(FC > 0, "Over100", "Under100")]
scpa_results[, pathway := gsub("HALLMARK_", "", Pathway)]
plot_data <- scpa_results[order(-qval), head(.SD, 12), by = direction]
plot_data <- plot_data[order(FC)]
plot_data[, pathway := factor(pathway, levels = pathway)]

ggplot(plot_data, aes(x = FC, y = pathway)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey70") +
  geom_segment(aes(x = 0, xend = FC, y = pathway, yend = pathway),
               color = "grey60", linewidth = 1) +
  geom_point(aes(size = logP, fill = FC), shape = 21, color = "black") +
  scale_fill_gradient2(low = "firebrick4", mid = "grey95", high = "steelblue",
                       midpoint = 0, name = "Fold Change") +
  scale_size_continuous(range = c(1, 9), name = "-log10(adjP)") +
  theme_classic() +
  labs(title = "GNG4 Subset: SCPA Differential Pathways",
       x = "Fold Change (FC)",  y = NULL) +
  annotate("text", x = -1, y = 0.65, label = "Under100", color = "firebrick4") +
  annotate("text", x = 1, y = 0.65, label = "Over100", color = "steelblue") +
  coord_cartesian(clip = "off") +
  theme(axis.text.y = element_text(size = 10, color = "black"),
    panel.grid.major.y = element_line(color = "grey92", linewidth = 0.5) ) +
  coord_cartesian(xlim = c(-4,3))
```

```
Coordinate system already present.
```

```
i Adding new coordinate system, which will replace the existing one.
```

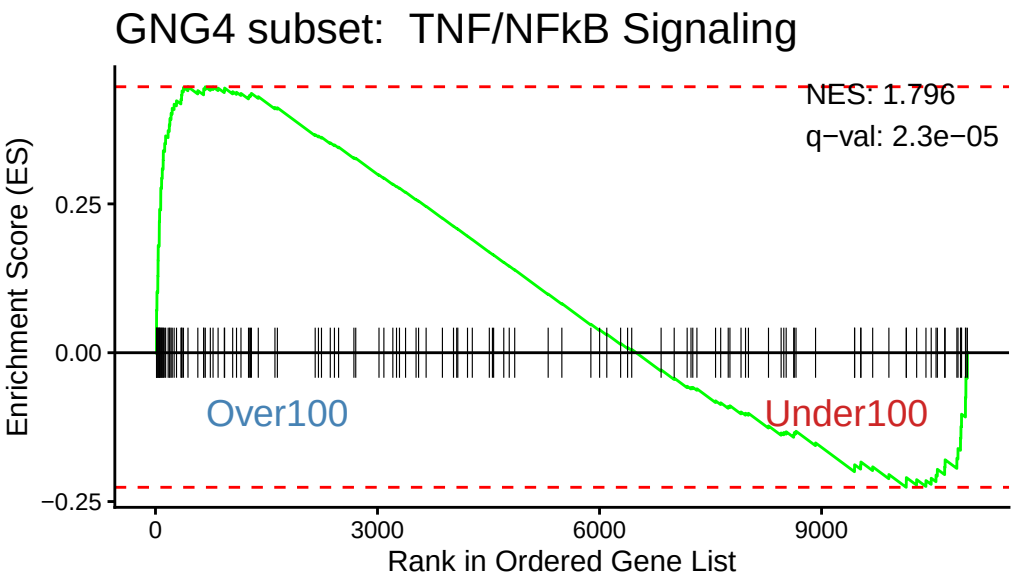



```
ggsave(filename = "Images/GNG4-SCPA-Lollipop.pdf", height=6, width=7)


Idents(gng4.cells) <- "titer.categ"
deg <- FindMarkers(gng4.cells, ident.1 = "Over100", ident.2 = "Under100",
                   test.use = "wilcox")
deg$gene <- rownames(deg)
setDT(deg)
deg[, rank_metric := sign(avg_log2FC) * -log10(p_val)]
ranks <- deg$rank_metric
names(ranks) <- deg$gene
ranks <- sort(ranks, decreasing = TRUE)
m_df <- msigdbr::msigdbr(species = "Homo sapiens", category = "H")
pathways_list <- split(m_df$gene_symbol, m_df$gs_name)
fgsea_res <- fgsea::fgsea(pathways = pathways_list["HALLMARK_TNFA_SIGNALING_VIA_NFKB"],
                          stats = ranks, minSize = 15, maxSize = 500)
```

```
Warning in preparePathwaysAndStats(pathways, stats, minSize, maxSize, gseaParam, : There are ties in the prerank
The order of those tied genes will be arbitrary, which may produce unexpected results.
```

```
stats_text <- paste0("NES: ", round(fgsea_res$NES, 3), "\n",
                     "q-val: ", format.pval(fgsea_res$padj, digits = 2))

fgsea::plotEnrichment(pathways_list[["HALLMARK_TNFA_SIGNALING_VIA_NFKB"]],
                                ranks) +
  labs(title = "GNG4 subset:  TNF/NFkB Signaling",
       x = "Rank in Ordered Gene List",
       y = "Enrichment Score (ES)") +
  theme_classic() +
  annotate("text", x = length(ranks) * 0.8, y = 0.4, label = stats_text, hjust=0) +
  annotate("text", x = length(ranks)*0.15, y = -0.1, label = "Over100",
```

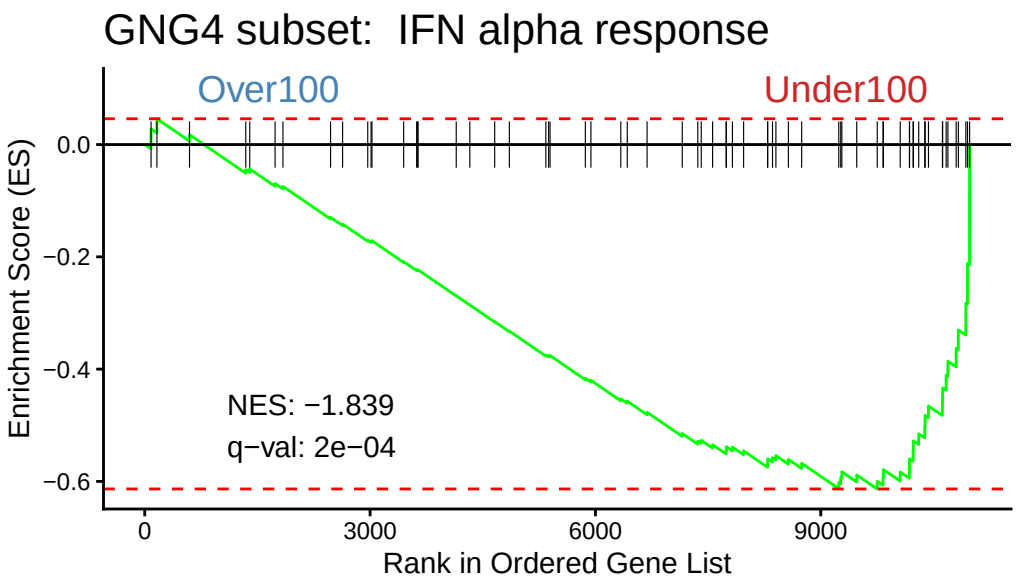
```
          color = "steelblue", size = 5) +
  annotate("text", x = length(ranks)*0.85, y = -0.1,
           label = "Under100", color = "firebrick3",  size = 5) +
  coord_cartesian(clip = "off") +
  theme( plot.title = element_text(size = 16),
    plot.margin = margin(t = 10, r = 10, b = 30, l = 10) )
```




```
# ggsave(filename = "Images/GNG4-GSEA-TNFNFkBsignaling.pdf", width=5, height=4)

fgsea_res <- fgsea::fgsea(pathways = pathways_list["HALLMARK_INTERFERON_ALPHA_RESPONSE"],
                          stats = ranks, minSize = 15, maxSize = 500)
```

```
Warning in preparePathwaysAndStats(pathways, stats, minSize, maxSize, gseaParam, : There are ties in the prerank
The order of those tied genes will be arbitrary, which may produce unexpected results.
```

```
stats_text <- paste0("NES: ", round(fgsea_res$NES, 3), "\n",
                     "q-val: ", format.pval(fgsea_res$padj, digits = 2))
fgsea::plotEnrichment(pathways_list[["HALLMARK_INTERFERON_ALPHA_RESPONSE"]],
                              ranks) +
  labs(title = "GNG4 subset:  IFN alpha response",
       x = "Rank in Ordered Gene List",
       y = "Enrichment Score (ES)") +
  theme_classic() +
  annotate("text", x = length(ranks) * 0.1, y = -0.5, label = stats_text, hjust=0) +
  annotate("text", x = length(ranks)*0.15, y = 0.1,
           label = "Over100", color = "steelblue", size = 5) +
  annotate("text", x = length(ranks)*0.85, y = 0.1,
           label = "Under100", color = "firebrick3",  size = 5) +
  coord_cartesian(clip = "off") +
  theme( plot.title = element_text(size = 16),
    plot.margin = margin(t = 10, r = 10, b = 30, l = 10) )
```




```
# ggsave(filename = "Images/GNG4-GSEA-IFNaResponse.pdf", width=5, height=4)

rm(cluster.sub, deg, top_genes, mat, col_annot, m_df, pathways_list, p_tnf)
```

```
Warning in rm(cluster.sub, deg, top_genes, mat, col_annot, m_df, pathways_list,
: object 'p_tnf' not found
```

```
tnfr2 <- subset(tfh, subset = tfh.subset.annotation == "TNFR2 subset")
over100_mat <- as.matrix( GetAssayData(tnfr2, assay = "SCT", slot = "data")[, tnfr2$titer.categ == "Over100"] )
under100_mat <- as.matrix( GetAssayData(tnfr2, assay = "SCT", slot = "data")[, tnfr2$titer.categ == "Under100"]
pathways <- msigdbr::msigdbr(species = "Homo sapiens", category = "H") %>%
  SCPA::format_pathways()
scpa_results <- SCPA::compare_pathways( samples = list(over100_mat, under100_mat),
                                    pathways = pathways)
```

```
Using single core processing. Specify 'parallel = TRUE' and `cores = x` arguments for parallel processing

Cell numbers in population 1 = 223

Cell numbers in population 2 = 144

- If greater than 500 cells, these populations will be downsampled

All 50 pathways passed the min/max genes threshold

Calculating pathway fold changes...

Performing a two-sample analysis with SCPA...
```

```
  |
  |                                                      |   0%
  |
  |=                                                     |   2%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==                                                    |   4%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===                                                   |   6%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |====                                                  |   8%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=====                                                 |  10%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |======                                                |  12%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=======                                               |  14%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value
```

```
  |
  |========                                               |  16%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=========                                              |  18%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==========                                             |  20%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===========                                            |  22%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |============                                           |  24%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=============                                          |  26%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==============                                         |  28%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===============                                        |  30%
```

```
Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================                                 |  32%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=================                                |  34%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==================                               |  36%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===================                              |  38%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |====================                             |  40%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=====================                            |  42%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |======================                           |  44%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value
```

```
  |
  |======================                          |  46%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=======================                         |  48%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |========================                        |  50%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=========================                       |  52%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==========================                      |  54%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===========================                     |  56%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |============================                    |  58%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=============================                   |  60%
```

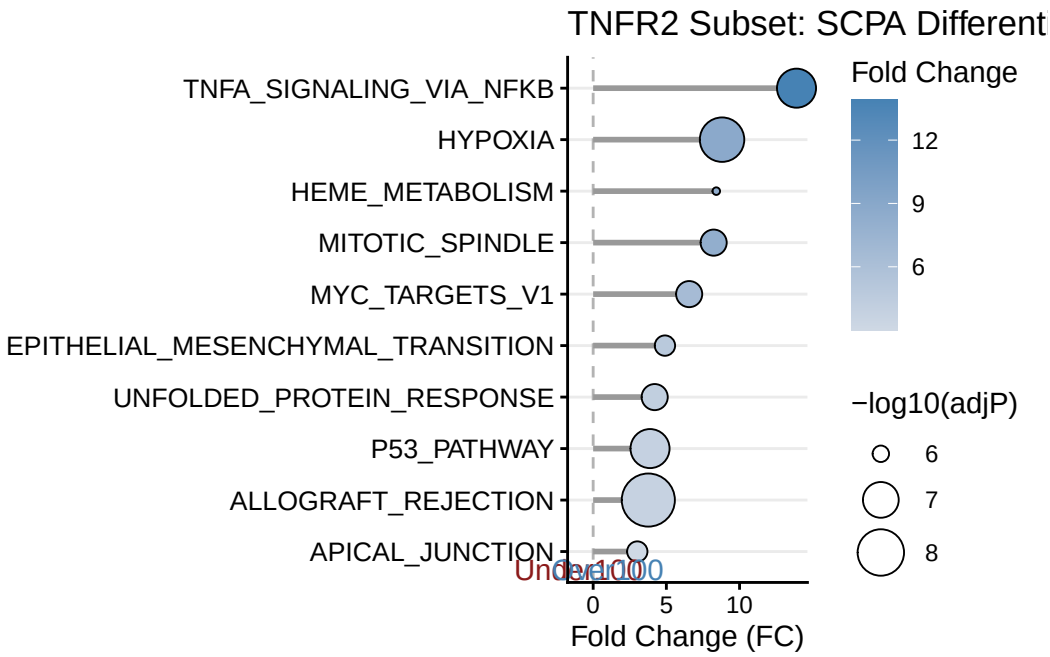
```
Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================                  |  62%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================                  |  64%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================                  |  66%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================                  |  68%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================                  |  70%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================                  |  72%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================                  |  74%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value
```

```
  |
  |=====================================            |  76%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |======================================           |  78%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=======================================          |  80%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |========================================         |  82%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=========================================        |  84%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==========================================       |  86%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===========================================      |  88%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |============================================     |  90%
```

```
Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=============================================    |  92%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==============================================   |  94%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=============================================== |  96%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================================ |  98%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=================================================| 100%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value
```

```r
scpa_results <- scpa_results %>%  mutate(adjPvalBH = p.adjust(Pval, method = "BH"),
                                         logP = -log10(adjPvalBH)) %>%
  arrange(desc(qval))

setDT(scpa_results)
scpa_results[, direction := ifelse(FC > 0, "Over100", "Under100")]
scpa_results[, pathway := gsub("HALLMARK_", "", Pathway)]
plot_data <- scpa_results[order(-qval), head(.SD, 10), by = direction]
plot_data <- plot_data[order(FC)]
plot_data[, pathway := factor(pathway, levels = pathway)]

ggplot(plot_data, aes(x = FC, y = pathway)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey70") +
  geom_segment(aes(x = 0, xend = FC, y = pathway, yend = pathway),
               color = "grey60", linewidth = 1) +
```

```
  geom_point(aes(size = logP, fill = FC), shape = 21, color = "black") +
  scale_fill_gradient2(low = "firebrick4", mid = "grey95", high = "steelblue",
                       midpoint = 0, name = "Fold Change") +
  scale_size_continuous(range = c(1, 9), name = "-log10(adjP)") +
  theme_classic() +
  labs(title = "TNFR2 Subset: SCPA Differential Pathways",
       x = "Fold Change (FC)",  y = NULL) +
  annotate("text", x = -1, y = 0.65, label = "Under100", color = "firebrick4") +
  annotate("text", x = 1, y = 0.65, label = "Over100", color = "steelblue") +
  coord_cartesian(clip = "off") +
  theme(axis.text.y = element_text(size = 10, color = "black"),
    panel.grid.major.y = element_line(color = "grey92", linewidth = 0.5) )
```




```
# ggsave(filename = "Images/TNFR2-SCPA-Lollipop.pdf", height=6, width=7)
```

```
gpr183 <- subset(tfh, subset = tfh.subset.annotation == "GPR183 subset")
over100_mat <- as.matrix( GetAssayData(gpr183, assay = "SCT", slot = "data")[, gpr183$titer.categ == "Over100"]
under100_mat <- as.matrix( GetAssayData(gpr183, assay = "SCT", slot = "data")[, gpr183$titer.categ == "Under100"
pathways <- msigdbr::msigdbr(species = "Homo sapiens", category = "H") %>%
  SCPA::format_pathways()
scpa_results <- SCPA::compare_pathways( samples = list(over100_mat, under100_mat),
                                  pathways = pathways)
```

```
Using single core processing. Specify 'parallel = TRUE' and `cores = x` arguments for parallel processing

Cell numbers in population 1 = 202

Cell numbers in population 2 = 249

- If greater than 500 cells, these populations will be downsampled
```

```
All 50 pathways passed the min/max genes threshold

Calculating pathway fold changes...

Performing a two-sample analysis with SCPA...


  |
  |                                                      |   0%
  |
  |=                                                     |   2%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==                                                    |   4%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===                                                   |   6%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |====                                                  |   8%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=====                                                 |  10%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |======                                                |  12%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=======                                               |  14%
```

```
Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |========                                              |  16%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=========                                             |  18%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==========                                            |  20%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===========                                           |  22%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |============                                          |  24%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=============                                         |  26%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==============                                        |  28%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value
```

```
  |
  |===============                                  |  30%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================                                 |  32%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=================                                |  34%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==================                               |  36%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===================                              |  38%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |====================                             |  40%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=====================                            |  42%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |======================                           |  44%
```

```
Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |======================                          |  46%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=======================                         |  48%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |========================                        |  50%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=========================                       |  52%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==========================                      |  54%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===========================                     |  56%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |============================                    |  58%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value
```

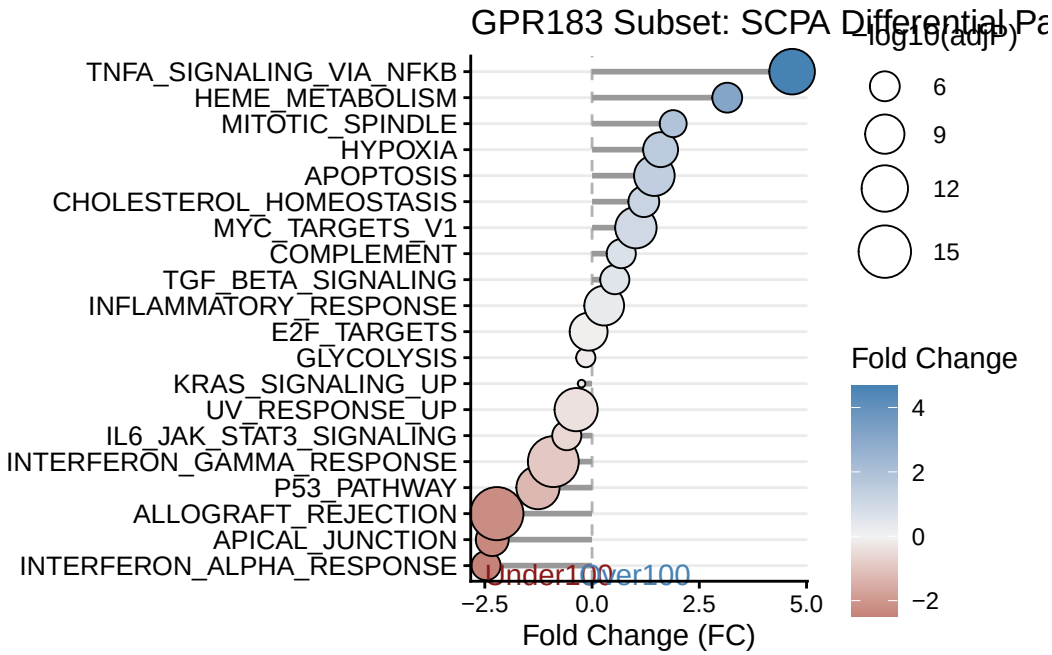
```
  |
  |============================                      |  60%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=============================                     |  62%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==============================                    |  64%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===============================                   |  66%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================                  |  68%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=================================                 |  70%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==================================                |  72%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===================================               |  74%
```

```
Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |====================================            |  76%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=====================================           |  78%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |======================================          |  80%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=======================================         |  82%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |========================================        |  84%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |=========================================       |  86%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==========================================      |  88%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value
```

```
  |
  |=============================================     |  90%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==============================================    |  92%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |===============================================   |  94%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================================  |  96%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |================================================= |  98%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value


  |
  |==================================================| 100%

Warning in initialize(value, ...): There must be an even number of elements
Adding a ghost value
```

```r
scpa_results <- scpa_results %>%  mutate(adjPvalBH = p.adjust(Pval, method = "BH"),
                                         logP = -log10(adjPvalBH)) %>%
  arrange(desc(qval))

setDT(scpa_results)
scpa_results[, direction := ifelse(FC > 0, "Over100", "Under100")]
scpa_results[, pathway := gsub("HALLMARK_", "", Pathway)]
plot_data <- scpa_results[order(-qval), head(.SD, 10), by = direction]
plot_data <- plot_data[order(FC)]
plot_data[, pathway := factor(pathway, levels = pathway)]
```

```
ggplot(plot_data, aes(x = FC, y = pathway)) +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey70") +
  geom_segment(aes(x = 0, xend = FC, y = pathway, yend = pathway),
               color = "grey60", linewidth = 1) +
  geom_point(aes(size = logP, fill = FC), shape = 21, color = "black") +
  scale_fill_gradient2(low = "firebrick4", mid = "grey95", high = "steelblue",
                       midpoint = 0, name = "Fold Change") +
  scale_size_continuous(range = c(1, 9), name = "-log10(adjP)") +
  theme_classic() +
  labs(title = "GPR183 Subset: SCPA Differential Pathways",
       x = "Fold Change (FC)",  y = NULL) +
  annotate("text", x = -1, y = 0.65, label = "Under100", color = "firebrick4") +
  annotate("text", x = 1, y = 0.65, label = "Over100", color = "steelblue") +
  coord_cartesian(clip = "off") +
  theme(axis.text.y = element_text(size = 10, color = "black"),
    panel.grid.major.y = element_line(color = "grey92", linewidth = 0.5) )
```



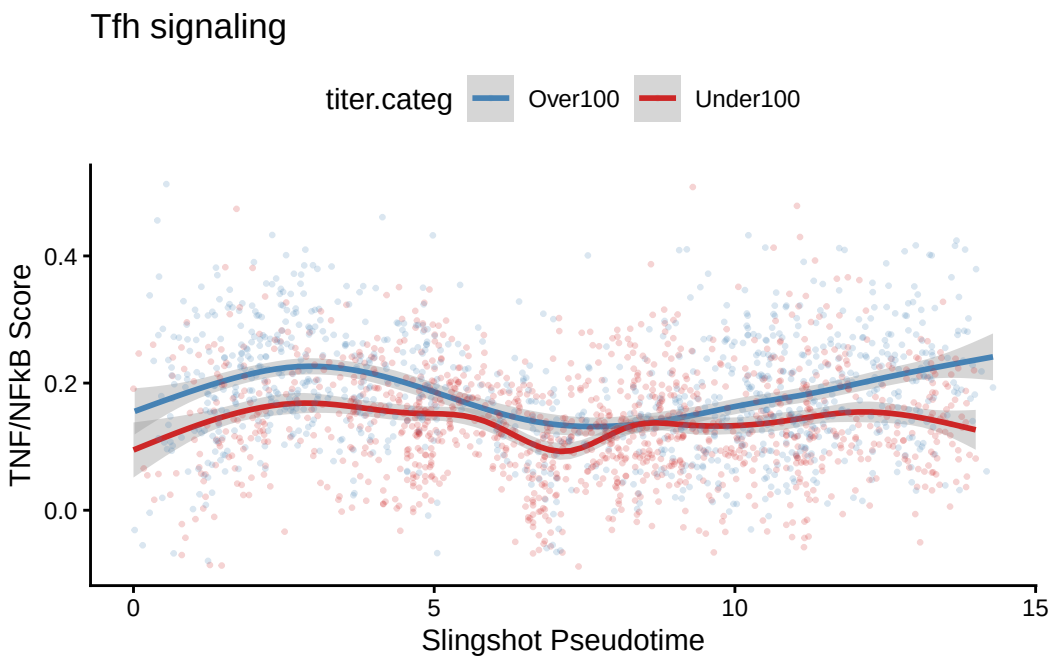

```
# ggsave(filename = "Images/GPR183-SCPA-Lollipop.pdf", height=6, width=7)
```

GRN analysis using decoupleR

```
hallmark <- GSEABase::getGmt("h.all.v2026.1.Hs.symbols.gmt")
genes <- GSEABase::geneIds(hallmark[["HALLMARK_TNFA_SIGNALING_VIA_NFKB"]])

tfh <- AddModuleScore(object = tfh, features = list(genes),
                      name = "HALLMARK_TNFA_SIGNALING_VIA_NFKB")
#
# ggplot(tfh@meta.data, aes(x = tfh.subset.annotation, y = HALLMARK_TNFA_SIGNALING_VIA_NFKB1,
#                           fill = titer.categ)) +
#   geom_violin(alpha = 0.5, color = "black", width = 0.7, position = position_dodge(0.8)) +
#   stat_summary(fun = median, fun.min = median, fun.max = median,
```

```
#                geom = "errorbar", width = 0.5, size = 0.5,
#                color = "black", position = position_dodge(0.8)) +
#   ggpubr::stat_compare_means(aes(group = titer.categ),
#                      label = "p.format", method = "wilcox.test", hide.ns = FALSE,
#                      label.y = 0.48) +
#   theme_classic() + coord_flip() +
#   scale_fill_manual(values = c("Under100" = "firebrick3", "Over100" = "steelblue")) +
#   labs(title = "Seurat module scores for TNF/NFkB",
#        subtitle = "Wilcoxon Rank Sum Test; Median shown",
#        x = "Tfh Subset", y = "Module Score") +
#   theme(axis.text = element_text(color = "black"), legend.position = "top",
#         axis.title.y = element_blank())


md <- as.data.table(tfh@meta.data)
md <- md[!is.na(Pseudotime_Lin1)] # Remove cells with no PseudoTime
ggplot(md, aes(x = Pseudotime_Lin1, y = HALLMARK_TNFA_SIGNALING_VIA_NFKB1,
               color = titer.categ)) +
  geom_smooth(method = "gam", se = TRUE) +
  geom_point(alpha = 0.2, size = 0.5) +
  scale_color_manual(values = c("Under100" = "firebrick3", "Over100" = "steelblue")) +
  theme_classic() +
  labs(title = "Tfh signaling", x = "Slingshot Pseudotime", y = "TNF/NFkB Score") +
  theme(legend.position = "top")
```

```
`geom_smooth()` using formula = 'y ~ s(x, bs = "cs")'
```




```
md$titer.categ <- as.factor(md$titer.categ)
gam_mod <- mgcv::gam(HALLMARK_TNFA_SIGNALING_VIA_NFKB1 ~ titer.categ +
               s(Pseudotime_Lin1, by = titer.categ), data = md)
summary(gam_mod)
```

```
Family: gaussian
Link function: identity

Formula:
HALLMARK_TNFA_SIGNALING_VIA_NFKB1 ~ titer.categ + s(Pseudotime_Lin1,
    by = titer.categ)

Parametric coefficients:
                     Estimate Std. Error t value Pr(>|t|)
(Intercept)          0.180598   0.002840   63.59   <2e-16 ***
titer.categUnder100 -0.040569   0.003839  -10.57   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
                                        edf Ref.df      F p-value
s(Pseudotime_Lin1):titer.categOver100  6.306  7.467 17.889  <2e-16 ***
s(Pseudotime_Lin1):titer.categUnder100 7.945  8.711  6.376  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.143   Deviance explained = 14.8%
GCV = 0.0074847  Scale est. = 0.0074299  n = 2220
```

```
library(decoupleR)
```

```
Attaching package: 'decoupleR'

The following object is masked from 'package:data.table':

    :=
```

```
library(OmnipathR)
```

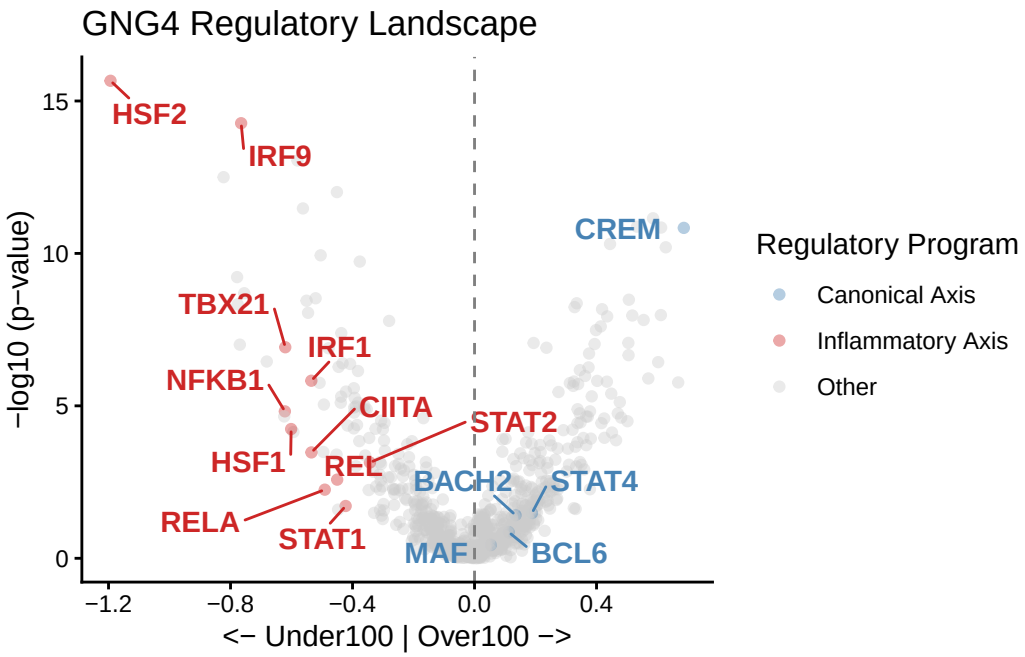
```
Attaching package: 'OmnipathR'

The following object is masked from 'package:readr':

    read_log
```

```
net <- decoupleR::get_collectri(organism = 'human', split_complexes = FALSE)
```

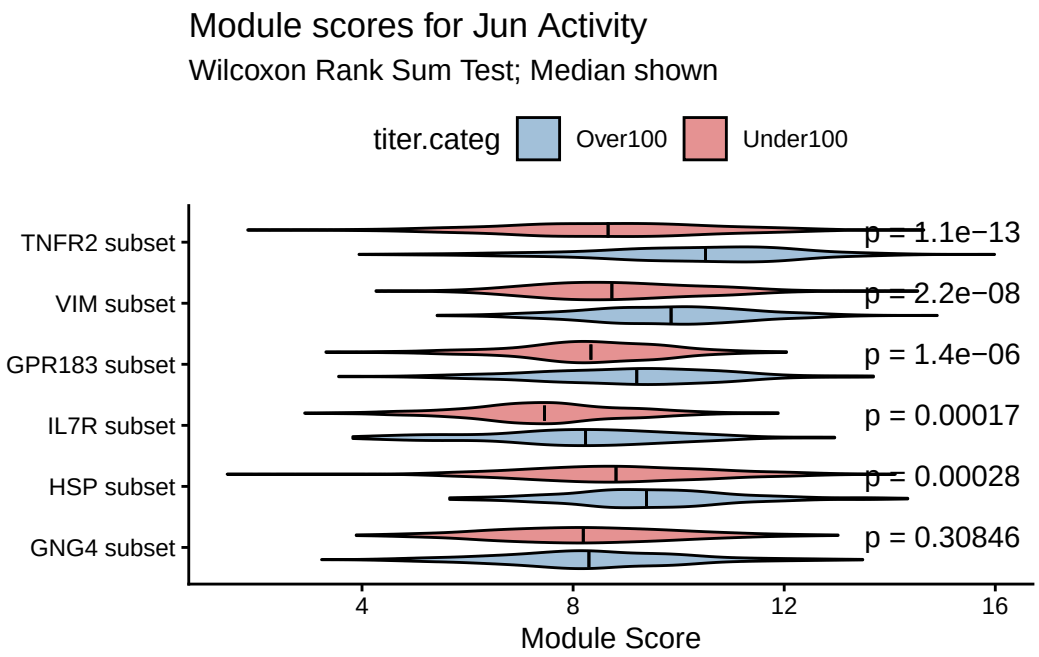
```
Warning in OmnipathR::import_tf_mirna_interactions(genesymbols = TRUE, resources = "CollecTRI", : 'OmnipathR::im
Use 'tf_mirna' instead.
See help("Deprecated")
```

```
tf_activity.collectri  <- decoupleR::run_ulm(mat = as.matrix(tfh@assays$RNA@data),
                                net = net, .source = 'source', .target = 'target',
                                .mor = 'mor', minsize = 5)
tfh$JUN_Activity <- tf_activity.collectri[tf_activity.collectri$source == "JUN", ]$score
tfh$FOS_Activity <- tf_activity.collectri[tf_activity.collectri$source == "FOS", ]$score
```

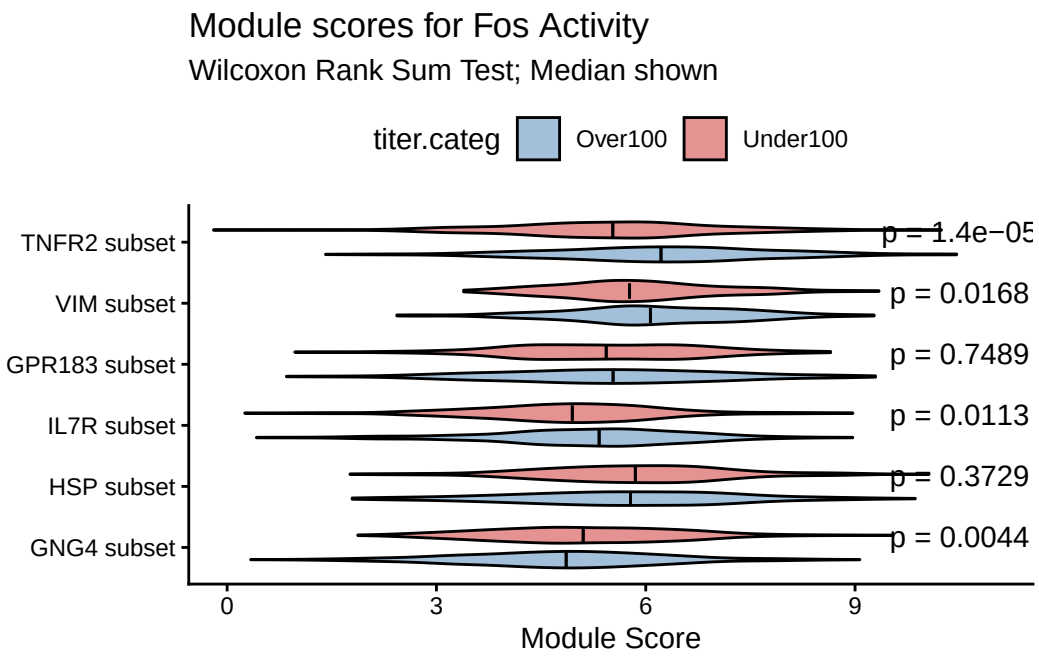
```r
md <- as.data.table(tfh@meta.data, keep.rownames = "condition")
setDT(tf_activity.collectri)
tf_activity.collectri <- merge(tf_activity.collectri,
                               md[, .(condition, titer.categ, tfh.subset.annotation)],
                               by = "condition" )

setDT(tf_activity.collectri)
tf.stats <- tf_activity.collectri[, .(LFC_Activity = mean(score[titer.categ == "Over100"]) - mean(score[titer.ca
                    p_val = wilcox.test(score ~ titer.categ)$p.value), by = source]
tf.stats[, fdr := p.adjust(p_val, method = "BH")]
setorder(tf.stats, -LFC_Activity)
subset.tf.stats <- tf_activity.collectri[, .(
  LFC_Activity = mean(score[titer.categ == "Over100"]) -
    mean(score[titer.categ == "Under100"]),
  p_val = wilcox.test(score ~ titer.categ)$p.value), by = .(source, tfh.subset.annotation)]

gng4.tf.top <- subset.tf.stats[tfh.subset.annotation == "GNG4 subset"][order(-LFC_Activity)]
plot_tfs <- gng4.tf.top[p_val < 0.05][order(LFC_Activity)]
plot_tfs <- rbind(head(plot_tfs, 15), tail(plot_tfs, 15))

StrongAb <- c("BACH2", "STAT4", "BCL6", "MAF", "CREM")
PoorAb <- c("STAT1", "STAT2", "IRF9", "IRF1", "CIITA", "TBX21", "RELA", "REL", "NFKB1", "HSF2", "HSF1")
gng4.tf.top[, label_group := "Other"]
gng4.tf.top[source %in% StrongAb, label_group := "Canonical Axis"]
gng4.tf.top[source %in% PoorAb, label_group := "Inflammatory Axis"]
ggplot(gng4.tf.top, aes(x = LFC_Activity, y = -log10(p_val), color = label_group)) +
  geom_point(alpha = 0.4, size = 1.5) +
  ggrepel::geom_text_repel(data = gng4.tf.top[label_group != "Other"],
                           aes(label = source), box.padding = 0.5,
                           max.overlaps = Inf, fontface = "bold",
                  size = 4, show.legend = FALSE) +
  scale_color_manual(values = c("Canonical Axis" = "steelblue",
                                "Inflammatory Axis" = "firebrick3",
                                "Other" = "grey80")) +
  theme_classic() +
  labs(title = "GNG4 Regulatory Landscape", x = "← Under100 | Over100 →",
       y = "-log10 (p-value)", color = "Regulatory Program") +
  geom_vline(xintercept = 0, linetype = "dashed", color = "grey50") +
  theme(legend.position = "right")
```




```
ggplot(tfh@meta.data, aes(x = tfh.subset.annotation, y = JUN_Activity,
                          fill = titer.categ)) +
  geom_violin(alpha = 0.5, color = "black", width = 0.7, position = position_dodge(0.8)) +
  stat_summary(fun = median, fun.min = median, fun.max = median,
               geom = "errorbar", width = 0.5, size = 0.5,
               color = "black", position = position_dodge(0.8)) +
  ggpubr::stat_compare_means(aes(group = titer.categ),
                     label = "p.format", method = "wilcox.test", hide.ns = FALSE,
                     label.y = 15) +
  theme_classic() + coord_flip() +
  scale_fill_manual(values = c("Under100" = "firebrick3", "Over100" = "steelblue")) +
  labs(title = "Module scores for Jun Activity",
       subtitle = "Wilcoxon Rank Sum Test; Median shown",
       x = "Tfh Subset", y = "Module Score") +
  theme(axis.text = element_text(color = "black"), legend.position = "top",
        axis.title.y = element_blank())
```




```
ggplot(tfh@meta.data, aes(x = tfh.subset.annotation, y = FOS_Activity,
                          fill = titer.categ)) +
  geom_violin(alpha = 0.5, color = "black", width = 0.7, position = position_dodge(0.8)) +
  stat_summary(fun = median, fun.min = median, fun.max = median,
               geom = "errorbar", width = 0.5, size = 0.5,
               color = "black", position = position_dodge(0.8)) +
  ggpubr::stat_compare_means(aes(group = titer.categ),
                     label = "p.format", method = "wilcox.test", hide.ns = FALSE,
                     label.y = 10.5) +
  theme_classic() + coord_flip(ylim = c(0,11)) +
  scale_fill_manual(values = c("Under100" = "firebrick3", "Over100" = "steelblue")) +
  labs(title = "Module scores for Fos Activity",
       subtitle = "Wilcoxon Rank Sum Test; Median shown",
       x = "Tfh Subset", y = "Module Score") +
  theme(axis.text = element_text(color = "black"), legend.position = "top",
        axis.title.y = element_blank())
```




```
rm(gam_mod, StrongAb, PoorAb, gng4.tf.top, tf.stats, subset.tf.stats)
```

Visium HD tonsil dataset

```
pos <- read_parquet("tissue_positions.parquet")
write.csv(pos, "tissue_positions.csv", row.names = FALSE)

#
# options(future.globals.maxSize = 16000 * 1024^2)
# data_dir <- "Visium_HD_Human_Tonsil_Fresh_Frozen_IF_binned_outputs/binned_outputs/square_016um"
# tonsil <- Load10X_Spatial(data.dir = data_dir, filename = "filtered_feature_bc_matrix.h5",
#   assay = "Spatial")
# tonsil <- SCTransform(tonsil, assay = "Spatial", verbose = FALSE)

tonsil <- qs2::qs_read(file = "Visium_HD_Human_Tonsil_Fresh_Frozen_IF_binned_outputs/CD4_Tonsil_08um.qs2", nthre
DefaultAssay(tonsil) <- 'SCT'
SpatialFeaturePlot(tonsil, features = 'BCL6', pt.size.factor = 5,
  alpha = c(0.1, 1) )
SpatialDimPlot(tonsil, pt.size.factor = 0.5) +  theme_bw() +
  theme(axis.title = element_text(size = 12), axis.text = element_text(size = 10))
markers <- c("MS4A1", "CD19", "CD3D", "CD3E", "LYZ", "KRT8")
DotPlot(tonsil, features = markers) + RotatedAxis()
SpatialFeaturePlot(tonsil, features = c("CD4", "CD8A"),
                   pt.size.factor = 1.2, ncol = 2, alpha = c(0.1, 1))

t_cell_clusters <- c(2, 7, 22)
DotPlot(tonsil, idents = c(2, 7, 22), features = c("CD3D", "CD4", "CD8A", "GNG4"))
VariableFeatures(tonsil, assay = "Spatial") <- VariableFeatures(tonsil[["sketch"]])
features_to_use <- VariableFeatures(tonsil[["sketch"]])
tonsil <- NormalizeData(tonsil, assay = "Spatial")
tonsil <- ScaleData(tonsil, assay = "Spatial", features = features_to_use)
tonsil <- RunPCA(tonsil, assay = "Spatial", reduction.name = "pca.full", npcs = 50)
```

```r
tonsil <- ProjectData( object = tonsil, assay = "Spatial", dims = 1:50,
  full.reduction = "pca",  sketched.assay = "sketch", sketched.reduction = "pca")

t_cells <- subset(tonsil, idents = c(2, 7, 22))
SpatialFeaturePlot(t_cells, features = c("CD4"), pt.size.factor = 5,
                   ncol = 2, alpha = c(0.1, 1))
t_cells <- SCTransform(t_cells, assay = "Spatial", verbose = TRUE)
t_cells <- RunPCA(t_cells, verbose = FALSE)
t_cells <- RunUMAP(t_cells, dims = 1:30)
t_cells <- FindNeighbors(t_cells, dims = 1:30)
t_cells <- FindClusters(t_cells, resolution = 0.8)
feats <- c("CD4", "GNG4", "PDCD1", "CXCL13", "ICOS", "MAF", "BCL6")
DotPlot(t_cells, features = feats) + RotatedAxis() + ggtitle("Tfh Cluster Identification")
```

```r
tonsil <- RunPCA(tonsil, assay = "SCT", verbose = FALSE)
tonsil <- FindNeighbors(tonsil, dims = 1:30)
tonsil <- FindClusters(tonsil, resolution = 0.5)
tonsil <- RunUMAP(tonsil, dims = 1:30)
DimPlot(tonsil, reduction = "umap", label = TRUE)
```

```r
sessionInfo()
```

```
R version 4.4.3 (2025-02-28 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default


locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/New_York
tzcode source: internal

attached base packages:
[1] stats4    stats     graphics  grDevices utils     datasets  methods
[8] base

other attached packages:
 [1] OmnipathR_3.14.0            decoupleR_2.12.0
 [3] slingshot_2.14.0            TrajectoryUtils_1.14.0
 [5] princurve_2.1.6             fgsea_1.32.4
 [7] ggpubr_0.6.0                VISION_3.0.2
 [9] miloR_2.2.0                 edgeR_4.4.2
[11] circlize_0.4.16             SeuratExtend_1.2.8
[13] SeuratExtendData_0.3.0      ggrepel_0.9.6
[15] data.table_1.17.0           rstatix_0.7.2
[17] limma_3.62.2                scuttle_1.16.0
[19] glmGamPoi_1.18.0            ggbiplot_0.6.2
[21] GSVA_2.0.7                  GSEABase_1.68.0
```

```
[23] graph_1.84.1                annotate_1.84.0
[25] XML_3.99-0.22               AnnotationDbi_1.68.0
[27] DESeq2_1.46.0               readxl_1.4.5
[29] SingleCellExperiment_1.28.1 SummarizedExperiment_1.36.0
[31] Biobase_2.66.0              GenomicRanges_1.58.0
[33] GenomeInfoDb_1.42.3         IRanges_2.40.1
[35] S4Vectors_0.44.0            BiocGenerics_0.52.0
[37] MatrixGenerics_1.18.1       matrixStats_1.5.0
[39] pheatmap_1.0.13             reshape2_1.4.4
[41] scRepertoire_2.2.1          patchwork_1.3.0
[43] Seurat_5.2.1                SeuratObject_5.0.2
[45] sp_2.2-0                    RColorBrewer_1.1-3
[47] viridis_0.6.5               viridisLite_0.4.2
[49] lubridate_1.9.4             forcats_1.0.1
[51] stringr_1.5.1               dplyr_1.1.4
[53] purrr_1.0.4                 readr_2.1.5
[55] tidyr_1.3.1                 tibble_3.2.1
[57] ggplot2_4.0.0               tidyverse_2.0.0

loaded via a namespace (and not attached):
  [1] igraph_2.1.4              ica_1.0-3
  [3] plotly_4.10.4             Formula_1.2-5
  [5] zlibbioc_1.52.0           tidyselect_1.2.1
  [7] rvest_1.0.5               bit_4.6.0
  [9] doParallel_1.0.17         clue_0.3-66
 [11] lattice_0.22-6            rjson_0.2.23
 [13] evmix_2.12                blob_1.3.0
 [15] S4Arrays_1.6.0            parallel_4.4.3
 [17] png_0.1-8                 cli_3.6.4
 [19] goftest_1.2-3             textshaping_1.0.0
 [21] pbmcapply_1.5.1           BiocNeighbors_2.0.1
 [23] stringdist_0.9.17         uwot_0.2.3
 [25] curl_6.2.1                mime_0.13
 [27] evaluate_1.0.3            ComplexHeatmap_2.22.0
 [29] stringi_1.8.4             backports_1.5.0
 [31] gsl_2.1-8                 httpuv_1.6.15
 [33] magrittr_2.0.3            rappdirs_0.3.3
 [35] splines_4.4.3             webutils_1.2.2
 [37] mclust_6.1.2              ggraph_2.2.1
 [39] logger_0.4.1              sctransform_0.4.1
 [41] ggbeeswarm_0.7.2          logging_0.10-108
 [43] DBI_1.2.3                 HDF5Array_1.34.0
 [45] withr_3.0.2               systemfonts_1.3.2
 [47] lmtest_0.9-40             tidygraph_1.3.1
 [49] SCPA_1.6.2                htmlwidgets_1.6.4
 [51] labeling_0.4.3            SparseArray_1.6.2
 [53] swagger_5.17.14.1         cellranger_1.1.0
 [55] sparsesvd_0.2-3           reticulate_1.41.0.1
 [57] zoo_1.8-13                XVector_0.46.0
 [59] knitr_1.50                UCSC.utils_1.2.0
 [61] RhpcBLASctl_0.23-42       timechange_0.3.0
 [63] foreach_1.5.2             grid_4.4.3
 [65] rhdf5_2.50.2              vegan_2.7-3
 [67] quantreg_6.1              R.oo_1.27.0
 [69] RSpectra_0.16-2           irlba_2.3.5.1
 [71] ggrastr_1.0.2             fastDummies_1.7.5
```

```
 [73] lazyeval_0.2.2            yaml_2.3.10
 [75] survival_3.8-3            SpatialExperiment_1.16.0
 [77] scattermore_1.2           crayon_1.5.3
 [79] RcppAnnoy_0.0.22          progressr_0.15.1
 [81] tweenr_2.0.3              later_1.4.1
 [83] ggridges_0.5.6            codetools_0.2-20
 [85] base64enc_0.1-3           GlobalOptions_0.1.2
 [87] KEGGREST_1.46.0           Rtsne_0.17
 [89] wordspace_0.2-9           shape_1.4.6.1
 [91] fastICA_1.2-7             foreign_0.8-88
 [93] pkgconfig_2.0.3           xml2_1.5.2
 [95] spatstat.univar_3.1-2     iNEXT_3.0.2
 [97] evd_2.3-7.1               spatstat.sparse_3.1-0
 [99] ape_5.8-1                 xtable_1.8-4
[101] car_3.1-3                 plyr_1.8.9
[103] httr_1.4.7                tools_4.4.3
[105] globals_0.16.3            beeswarm_0.4.0
[107] htmlTable_2.4.3           broom_1.0.7
[109] checkmate_2.3.2           nlme_3.1-167
[111] selectr_0.5-1             MatrixModels_0.5-3
[113] assertthat_0.2.1          digest_0.6.37
[115] permute_0.9-10            numDeriv_2016.8-1.1
[117] Matrix_1.7-2              farver_2.1.2
[119] tzdb_0.5.0                rpart_4.1.24
[121] glue_1.8.0                cachem_1.1.0
[123] nbpMatching_1.5.6         polyclip_1.10-7
[125] Hmisc_5.2-5               generics_0.1.3
[127] Biostrings_2.74.1         ggalluvial_0.12.6
[129] presto_1.0.0              parallelly_1.42.0
[131] multicross_2.1.0          statmod_1.5.0
[133] RcppHNSW_0.6.0            ragg_1.3.3
[135] crossmatch_1.3-1          ScaledMatrix_1.14.0
[137] carData_3.0-5             pbapply_1.7-2
[139] iotools_0.4-0             vroom_1.6.5
[141] spam_2.11-1               utf8_1.2.4
[143] graphlayouts_1.2.2        gtools_3.9.5
[145] ggsignif_0.6.4            gridExtra_2.3
[147] shiny_1.10.0              GenomeInfoDbData_1.2.13
[149] plumber_1.3.3             R.utils_2.13.0
[151] rhdf5filters_1.18.1       memoise_2.0.1
[153] rmarkdown_2.29            scales_1.4.0
[155] R.methodsS3_1.8.2         future_1.34.0
[157] RANN_2.6.2                Cairo_1.7-0
[159] spatstat.data_3.1-6       rstudioapi_0.17.1
[161] cluster_2.1.8             msigdbr_25.1.1
[163] spatstat.utils_3.1-3      hms_1.1.3
[165] fitdistrplus_1.2-2        cowplot_1.1.3
[167] colorspace_2.1-1          rlang_1.1.5
[169] ggdendro_0.2.0            quadprog_1.5-8
[171] DelayedMatrixStats_1.28.1 sparseMatrixStats_1.18.0
[173] truncdist_1.0-2           dotCall64_1.2
[175] ggforce_0.4.2             mgcv_1.9-1
[177] xfun_0.51                 iterators_1.0.14
[179] abind_1.4-8               Rhdf5lib_1.28.0
[181] promises_1.3.2            RSQLite_2.4.6
[183] DelayedArray_0.32.0       harmony_1.2.3
```

```
[185] compiler_4.4.3            prettyunits_1.2.0
[187] beachmat_2.22.0           SparseM_1.84-2
[189] listenv_0.9.1             Rcpp_1.0.14
[191] BiocSingular_1.22.0       tensor_1.5
[193] MASS_7.3-64               progress_1.2.3
[195] BiocParallel_1.40.0       cubature_2.1.4-1
[197] babelgene_22.9            spatstat.random_3.3-2
[199] R6_2.6.1                  fastmap_1.2.0
[201] fastmatch_1.1-6           vipor_0.4.7
[203] ROCR_1.0-11               rsvd_1.0.5
[205] nnet_7.3-20               gtable_0.3.6
[207] phangorn_2.12.1           KernSmooth_2.23-26
[209] miniUI_0.1.1.1            deldir_2.0-4
[211] htmltools_0.5.8.1         bit64_4.6.0-1
[213] spatstat.explore_3.3-4    lifecycle_1.0.5
[215] S7_0.2.0                  zip_2.3.2
[217] vctrs_0.6.5               VGAM_1.1-14
[219] spatstat.geom_3.3-6       future.apply_1.11.3
[221] pracma_2.4.6              pillar_1.11.1
[223] magick_2.9.1              locfit_1.5-9.12
[225] jsonlite_1.9.1            GetoptLong_1.0.5
```

```
obj1.clean$predicted.celltype.l1 <- obj1.clean$Comments <- obj1.clean$Randomized <- obj1.clean$V2weightkg <- obj
  obj1.clean$Notes <- obj1.clean$scDblFinder.class <-   obj1.clean$HBsAb2 <- obj1.clean$HBsAb2date <-  NULL
# saveRDS(obj1.clean, "20260311_obj1.clean_subclustered.RDS")

vis.cd4 <- Vision(obj1.clean, signatures = "h.all.v2026.1.Hs.symbols.gmt")
vis.cd4 <- analyze(vis.cd4)
vis.cd4 <- runHotspot(vis.cd4, model = "danb")
viewResults(vis.cd4)
saveRDS(vis.cd4, "cd4_vision_hotspot_results.rds")
obj1.clean <- AddMetaData(obj1.clean, metadata = as.data.frame( vis.cd4@ModuleHotspotScores ))
VlnPlot2(obj1.clean, features = "HOTSPOT_2", group.by = "titer.categ", pt.size = 0.1, stat.method = "wilcox.test
```